Space Exploration & Aerospace Engineering

Part IX

Human Spaceflight and Habitation

Keeping people alive and productive in space is the hardest engineering problem of all. Part IX explains the human body in space, life support, crewed spacecraft, space stations and the spacesuits that let astronauts work outside.



Chapters 30–33 · pages 526–609 of the complete manual
First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

Space Exploration & Aerospace Engineering

A complete, illustrated guide to the space ecosystem — from stratospheric balloons to interstellar probes — explaining how every major class of space system works, how it is engineered, and where artificial intelligence is changing it.

First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

Typography and production

Set in Source Serif 4, Inter and JetBrains Mono. Equations typeset with MathJax. Produced from Markdown sources with an automated pipeline (pandoc, Chromium, pypdf); the sources and build scripts accompany the PDF.

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This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example “Chapter 24”) and to the appendices refer to the complete book.



PART IX

Human Spaceflight and Habitation

Keeping people alive and productive in space is the hardest engineering problem of all. Part IX explains the human body in space, life support, crewed spacecraft, space stations and the spacesuits that let astronauts work outside.

30 Human Spaceflight Fundamentals and Life Support

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Image: NASA/JSC

CHAPTER

30 Human Spaceflight Fundamentals and Life Support



Image: NASA/JSC

A human being is the most demanding payload a spacecraft can carry. Every day each crew member consumes almost a kilogram of oxygen and several kilograms of water and food, and produces carbon dioxide, heat and waste that must be removed. Weightlessness weakens bones, muscles, the heart and even the eyes, while cosmic radiation slowly raises the risk of cancer. This chapter explains how the body adapts to spaceflight and how crews are protected: the radiation environment and dose limits, the design of cabin atmospheres, the metabolic budget, and the environmental control and life support systems (ECLSS) that recycle air and water on the International Space Station. It closes with food, medicine, psychology, the history of fatal accidents and the safety standards that govern crewed vehicles in 2026.

IN THIS CHAPTER

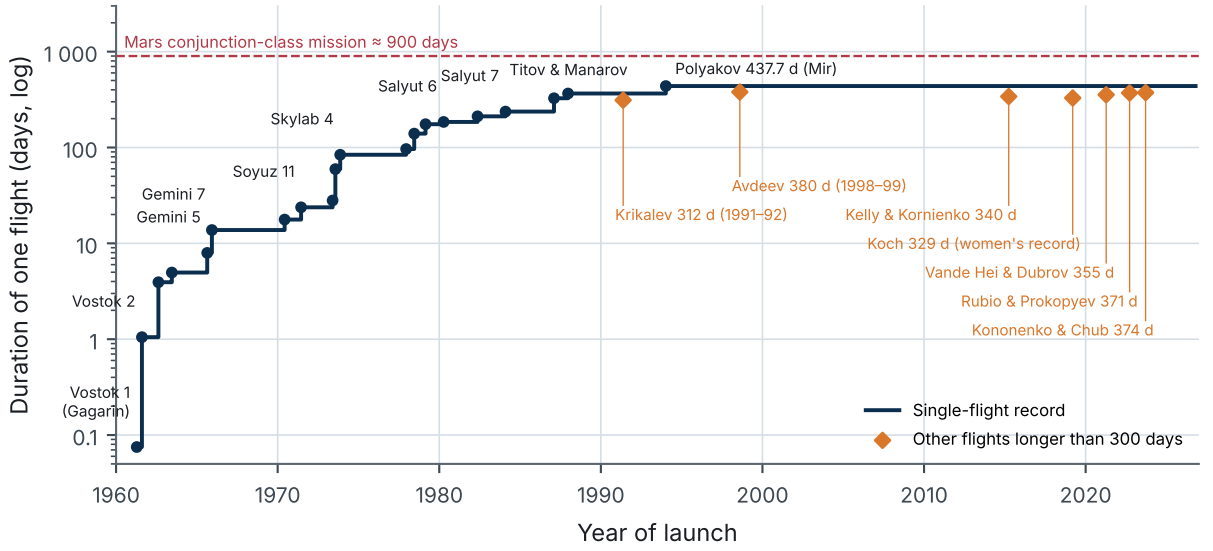
30.1 Humans in Space: How Long and How Far
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30.3 Radiation: The Long-Term Hazard
30.4 Designing the Cabin Atmosphere
30.5 The Human Metabolic Budget
30.6 Environmental Control and Life Support Systems
30.7 Living and Working in Space
30.8 Crew Safety and Risk

On 12 April 1961, Yuri Gagarin completed one orbit of the Earth in Vostok 1 and landed 108 minutes after launch. By mid-2026 about 780 people from 55 countries had flown above 100 km — a crew small enough to fit into a single airliner, but one that has lived continuously in orbit since November 2000. People are sent into space because they bring what machines still lack: judgement in unforeseen situations, dexterity for repairs and assembly, and a field scientist's eye. Five Shuttle crews serviced and upgraded the Hubble Space Telescope, and Apollo astronauts chose the most telling rocks on the Moon.

The price is steep. A crewed spacecraft must provide a pressurised, breathable, temperature-controlled bubble. It must remove carbon dioxide, heat, moisture and waste, supply water and food, and protect against radiation and fire. It must keep the crew healthy and bring them home, all with far higher reliability than an uncrewed spacecraft. This chapter covers the biology and the life support engineering; the vehicles themselves are described in Chapter 31, space stations and habitats in Chapter 32, and spacesuits in Chapter 33.

30.1 Humans in Space: How Long and How Far

Human spaceflight has progressed in steps of duration (Figure 30.1). Gemini 7 showed in 1965 that people could survive the two weeks needed for a lunar mission. Skylab and the Soviet Salyut and Mir stations pushed single flights from one month to over a year. The record of **437.7 days**, set by the Russian physician Valeri Polyakov on Mir in 1994–95, still stands in 2026. The longest flights on the International Space Station (ISS) are those of Oleg Kononenko and Nikolai Chub (374 days, 2023–24). Next come Sergey Prokopyev, Dmitri Petelin and Frank Rubio (371 days, 2022–23), whose flight is the US record. It was extended when their Soyuz MS-22 sprang a coolant leak. In June 2024 Kononenko became the first person to spend 1 000 days in space, finishing his fifth flight with 1 111 days in total. Peggy Whitson holds the US and women's cumulative record of 695 days.



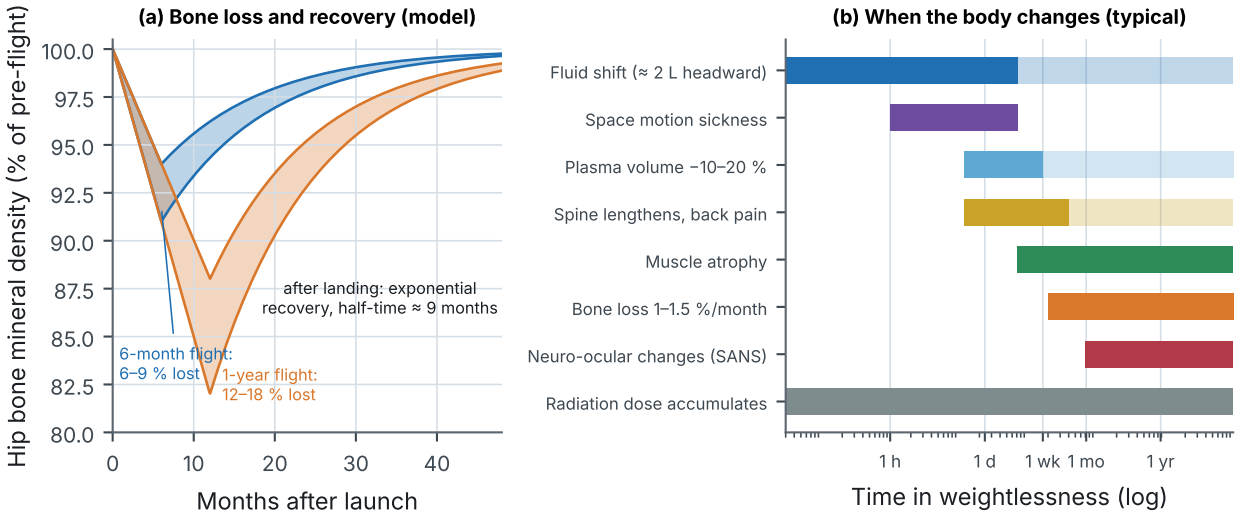
Record progression from mission durations; Kononenko holds the cumulative record, 1 111 days over five flights (2024).

FIGURE 30.1 Single-flight duration record from Vostok 1 to Polyakov's 437.7 days on Mir, with the other flights that have lasted longer than 300 days. A crewed Mars mission of the conjunction class would last about twice as long as the record.

A round trip to Mars would last 500–1 000 days with no resupply and no quick return. Everything in this chapter — from bone loss to recycling — becomes harder as missions grow longer and further from Earth.

30.2 The Human Body in Weightlessness

In orbit the crew falls freely with the spacecraft, so nothing presses the body against a floor (Chapter 2). Tissues tuned to 1 g respond within minutes to months (Figure 30.2). Most changes are adaptations rather than illnesses, but they become problems on return to gravity — on Earth, the Moon or Mars.



(a) Linear loss of 1–1.5 %/month in flight, exponential recovery with 9-month half-time (Sibonga et al. 2007); modern resistive exercise reduces the loss. (b) Dark: onset/main phase; pale: new steady state.

FIGURE 30.2 How the body changes in weightlessness. (a) A simple model of bone loss in the hip during 6- and 12-month flights and of recovery after landing. (b) Typical onset and duration of the main changes.

30.2.1 Fluids, Heart and Eyes

On Earth, gravity pulls blood and tissue fluid towards the legs. In weightlessness about two litres shift towards the chest and head within hours: faces become puffy, legs thin and noses feel blocked. The body treats the extra fluid in the chest as an overload and excretes it, so blood plasma volume falls by roughly 10–20 % within days, and the reflexes that control blood pressure weaken. After landing many astronauts suffer **orthostatic intolerance** — dizziness or fainting when standing up — so crews drink salt water before entry, wear compression garments and are often helped out of the capsule.

A more puzzling effect, first described in 2011, concerns the eyes. **Spaceflight-associated neuro-ocular syndrome (SANS)** is a set of changes to the eye. The optic disc swells where the nerve enters the eye, and folds form in the choroid layer behind the retina. The back of the eyeball flattens, and vision shifts towards long-sightedness. Imaging studies suggest that most long-duration crew members show some of these signs; in one prospective ISS study about two-thirds had optic-disc swelling by the 150th day of flight. The likely cause is the headward fluid shift, which keeps pressure in the veins of the head slightly raised all the time. Crews monitor their eyes with optical coherence tomography and ultrasound (Figure 30.4), and lower-body negative-pressure devices that draw fluid back towards the legs are being tested. Some changes persist for years, making SANS a leading health concern for Mars missions.

30.2.2 Balance and Motion Sickness

The inner ear's otolith organs sense linear acceleration and the direction of gravity. In free fall they send signals that conflict with what the eyes see. More than half of all space travellers suffer **space motion sickness** — nausea, vomiting, headache and lethargy — during the first two or three days. Spacewalks are not scheduled during this period, because vomiting inside a helmet could be fatal. The brain relearns to rely on vision and touch, but must readapt on return: astronauts walk unsteadily for days after landing — a serious matter for a crew stepping onto Mars after months in transit. Astronauts also grow up to about 3 % taller as the discs between the vertebrae expand, which often causes back pain.

30.2.3 Bone and Muscle

Bone is constantly rebuilt in response to the loads placed on it. Without body weight, the hip and lumbar spine lose mineral at **1–1.5 % per month** — about as much in one month as an elderly person loses in a year. The calcium released raises the risk of kidney stones. After landing, bone is rebuilt only slowly. A study of 45 crew members on 4–6-month flights found that it took about nine months to recover half of the loss. It also found that recovery was largely complete only after about three years. Figure 30.2 (a) shows what this implies for a one-year mission without effective exercise: a loss of 12–18 % in the hip, with years needed to recover.

Muscles that keep us upright — the calves, thighs and back — waste fastest. Even with exercise, ISS crews in a 2009 study lost about 13 % of calf muscle volume and almost a third of calf muscle power over six months. The main countermeasure is exercise, about two hours every day, on three machines:

- the **Advanced Resistive Exercise Device (ARED)**, which uses vacuum cylinders to provide up to about 2.7 kN (600 lbf) of resistance for squats, dead-lifts and heel raises (Figure 30.3);
- the **T2 treadmill**, on which the astronaut is held down by a harness and bungee cords;
- a **cycle ergometer** for aerobic exercise (CEVIS, replaced around 2023 by the smaller FERGO).

Since ARED arrived in late 2008, crews who also eat enough and take vitamin D have lost markedly less bone than earlier crews, and drugs that slow bone breakdown (bisphosphonates) have been tested in flight. Exercise has an engineering cost: more oxygen, heat and humidity, vibration that must be isolated from the structure, and equipment that a small Mars transit vehicle can barely accommodate.



FIGURE 30.3 NASA astronaut Nick Hague exercising on the Advanced Resistive Exercise Device in the ISS Tranquility module, October 2024. Image: NASA/JSC.



FIGURE 30.4 NASA astronaut Jonny Kim, assisted by JAXA astronaut Takuya Onishi, performing an eye ultrasound scan on the ISS in May 2025 to look for signs of SANS. Image: NASA/JSC.

30.2.4 Immune System, Sleep and the Twins Study

Spaceflight also disturbs the immune system: dormant herpes viruses reactivate and are shed in the saliva of many crew members, and skin rashes are common. Sleep is short — about six hours a night on average for Shuttle and ISS crews, despite 8.5 hours being scheduled — disturbed by noise, a sunrise every 92 minutes and workload.

The most detailed picture of these changes came from NASA's **Twins Study** (2015–16). Scott Kelly spent 340 days on the ISS while his identical twin Mark, also an astronaut, stayed on Earth as a genetic control (Figure 30.5). In flight, Scott's telomeres — the protective ends of the chromosomes — unexpectedly lengthened, then shortened rapidly after his return. Thousands of genes changed their activity, most returning to normal within six months. His cognitive test performance also dropped after landing. A 2024 package of 44 papers on the three-day private Inspiration4 flight found similar molecular changes even on a short mission, most of which faded within months. Spaceflight is a whole-body stress that people tolerate well for months, but whose long-term effects are still being measured.

Table 30.1 summarises the main effects and their countermeasures.

TABLE 30.1 Main effects of weightlessness on the human body and their countermeasures.

Body system	Effect in weightlessness	Typical size	Countermeasures
Fluids, heart	Headward fluid shift; fainting on return	≈ 2 L shifted; plasma −10–20 %	Fluid loading before entry, compression garments
Balance	Motion sickness; unsteady after landing	Over half of crew, first 2–3 days	Medication, post-flight rehabilitation
Bone	Loss from hip and spine; kidney stones	1–1.5 % per month	ARED exercise, diet, vitamin D, drugs
Muscle	Atrophy of postural muscles	Calf power ≈ −30 % in 6 months	Treadmill, ARED, ergometer, ≈ 2 h/day
Eyes (SANS)	Optic-disc swelling, choroidal folds	Signs in most long-duration crew	Monitoring; negative-pressure devices studied
Immune, sleep	Virus reactivation; ≈ 6 h of sleep	Many crew members	Lighting schedules, sleep stations

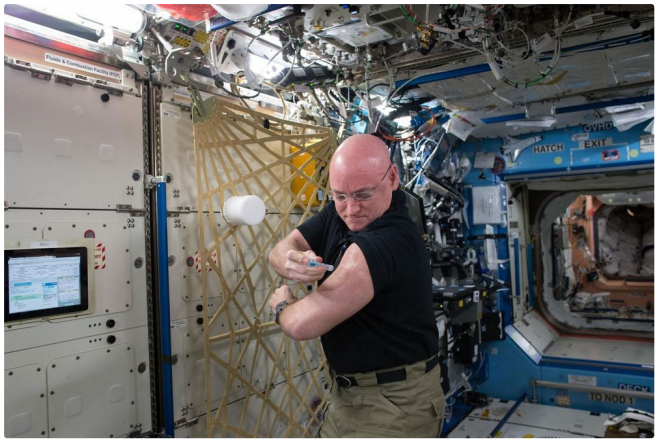


FIGURE 30.5 Scott Kelly giving himself a flu vaccination aboard the ISS in September 2015, part of the Twins Study investigation of how the immune system responds in flight. Image: NASA/JSC (Scott Kelly).

30.3 Radiation: The Long-Term Hazard

Radiation is the one hazard that grows steadily with every day outside the Earth's protection. Chapter 2 describes the sources and Chapter 22 the solar storms; for a crew they mean three things:

- **Galactic cosmic rays (GCRs)**: atomic nuclei at close to the speed of light, arriving continuously and almost impossible to stop. The rare heavy nuclei such as iron (**HZE** particles, for high charge Z and energy E) leave dense tracks of damage through cells.
- **Solar particle events (SPEs)**: rare bursts of protons that are easier to shield, but can deliver a dangerous dose within hours to an unprotected crew.
- **Trapped particles** in the Van Allen belts, which dominate the dose in low Earth orbit and during transits of the belts.

30.3.1 Dose, Risk and Limits

The energy deposited per kilogram of tissue is the **absorbed dose**, in grays (1 Gy = 1 J/kg). Weighting it for the greater damage done by heavy ions and for the sensitivity of each organ gives the **effective dose**, in sieverts (Sv). Doses above about 1 Gy within hours cause acute radiation sickness. At the lower doses of spaceflight, the concern is an increased lifetime risk of cancer, and possible damage to the heart and central nervous system.

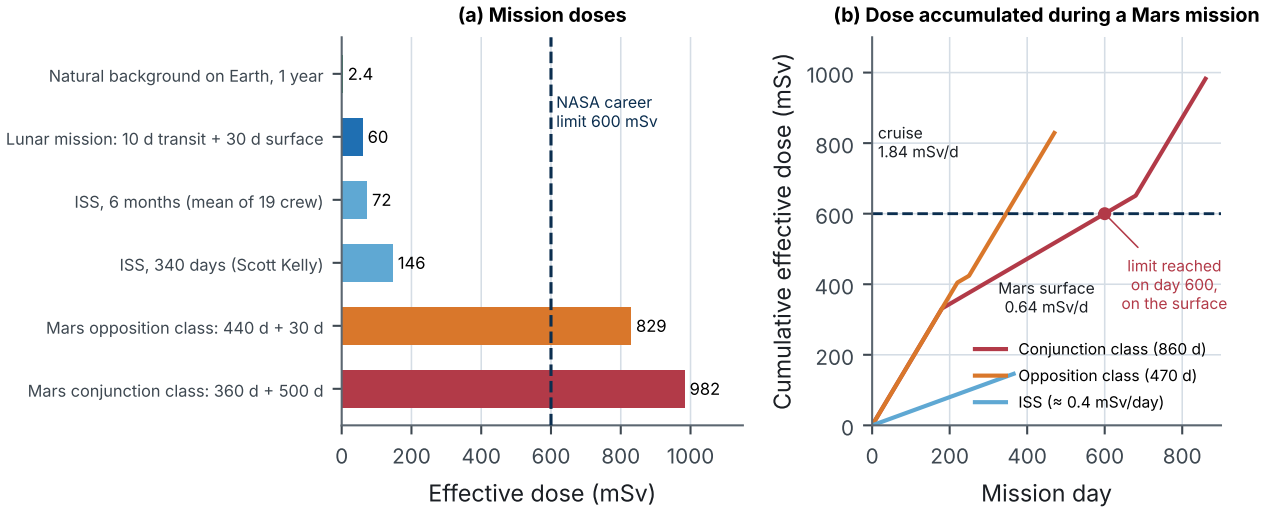
Table 30.2 lists measured dose rates. A six-month ISS mission gives about 70–160 mSv, depending on the solar cycle; Scott Kelly received 146 mSv in 340 days. Outside the magnetosphere the rate jumps. The Radiation Assessment Detector on the Curiosity rover measured 1.84 mSv per day during its cruise to Mars in 2011–12. On the surface, where the planet blocks half the sky, it measured 0.64 mSv per day. On the Moon, China's Chang'e 4 lander measured about 1.4 mSv per day.

In 2022 NASA adopted a single **career limit of 600 mSv** effective dose for all astronauts, regardless of age or sex. It corresponds to a calculated mean 3 % risk of dying from radiation-induced cancer for the most sensitive group (a 35-year-old woman). It replaced earlier limits that allowed older men a higher dose than young women. For comparison, international recommendations limit radiation workers on Earth to an average of 20 mSv per year. NASA also

requires shielding to keep the dose from a large SPE — modelled on the October 1989 event — below 250 mSv. Figure 30.6 compares mission doses with the limit. Low-Earth-orbit and lunar missions fit comfortably, but a Mars mission does not.

TABLE 30.2 Measured radiation dose rates for crews.

Environment	Dose rate	Notes
Earth's surface	≈ 0.007 mSv/day	2.4 mSv per year world average
ISS (400 km, 51.6°)	0.4–0.9 mSv/day	Mostly trapped protons and GCRs
Lunar surface	≈ 1.4 mSv/day	Chang'e 4, 2019 (solar minimum)
Interplanetary cruise	≈ 1.8 mSv/day	Curiosity RAD, 2011–12
Mars surface	≈ 0.64 mSv/day	Curiosity RAD, 2012–13
Large SPE in deep space	Up to hundreds of mSv in a day	Depends strongly on shielding



Rates: MSL/RAD cruise 1.84 mSv/day and Mars surface 0.64 mSv/day (2012–13); lunar surface 1.37 mSv/day (Chang'e 4 LND); ISS from Cucinotta et al. 2008 and the Twins Study. No large solar particle event included.

FIGURE 30.6 Radiation doses for crews. (a) Estimated effective doses for different missions compared with NASA's 600 mSv career limit. (b) Dose accumulated during conjunction-class and opposition-class Mars missions: the limit is reached during the surface stay.

30.3.2 Shielding and Storm Shelters

Shielding is specified by its **areal density** — mass per unit area, in g/cm² — because what matters is how much material a particle must cross. A spacecraft wall provides only a few g/cm²; racks, water and stowage inside add more. For SPE protons, shielding works well: the lower-energy protons stop, and NASA estimates that typical aluminium or polyethylene shielding halves the average SPE dose. For GCRs it barely helps — only about 7 %. This is because the most energetic nuclei pass through any practical thickness and break up into showers of **secondary** particles, including neutrons, that add their own dose. Hydrogen-rich materials such as polyethylene and water work better than aluminium because they fragment incoming ions without producing many secondaries.

The practical strategy is to accept the GCR dose, limit mission duration, and provide a **storm shelter** for SPEs. This is a small volume, surrounded by water, food and stowage, where the crew retreats for the hours to days of an event. Warnings come from the space-weather forecasters described in Chapter 22. During Artemis I in 2022 two female-torso phantoms, Helga and Zohar, flew in Orion's passenger seats carrying thousands of dosimeters (Figure 30.7). Dose rates

differed by a factor of four between the least and most shielded positions. Turning the spacecraft through 90° halved the dose rate in the inner belt. A well-placed shelter could keep the dose from a very large SPE below about 150 mSv. On Artemis II in April 2026 a planned in-flight shelter demonstration was shortened to prepare for entry. Building the shelter from stowage, already practised on the ground, was skipped. Only the ventilation of an astronaut sheltering in a locker was tested.



FIGURE 30.7 The radiation phantoms Helga (below) and Zohar (above) installed in Orion's passenger seats before Artemis I in August 2022. Zohar wears the AstroRad radiation-protection vest. Image: NASA/KSC (NASA/Frank Michaux).

EXAMPLE: RADIATION DOSE FOR A MARS MISSION

A conjunction-class mission spends 180 days travelling to Mars, 500 days on the surface and 180 days returning. Use the Curiosity RAD dose rates: 1.84 mSv/day in cruise and 0.64 mSv/day on the surface. Ignore SPEs, which the storm shelter is designed to handle.

1. Cruise dose: 360 days × 1.84 mSv/day = **662 mSv** — already above the 600 mSv career limit.
2. Surface dose: 500 days × 0.64 mSv/day = **320 mSv**.
3. Total: 982 mSv, **164 %** of the limit. Following the dose day by day (Figure 30.6 b), the limit is reached on mission day 600, after 420 of the 500 days on the surface.
4. An opposition-class mission (220 days out, 30 days on Mars, 220 days back) gives 1.84 × 440 + 0.64 × 30 = **829 mSv**. It is shorter, but still over the limit because more time is spent in deep space.
5. To stay within 600 mSv with a 500-day surface stay, the total transit time would have to be below (600 − 320)/1.84 = **152 days** — 76 days each way, beyond chemical propulsion. Thicker shielding hardly helps against GCRs; the options are faster propulsion (Chapter 42), flying near solar maximum when GCRs are weaker, or a deliberate decision to accept a higher risk.

30.4 Designing the Cabin Atmosphere

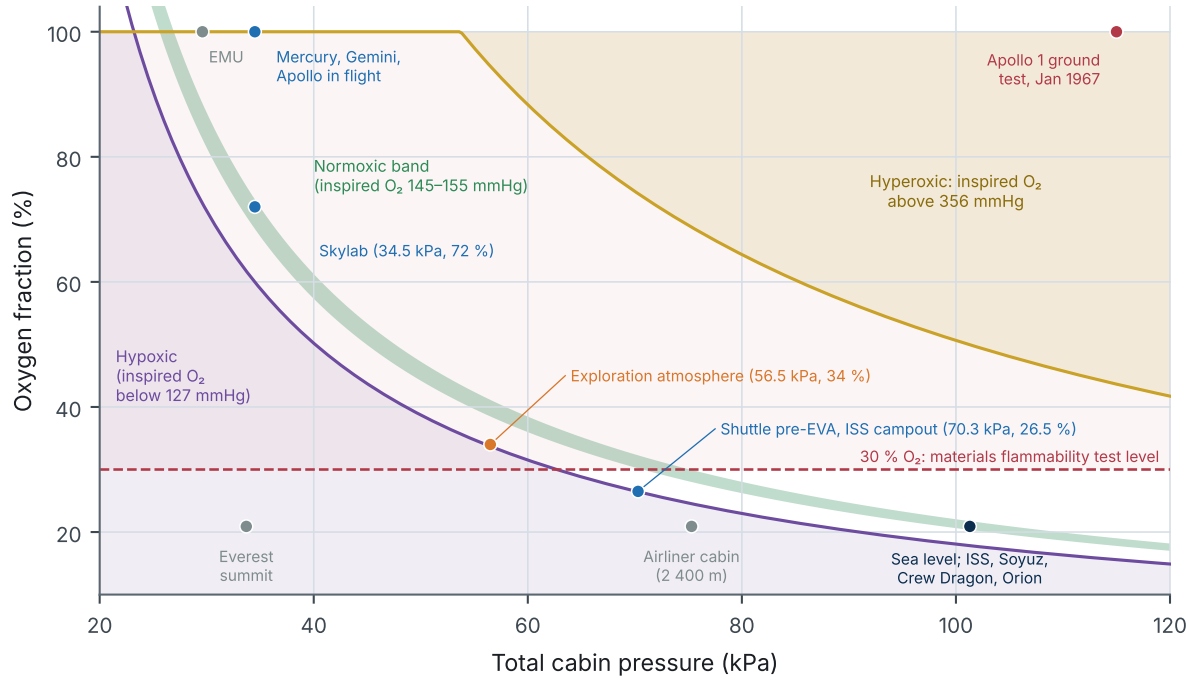
Engineers choose the cabin's total pressure and oxygen fraction, balancing four hazards: **hypoxia** (too little oxygen), **oxygen toxicity** (too much), **fire**, and **decompression sickness** when moving to a spacesuit at lower pressure. Lower pressure also reduces hull loads and leakage.

30.4.1 Oxygen and Hypoxia

What the lungs care about is the partial pressure of oxygen in the air reaching them. Air entering the lungs is saturated with water vapour at body temperature, which takes up 47 mmHg (6.3 kPa) regardless of altitude, so the **inspired oxygen partial pressure** is

$$P_{IO_2} = (P_{total} - 6.3\ \text{kPa}) \times F_{O_2}$$

where F_{O_2} is the oxygen fraction. At sea level this gives 149 mmHg (19.9 kPa). NASA's standard for crewed spacecraft (NASA-STD-3001, Volume 2) treats 145–155 mmHg as normal and limits mild hypoxia to no lower than 127 mmHg. It sets 356 mmHg as the highest value tolerable indefinitely. This is because breathing very high oxygen pressures damages the lungs and, at extreme levels, causes seizures. On the summit of Everest the inspired value is only 43 mmHg. Figure 30.8 maps these limits onto the two design variables and shows the choices made for real spacecraft.



Inspired O_2 partial pressure = (total pressure − 47 mmHg water vapour) × O_2 fraction; limits from NASA-STD-3001 Vol. 2 (Rev F, 2026). Lower total pressure means less nitrogen and shorter pre-EVA oxygen prebreathe.

FIGURE 30.8 The cabin atmosphere design space. Curves of constant inspired oxygen pressure show the hypoxic, normal and hyperoxic regions; the dashed line marks the 30 % oxygen level at which spacecraft materials are commonly tested for flammability.

30.4.2 Fire

Materials burn faster as the oxygen fraction rises, and in weightlessness, with no buoyant convection, flames become spherical and slow but can persist where on Earth they would go out. The danger was established tragically on 27 January 1967, when a fire swept through the Apollo 1 command module during a ground test, killing Gus Grissom, Ed White and Roger Chaffee. The cabin held pure oxygen at about 115 kPa and contained flammable nylon netting and Velcro. A spark from wiring started a fire that spread in seconds, and the inward-opening hatch could not be opened against the cabin pressure. NASA changed the launch atmosphere to 60 % oxygen and 40 % nitrogen (vented to pure oxygen at 34.5 kPa during ascent), replaced flammable materials, and fitted a quick-opening hatch.

Today, materials for US crewed vehicles must pass an upward flame-spread test, commonly run at 30 % oxygen and 70.3 kPa, the worst case expected in flight. NASA's Saffire experiments (2016–24) burned large samples inside departing Cygnus cargo ships, providing the first data on how big fires grow in weightlessness.

30.4.3 Decompression Sickness

Nitrogen dissolved in body tissues forms bubbles if the surrounding pressure falls quickly — **decompression sickness (DCS)**, the divers' "bends". Spacesuits work at low pressure to stay flexible: the US suit at 29.6 kPa of pure oxygen, the Russian Orlan at about 40 kPa (Chapter 33). Consider a crew member moving from a sea-level cabin, with 80 kPa of nitrogen in the tissues, into a suit at 29.6 kPa. The ratio of tissue nitrogen pressure to suit pressure would be 2.7 — far too high. Astronauts therefore **prebreathe** pure oxygen for hours before a spacewalk, sometimes with exercise or a night at reduced pressure (the "campout").

A lower-pressure cabin with more oxygen shortens prebreathe dramatically. NASA's **exploration atmosphere** — 56.5 kPa (8.2 psi) with 34 % oxygen — leaves about 37 kPa of nitrogen in the tissues, a ratio of only 1.26 to the suit. So about 15 minutes of in-suit prebreathe would be enough for frequent surface spacewalks. Its inspired oxygen pressure, 128 mmHg, is like living at about 1 200 m altitude. The penalty is fire: a 2026 NASA assessment found that most spacecraft subsystems would need material changes at 34 % oxygen, and far fewer at 70.3 kPa (10.2 psi). As of 2026 the atmosphere for lunar landers and surface habitats is still being evaluated. Table 30.3 compares the choices.

TABLE 30.3 Cabin atmospheres of crewed spacecraft. The last column is the tissue nitrogen pressure after long exposure divided by the suit pressure; higher values need longer oxygen prebreathe.

Vehicle or regime	Total pressure	O_2 fraction	O_2 partial pressure	N_2 ratio to 29.6 kPa suit
Mercury, Gemini, Apollo (in flight)	34–38 kPa	100 %	34–38 kPa	≈ 0
Skylab (1973–74)	34.5 kPa	≈ 72 %	24.8 kPa	0.33
ISS, Soyuz, Shuttle, Crew Dragon, Orion	101.3 kPa	21 %	21.3 kPa	2.70
Shuttle pre-EVA; ISS campout	70.3 kPa	26.5 %	18.6 kPa	1.75
Exploration atmosphere (studied)	56.5 kPa	34 %	19.2 kPa	1.26

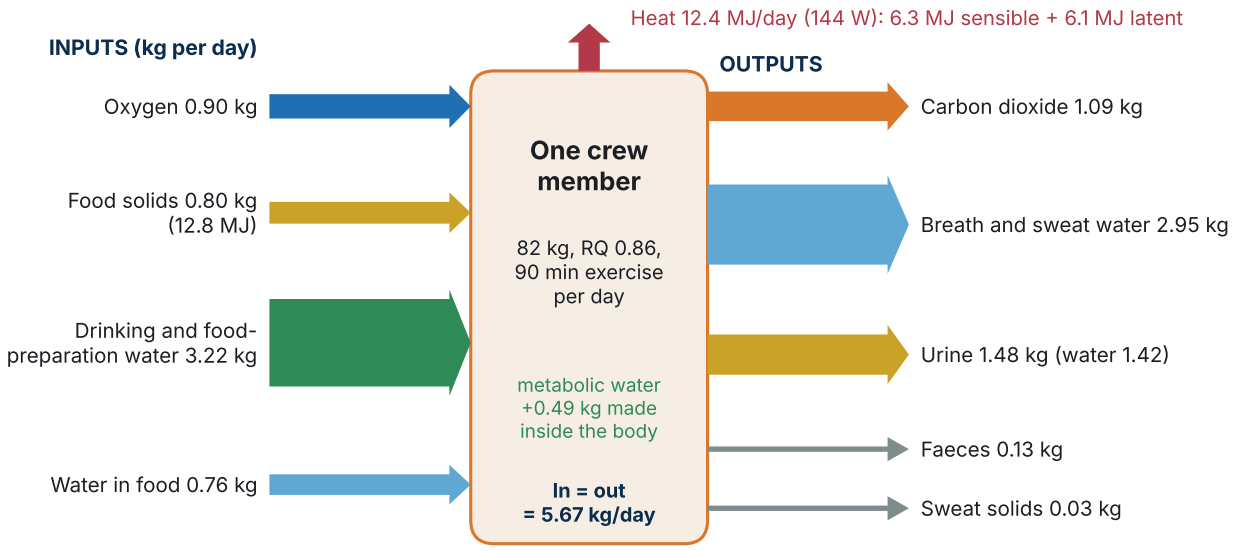
Early US spacecraft used pure oxygen at about 35 kPa to save mass: thinner walls, simpler life support, no nitrogen tanks. Soviet spacecraft used sea-level air, a choice the ISS inherited at the cost of long prebreathe.

30.5 The Human Metabolic Budget

A crew member burns food with oxygen. NASA's *Baseline Values and Assumptions Document* (BVAD) defines a reference astronaut of 82 kg who exercises for 90 minutes a day (Table 30.4). The ratio of carbon dioxide produced to oxygen consumed — the **respiratory quotient** — is about 0.86, so

more carbon dioxide is exhaled, by mass, than oxygen inhaled. The body also makes about half a kilogram of water a day from food and loses about three kilograms as vapour that the cabin air system must condense.

Two lessons stand out from Figure 30.9. Water dominates — about 4.5 kg of it passes through a person every day, four-fifths of the total — which is why water recycling pays off fastest. And although oxygen is a small mass, it must be supplied, and carbon dioxide removed, continuously: a few hours of failure can be dangerous.



Arrow width proportional to mass per day. Food as supplied with packaging: 1.8–2.4 kg/day.

FIGURE 30.9 Daily mass balance of one crew member. Inputs and outputs balance at 5.67 kg per day; most of the mass flowing through a person is water.

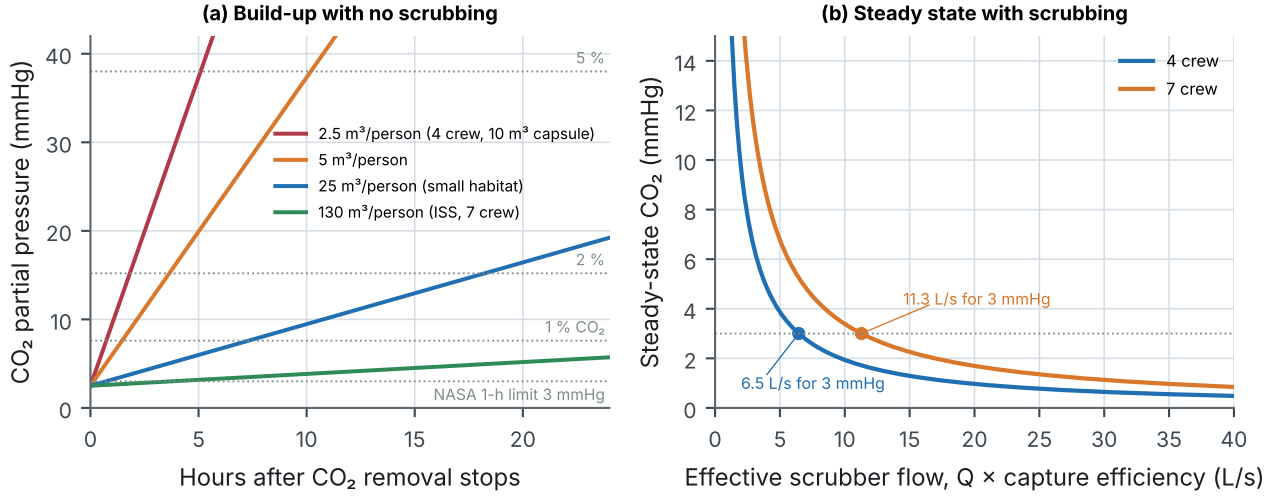
TABLE 30.4 The metabolic budget of one crew member (NASA BVAD, 2022).

Item	kg per crew member per day	Notes
Oxygen consumed	0.90	Older design value 0.84
Carbon dioxide produced	1.09	Older design value 1.00
Food solids (dry)	0.80	12.8 MJ of energy
Food as supplied on ISS	2.39	Including water and 0.43 kg packaging
Drinking and food-preparation water	3.22	Plus 0.76 kg water in food
Breath and sweat water	2.95	Removed as humidity condensate
Urine	1.48	1.42 kg water
Faeces	0.13	0.10 kg water
Metabolic heat	12.4 MJ	144 W average; half as latent heat

30.5.1 Carbon Dioxide

Carbon dioxide is toxic well before it displaces oxygen. Concentrations of a few per cent cause headaches, rapid breathing and impaired thinking, and above about 5–10 % they become incapacitating. ISS crews reported headaches at partial pressures of about 3–5 mmHg. Analysis showed that the odds of a headache doubled with every 1 mmHg increase. NASA-STD-3001 now limits the one-hour average to 3 mmHg (0.4 kPa).

Small cabins leave little margin (Figure 30.10 a). If the scrubber stops at 2.5 mmHg in a capsule with 2.5 m^3 of air per person, the level climbs by 7 mmHg per hour. It reaches 1 % in about 45 minutes and 2 % in under two hours. On the ISS, the pressurised volume of about 900 m^3 gives some 130 m^3 per person for a crew of seven. There, the same rise takes about a day and a half. With scrubbing, the steady-state level is inversely proportional to the air flow the scrubber cleans (panel b).



Ideal gas at 22 °C, well-mixed cabin, 1.0 kg CO_2 per person per day. (a) $dp/dt = \dot{n}RT/V$; (b) $p = \dot{n}RT/(Q\eta)$.

FIGURE 30.10 Carbon dioxide in a closed cabin. (a) Rise in partial pressure after carbon dioxide removal stops, for different free gas volumes per crew member. (b) Steady-state partial pressure as a function of the effective scrubber flow.

EXAMPLE: SIZING A LITHIUM HYDROXIDE SCRUBBER

A four-person capsule must remove carbon dioxide for a five-day free flight with non-regenerable lithium hydroxide (LiOH). Each crew member produces 1.0 kg of CO_2 per day.

1. CO_2 to remove: $4 \times 5 \times 1.0 =$ **20 kg**.
2. Chemistry: $2\,LiOH + CO_2 \rightarrow Li_2CO_3 + H_2O$. Two moles of LiOH (2 × 23.95 g) absorb one mole of CO_2 (44.01 g). So the theoretical capacity is 0.92 kg of CO_2 per kg of LiOH, and the minimum LiOH mass is 20/0.92 = **21.8 kg**.
3. Real beds are never fully used, because the gas breaks through before the last granules react. Shuttle practice was a 7 kg canister for four crew-days, or 1.75 kg per crew-day: **35 kg**, five canisters.
4. By-products: the reaction releases $18/44 \times 20 =$ **8.2 kg of water** that the humidity system must remove, and about 2.1 MJ of heat per kilogram of CO_2 — some 25 W per crew member.
5. Air flow: to hold 3 mmHg (0.16 mol of CO_2 per m^3 at 22 °C) the bed must remove 91 mol/day = 1.05×10^{-3} mol/s. If it captures 90 % per pass, the fan must move $1.05 \times 10^{-3} / (0.16 \times 0.9) \approx$ **7.2 L/s** through it (Figure 30.10 b).

30.6 Environmental Control and Life Support Systems

The **environmental control and life support system (ECLSS)** controls pressure and oxygen, and removes carbon dioxide and trace contaminants (**air revitalisation**). It also controls temperature and humidity, supplies and recovers water, manages waste, and detects and suppresses fire.

Figure 30.11 shows how these are linked on the ISS, the most complete system ever flown, and Table 30.5 compares the technologies.

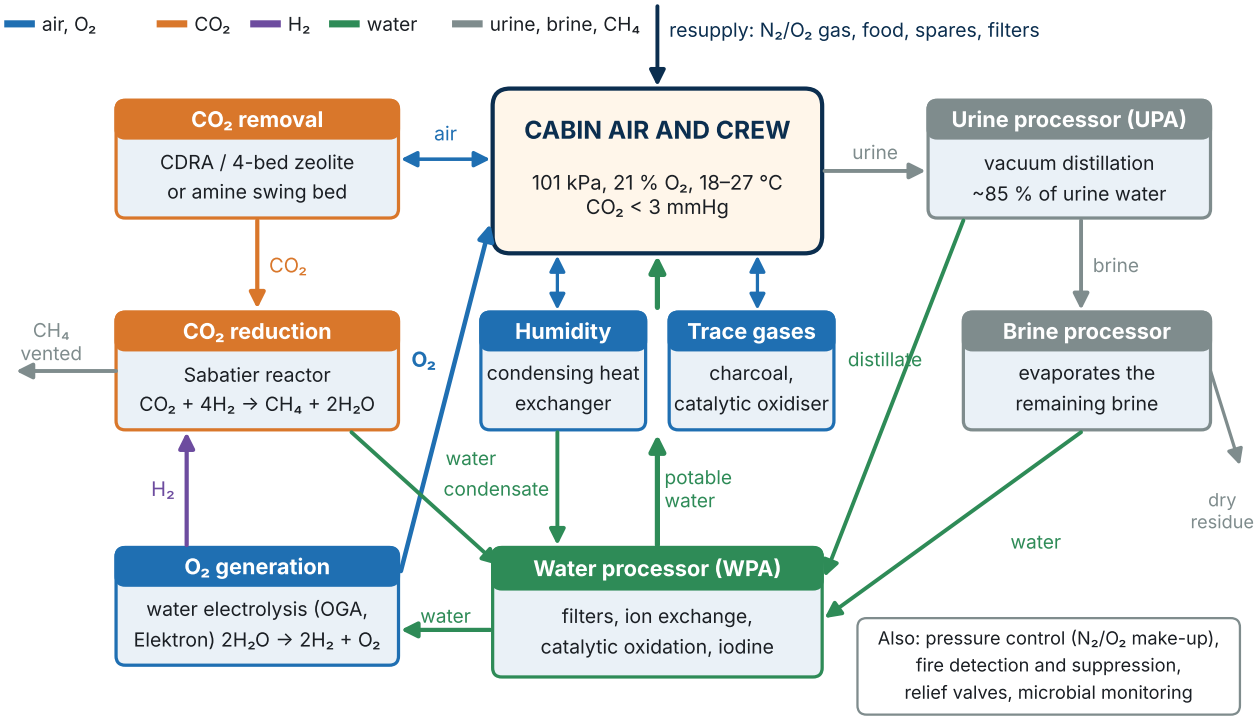


FIGURE 30.11 Simplified ECLSS flow diagram of the ISS US segment. Air is scrubbed of CO_2, which a Sabatier reactor combines with hydrogen from the oxygen generator to make water; urine, humidity condensate and recovered brine water are purified for drinking and for electrolysis.

30.6.1 Removing Carbon Dioxide

Short missions use **lithium hydroxide** canisters that are thrown away when full: Apollo, the Shuttle and Crew Dragon all used them. The most famous are those of Apollo 13. After the oxygen-tank explosion in April 1970 the three astronauts sheltered in the lunar module, whose round canisters were sized for two people for two days; the command module's spare canisters were square. Engineers in Houston devised a way to connect them with plastic bags, cardboard and tape, and the crew built the "mailbox" from instructions read up to them (Figure 30.12). Soyuz uses potassium superoxide, which absorbs carbon dioxide and releases oxygen, with LiOH as a supplement.

Long missions need **regenerable** systems. The ISS Carbon Dioxide Removal Assembly (**CDRA**) passes cabin air through beds of zeolite — a mineral with molecular-sized pores — that adsorb carbon dioxide. A full bed is heated and exposed to vacuum to drive the gas out while a second bed takes over. Desiccant beds first remove water, which would block the zeolite, and return it to the cabin. The Russian **Vozdukh** works on the same swing principle with three beds of an amine sorbent. A redesigned Four Bed CO_2 Scrubber, delivered in 2021, now serves as a primary system. Orion uses an **amine swing bed**, regenerated by exposure to vacuum every few minutes — compact, but it vents some water overboard.

30.6.2 Making Oxygen

Short missions carry oxygen as high-pressure gas or cryogenic liquid. The ISS makes its own by **electrolysis** of water: the US Oxygen Generation Assembly (Figure 30.13) and the Russian Elektron split water into oxygen for the cabin and hydrogen. Each kilogram of oxygen requires 1.13 kg of water and, even for an ideal cell, about 5 kWh of electrical energy; real electrolysers need more. Backup oxygen comes from tanks delivered by cargo vehicles and from **solid-fuel oxygen generators** — "candles" of lithium perchlorate that release oxygen when burned. A candle on Mir caught fire in February 1997, filling the station with smoke.

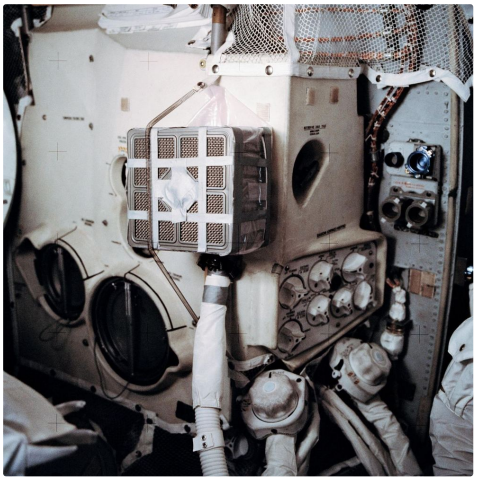


FIGURE 30.12 The "mailbox" built by the Apollo 13 crew to use the command module's square lithium hydroxide canisters in the lunar module, April 1970. Image: NASA/JSC.

30.6.3 Closing the Oxygen Loop

Electrolysis consumes water, and carbon dioxide removed from the cabin still contains oxygen. Since 2010 the ISS has recovered part of it with a **Sabatier reactor**, which combines carbon dioxide with hydrogen from the oxygen generator over a heated catalyst:

$$\mathrm{CO_2 + 4\,H_2 \rightarrow CH_4 + 2\,H_2O}$$

The water returns to electrolysis. The methane is vented, taking half of the hydrogen with it, and because there is not enough hydrogen to convert all the carbon dioxide, the Sabatier closes only about 50 % of the oxygen loop. Future systems may recover hydrogen from the methane or use the **Bosch reaction** ($\mathrm{CO_2 + 2\,H_2 \rightarrow C + 2\,H_2O}$), which leaves solid carbon instead.



FIGURE 30.13 NASA astronaut Nichole Ayers servicing components of the Oxygen Generation System rack in the ISS Destiny laboratory, May 2025. Image: NASA/JSC.



FIGURE 30.14 NASA astronaut Jessica Watkins servicing the Water Recovery System rack in the Tranquility module, June 2022. Image: NASA/JSC.

30.6.4 Trace Contaminants, Temperature and Humidity

People, materials and equipment release hundreds of trace gases — ammonia, alcohols, aldehydes, siloxanes, carbon monoxide — that would accumulate in a sealed cabin. Activated charcoal adsorbs most of them and a heated catalytic oxidiser burns the rest, while filters remove dust and microbes. Fans keep the air moving, because without convection exhaled carbon dioxide would collect around a sleeping astronaut's head. A **condensing heat exchanger**, cooled by the internal water loop, keeps the cabin at about 18–27 °C. It also removes the 3 kg per person per day of water vapour, which joins the water recovery system as condensate.

30.6.5 Recovering Water

The ISS **Water Recovery System** (Figure 30.14) has two parts. The **Urine Processor Assembly** distils pretreated urine in a rotating drum at low pressure, recovering about 85 % of its water. A newer **Brine Processor Assembly** extracts most of the rest from the concentrated brine. The **Water Processor Assembly** treats the distillate and humidity condensate with filters and ion-exchange beds, oxidises the remaining organic compounds in a catalytic reactor, and adds iodine against microbes. In June 2023 NASA announced that the system had reached its goal of recovering **98 %** of the water, up from 93–94 % before the brine processor. The Russian segment recycles humidity condensate and uses silver ions as a biocide.

30.6.6 Waste, Hygiene and Fire

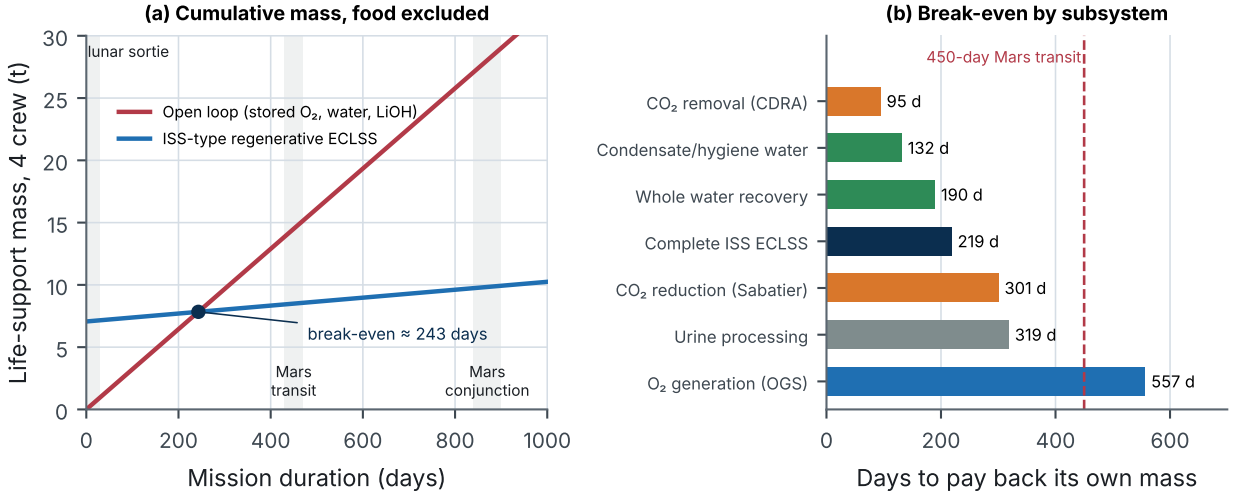
Toilets use air flow instead of gravity to carry waste away; the compact Universal Waste Management System, installed on the ISS in 2020, also flies on Orion. Faeces are stored and burned up in departing cargo vehicles, and crews wash with wet towels and rinseless shampoo. Smoke detectors sit in the ventilation ducts. From 2015 the US segment's carbon dioxide extinguishers were replaced by water-mist units, because carbon dioxide would create an atmosphere the crew's emergency masks cannot filter. In a fire the crew stops the ventilation, removes power, dons oxygen masks and, if necessary, retreats to the docked vehicles.

30.6.7 Open and Closed Loops

Recycling hardware is heavy, consumes power and needs spares, while stored consumables cost mass every day, so the choice depends on mission duration (Figure 30.15). An analysis looked at ISS hardware for a four-person, 450-day Mars transit. It found that the complete regenerative system, with three sets of spares, has a mass of 7 062 kg and replaces about 32 kg of supplies per day. It pays for itself after about 220 days — about 240 days if the residual resupply of oxygen and filters is counted — and halves the life support mass of a 450-day transit. Carbon dioxide removal pays off within about three months, but the oxygen generator, with its heavy spares, takes 557 days.

TABLE 30.5 ECLSS technologies of crewed spacecraft.

Function	Open loop	ISS US segment	ISS Russian segment	Capsules
CO_2 removal	LiOH canisters	CDRA, Four Bed CO_2 Scrubber (zeolite)	Vozdukh (amine beds)	LiOH (Dragon); amine swing bed (Orion); KO_2 (Soyuz)
O_2 supply	Gas or liquid tanks	Electrolysis (OGA)	Electrolysis (Elektron); candles	Tanks
CO_2 reduction	None, CO_2 discarded	Sabatier (since 2010)	None	None
Water	Stored, or fuel-cell water (Apollo, Shuttle)	Urine and brine processors, WPA: 98 %	Condensate recovery	Stored
Trace gases	Charcoal	Charcoal and catalytic oxidiser	Similar system	Charcoal, filters



ISS hardware with three spare sets (7 062 kg) replacing 32.2 kg/day of supplies for 4 crew (Jones, ICES-2017-85); (a) adds 3.2 kg/day residual resupply (O_2 not recovered by Sabatier, filters). Tankage neglected.

FIGURE 30.15 Open versus closed life support. (a) Cumulative mass of life support supplies for four crew with open-loop storage and with an ISS-type regenerative system. (b) Days needed for each ISS subsystem to save its own mass, including three sets of spares.

The ultimate step is a **bioregenerative** system, in which plants and microbes recycle air, water and food. The Soviet BIOS-3 facility in Krasnoyarsk supported three people for 180 days in 1972–73, with wheat and vegetables providing part of the food. China's Lunar Palace 1 ran for 370 days in 2017–18 with rotating crews of volunteers and reported about 98 % closure of the material cycle. ESA's MELiSSA programme is testing a loop of bacterial and plant compartments in Barcelona. Because one person's oxygen and food would need several tens of square metres of lit

crops, plants will long supplement packaged food rather than replace physico-chemical systems. NASA's Veggie and Advanced Plant Habitat chambers on the ISS have grown lettuce, mustard greens, radishes and chile peppers (Figure 30.16).



FIGURE 30.16 Chile peppers growing in the Advanced Plant Habitat on the ISS in November 2021, during Expedition 66. Image: NASA/JSC (Mark Vande Hei).

EXAMPLE: CONSUMABLES FOR A 30-DAY LUNAR SURFACE MISSION

Three astronauts spend 30 days on the Moon with open-loop life support. Use the BVAD values: oxygen 0.89 kg, water 3.2 kg and food 1.83 kg (packaged) per crew member per day. For LiOH, use the Shuttle rate of 1.75 kg per crew-day, scaled up for a CO_2 output of 1.085 kg. This assumes that the canister rating corresponds to the older 1.0 kg design value.

1. Crew-days: $3 \times 30 =$ **90**.
2. Oxygen: $90 \times 0.89 =$ **80 kg**.
3. Water (drinking, food preparation, minimal hygiene): $90 \times 3.2 =$ **288 kg**.
4. Food: $90 \times 1.83 =$ **165 kg**.
5. LiOH: $90 \times 1.75 \times 1.085 =$ **171 kg**.
6. Total: **704 kg**, or about **775 kg** with a 10 % margin — before tanks, cabin leakage and gas lost through the airlock on each spacewalk.

An ISS-type regenerative system would weigh several tonnes, so open loop is clearly right for 30 days. Water dominates, so a condensate recycler would be the first system worth adding for longer stays.

30.7 Living and Working in Space

30.7.1 Food

Space food must be safe, nutritious, light and appetising for months. ISS menus offer some 200 items: thermostabilised pouches, freeze-dried dishes rehydrated with hot water, and irradiated meat. There are also natural-form foods such as nuts and tortillas (which do not make crumbs), and powdered drinks. Salt and pepper come as liquids. Meals are a social event (Figure 30.17). The ISS food supply has grown to 2.39 kg per person per day including packaging, which makes up almost a fifth of it. For Mars, food must keep its nutrients for up to five years, longer than current foods do.



FIGURE 30.17 Expedition 31 crew members sharing a meal near the galley of the Zvezda service module, May 2012. Image: NASA/JSC.



FIGURE 30.18 The four volunteers of NASA's CHAPEA Mission 2 inside the 3D-printed simulated Mars habitat at Johnson Space Center, November 2025. Image: NASA/JSC.

30.7.2 Habitability and Medical Care

Each ISS crew member has a sleeping station the size of a telephone booth, with a sleeping bag tied to the wall and a door to shut out the noise of fans and pumps. Habitable volume, noise, lighting, privacy and window views all affect performance and morale on long missions (Chapter 32).

ISS medical care relies on two trained **crew medical officers**, a medical kit with ultrasound scanner, defibrillator and drugs, and flight surgeons on the ground; serious illness means returning home in a docked capsule. In January 2026 NASA's Crew-11 returned about a month early, the first time an ISS mission was ended for medical reasons; the affected crew member was reported stable. A Mars crew, with radio delays of up to 22 minutes each way and months from home, must diagnose and treat emergencies themselves.

30.7.3 Psychology and Isolation

Isolation, confinement, monotony and separation from family are among the hardest problems of long missions. Agencies select crews for stability and teamwork and provide psychological support. **Analogue missions** on Earth test this: Mars-500 confined six men in a mock spacecraft in Moscow for 520 days in 2010–11, and Hawaii's HI-SEAS ran missions of up to a year on Mauna Loa. NASA's **CHAPEA** (Crew Health and Performance Exploration Analog) placed four volunteers in a 160 m^2 3D-printed habitat at Johnson Space Center for 378 days in 2023–24. They faced simulated equipment failures, resource limits and communication delays. The second CHAPEA crew entered in October 2025 and is due to emerge at the end of October 2026 (Figure 30.18).

30.8 Crew Safety and Risk

30.8.1 Lessons Paid For in Lives

Eighteen people have died in four accidents on orbital missions, and three more in the Apollo 1 ground test (Table 30.6). Each accident changed how crewed spacecraft are designed and managed. The Soyuz 11 crew died during re-entry in 1971 when a valve opened as the modules separated and the cabin lost its air. They were not wearing pressure suits, and every Soyuz crew since has worn the Sokol suit for launch and landing. Both Shuttle losses were preceded by warning signs — eroded booster seals, foam strikes on earlier flights — that had come to be accepted as normal, a pattern known as the "normalisation of deviance".

Many more crews have survived serious failures, such as the Apollo 13 explosion (1970). Launch escape systems saved the crews of the Soyuz T-10-1 pad fire (1983) and the Soyuz MS-10 booster failure (2018). Crews also survived the Mir fire and collision (1997) and the Starliner thruster problems of 2024, after which its crew came home on a Crew Dragon. They show why crewed vehicles need **abort** capability in every phase, **safe havens**, and margins for the unexpected.

30.8.2 Human-Rating Requirements

NASA's requirements are set out in its human-rating process (NPR 8705.2C, with technical requirements in NASA-STD-8719.29 since 2023) and in **NASA-STD-3001**. Volume 1 of the standard covers crew health, including radiation limits, and Volume 2 (Revision F, July 2026) covers habitability and the cabin environment. One key principle is that the system must tolerate at least one failure without catastrophe, unless a component is designed for minimum risk (like structures and pressure vessels). The actual level of failure tolerance (one, two or more) is set by safety analysis. Other principles are that the crew can abort and escape, and that the crew can monitor and control the vehicle manually.

Risk is also set as a number. NASA's Commercial Crew Program required that the probability of **loss of crew** on a mission of up to 210 days to the ISS be no worse than 1 in 270. For ascent and entry, it had to be no worse than 1 in 500. NASA later estimated that the first Shuttle flight in 1981 carried a risk of about 1 in 12, and the mature Shuttle about 1 in 90 — consistent with two losses in 135 flights. Reliability and risk analysis (Chapter 4) identifies the dominant contributors, but experience shows that the largest risks are often the ones nobody modelled.

TABLE 30.6 Fatal accidents in human spaceflight.

Accident	Date	Crew lost	Cause	Main consequences
Apollo 1 (ground test)	27 Jan 1967	3	Fire in pure oxygen at 115 kPa; inward-opening hatch	Materials, 60/40 launch atmosphere, quick-opening hatch
Soyuz 1	24 Apr 1967	1	Main parachute failed to deploy; reserve tangled	Parachute system redesign
Soyuz 11	30 Jun 1971	3	Valve opened at module separation; cabin depressurised	Pressure suits for launch and entry
Challenger (STS-51-L)	28 Jan 1986	7	Solid-booster joint seal failed in cold weather	Booster redesign, crew escape system, management reform

Accident	Date	Crew lost	Cause	Main consequences
Columbia (STS-107)	1 Feb 2003	7	Foam strike broke the wing's leading edge; vehicle broke up on entry	Inspection and repair in orbit, Shuttle retirement in 2011

AI FOR CREW HEALTH AND LIFE SUPPORT

A Mars crew will have to handle medical emergencies without real-time help from the ground. In 2025 NASA and Google tested a prototype clinical decision-support assistant for long-duration crews. In simulated cases it reached the correct diagnosis 74–88 % of the time, depending on the condition. Similar tools could guide an astronaut through an ultrasound scan. Life support is a natural target for machine learning too. ECLSS telemetry — pressures, flows, temperatures, valve positions — can train models that detect developing faults before they become failures. In 2025 researchers demonstrated supervised anomaly detection and diagnosis on data from the European Columbus module's ECLSS. Such **predictive maintenance** would let a crew far from Earth replace the right part before it fails and carry fewer spares (Chapter 46).

KEY TAKEAWAYS

- About 780 people have flown in space; the longest single flight is 437.7 days and the longest cumulative time 1 111 days, while a Mars mission would last up to 900 days.
- Weightlessness shifts fluids headward and causes motion sickness, eye changes (SANS), bone loss of 1–1.5 % per month and muscle atrophy; about two hours of daily exercise is the main countermeasure.
- Deep-space crews receive about 1.8 mSv per day from cosmic rays; a Mars mission would exceed NASA's 600 mSv career limit. Storm shelters handle solar particle events, but shielding barely reduces the cosmic-ray dose.
- Cabin atmospheres balance hypoxia, fire and decompression sickness: the ISS uses sea-level air, while NASA has studied 56.5 kPa with 34 % oxygen for lunar surface operations.
- Each crew member consumes about 0.9 kg of oxygen, 3–4 kg of water and 1.8–2.4 kg of packaged food per day and produces 1.1 kg of carbon dioxide and 144 W of heat.
- The ISS ECLSS removes CO_2 with zeolite beds, makes oxygen by electrolysis, recovers half of the oxygen with a Sabatier reactor and recycles 98 % of its water. It pays back its mass after about 7–8 months, so open loops suit short missions and recycling suits long ones.
- Four accidents on orbital missions and the Apollo 1 fire killed 21 people; Commercial Crew vehicles must meet a loss-of-crew risk no worse than 1 in 270 per mission.

REVIEW QUESTIONS

1. Why do astronauts suffer orthostatic intolerance after landing? Give two countermeasures.
2. A crew member loses hip bone at 1.2 % per month for 9 months. Estimate the loss at landing and how much is recovered a year later (9-month half-time).
3. Why does shielding reduce the dose from solar particle events much more than that from galactic cosmic rays? Why is polyethylene better than aluminium?
4. Using the dose rates in Table 30.2, calculate the dose for a 180-day lunar surface mission (with 10 days of transit) and compare it with the 600 mSv career limit.
5. Calculate the inspired oxygen partial pressure, in mmHg, for a cabin at 70.3 kPa with 26.5 % oxygen. Is it within NASA's normoxic range?
6. Why is a low-pressure, oxygen-rich cabin attractive for a lunar lander that supports frequent spacewalks, and what is its main disadvantage?

7. A four-person crew sits in a capsule with 10 m^3 of free air when its scrubber fails at a CO_2 partial pressure of 2.5 mmHg. How long until the partial pressure reaches 15 mmHg?
8. How do the CDRA, the oxygen generator and the Sabatier reactor work together, and why does the Sabatier close only about half of the oxygen loop?
9. Using Figure 30.15, explain why Apollo carried stored oxygen and water while the ISS recycles them. Which subsystem pays back its mass fastest?
10. What did the Apollo 1, Soyuz 11 and Columbia accidents each teach designers of crewed spacecraft?

31 CHAPTER Crewed Spacecraft



Image: NASA/JSC

A crewed spacecraft is a small, self-contained world that must keep people alive, carry them away from an exploding rocket within seconds, and bring them home through a fireball at up to 11 km/s. This chapter traces the vehicles that have carried people since 1961 — from Vostok and Mercury through Apollo, Soyuz and the Space Shuttle to Shenzhou, Crew Dragon, Starliner and Orion, which carried four astronauts around the Moon in April 2026 — and the vehicles now being built in India and China and for lunar landings. It explains how they are engineered: pressure vessels, windows, seats and acceleration limits, avionics, propulsion, docking systems, launch abort, entry and parachutes, and how their safety is estimated, demonstrated and certified.

IN THIS CHAPTER

31.1 Sixty-Five Years of Crewed Spacecraft
31.2 Anatomy of a Crewed Spacecraft
31.3 Launch Abort
31.4 Coming Home: Entry, Descent and Landing
31.5 Reliability and Safety

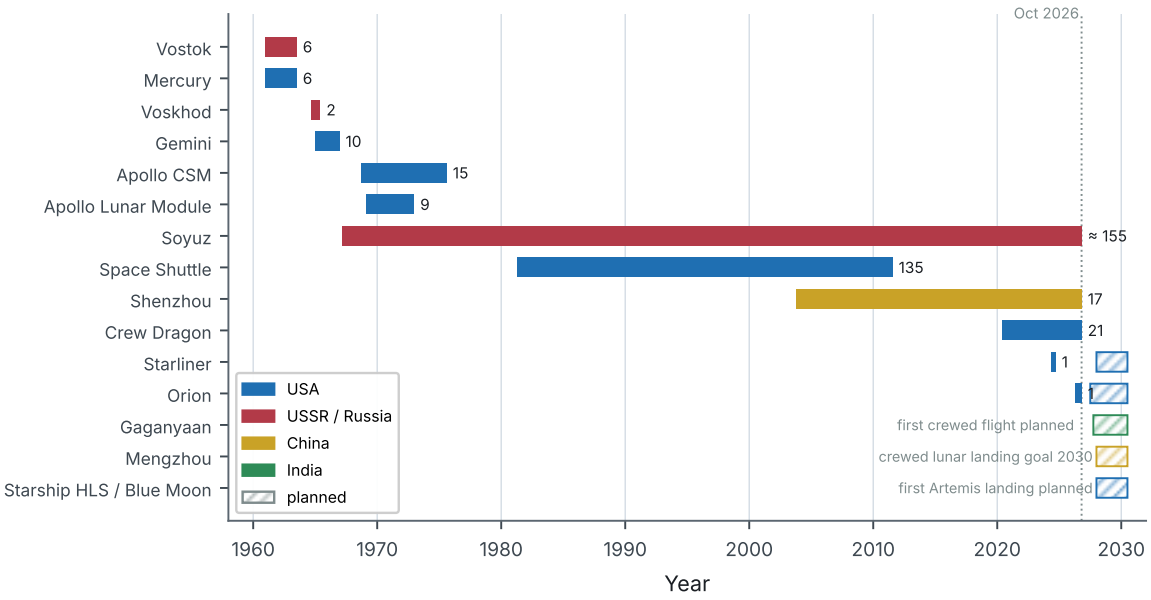
A **crewed spacecraft** carries people into space and back. Unlike a robotic spacecraft, its payload cannot be replaced and tolerates only a narrow range of pressure, temperature, acceleration and air composition. Three requirements set it apart. It must keep its crew alive (life support, Chapter 30), and it must be able to escape from a failing launch vehicle at any moment of the ascent. It must also bring its crew back to Earth gently enough for them to walk away.

Since Yuri Gagarin's flight in April 1961, fewer than twenty types of vehicle have carried people into space. Only four were in service in October 2026: Russia's Soyuz, China's Shenzhou, SpaceX's Crew Dragon and NASA's Orion. Boeing's Starliner was grounded after its troubled first crewed

flight, and India's Gaganyaan and China's Mengzhou were preparing for their first orbital flights. This chapter describes these vehicles and the engineering behind them; space stations are covered in Chapter 32, spacesuits in Chapter 33 and launch vehicles in Chapters 34 and 36.

31.1 Sixty-Five Years of Crewed Spacecraft

Figure 31.1 shows every orbital crewed spacecraft and how long it served, and Table 31.1 compares them. Two waves stand out: the 1960s race to the Moon, and a second wave from 2020 driven by commercial contracts and by China.



Numbers: crewed launches of each vehicle to October 2026 (Soyuz includes the 1975 and 1983 launch aborts; Apollo CSM includes Skylab and ASTP). Planned dates as announced in 2026.

FIGURE 31.1 Crewed spacecraft from 1961 to 2026 and those planned for the late 2020s. The numbers give the crewed launches of each type. Soyuz has flown for almost sixty years; the Space Shuttle flew 135 times in 30 years.

31.1.1 The First Generation: Vostok, Mercury, Voskhod and Gemini

The Soviet **Vostok** was a 2.3 m sphere, which needed no attitude control during its ballistic 8–9 g entry. It landed too hard for safety, so the cosmonaut ejected at about 7 km and came down under a separate parachute. Six Vostoks flew in 1961–63, carrying Gagarin and Valentina Tereshkova, the first woman in space.

NASA's **Mercury** capsule weighed only about 1.4 t so that it could ride the Redstone and Atlas missiles of the day. It set the pattern of every American capsule since: a cone with a broad, blunt heat shield at the base (Chapter 10), and a crew lying on contoured couches facing it. Mercury flew twice suborbitally in 1961, starting with Alan Shepard on 5 May, and four times in orbit from John Glenn's flight on 20 February 1962.

Voskhod was a Vostok without its ejection seat — and so without any means of escape early in the ascent. It carried three cosmonauts in 1964 and two in 1965, when Alexei Leonov made the first spacewalk. **Gemini** carried two astronauts and was the first spacecraft built to be *flown*: thrusters, an onboard computer and radar let it change orbit, rendezvous and dock. Between

March 1965 and November 1966 ten Gemini crews demonstrated the techniques Apollo needed — rendezvous (Figure 31.2), docking, spacewalks, fuel-cell power, guided lifting entry and flights of up to 14 days.



FIGURE 31.2 Gemini 7 photographed from Gemini 6A during the first space rendezvous, 15 December 1965, with the two spacecraft about 11 m apart. The white section is the adapter module carrying propellant, fuel cells and radiators. Image: NASA/JSC.

31.1.2 Apollo: The Command, Service and Lunar Modules

Apollo divided a lunar mission between three modules. The **command module (CM)**, a 3.9 m cone of 5.6 t, carried three astronauts and alone returned. The **service module** carried fuel cells, oxygen tanks, radiators and a single 91 kN engine, which had to work to bring the crew home from lunar orbit and was therefore pressure-fed, with redundant valves. The **Lunar Module (LM)** was the first spacecraft designed to fly only in vacuum. It had a descent stage with a throttleable engine and an ascent stage that carried two astronauts back to the CM in lunar orbit (Figure 31.3; landing is described in Chapter 27).

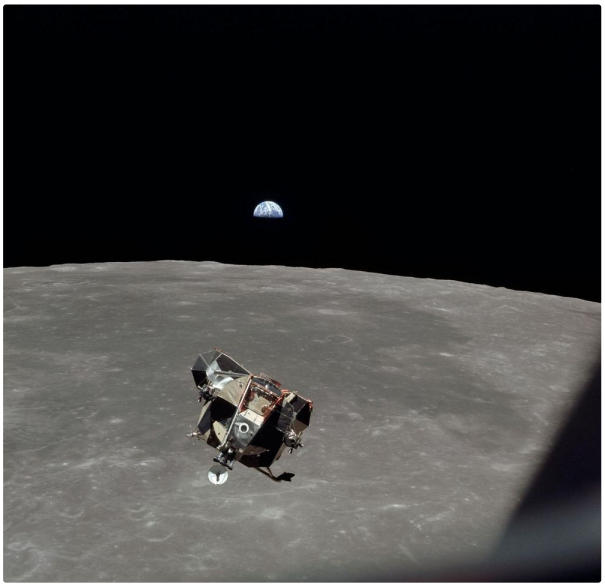


FIGURE 31.3 The ascent stage of Apollo 11's Lunar Module Eagle returns from the surface to rendezvous with the command module Columbia in lunar orbit, July 1969. Image: NASA/KSC.

The command and service module flew 15 crewed missions: 11 Apollo flights from 1968 to 1972, three Skylab crews and the Apollo–Soyuz Test Project of 1975. When an oxygen tank in Apollo 13's service module exploded in April 1970, the crew survived for almost four days in the Lunar Module, using it as a lifeboat.

31.1.3 Soyuz and the Space Shuttle

Soyuz, first flown with a crew in April 1967, is the longest-serving crewed spacecraft. A spherical **orbital module** serves as living space and airlock, the crew launch and land in the bell-shaped **descent module**, and an unpressurised **service module** carries the engine, solar arrays and radiators. Only the 2.2 m descent module returns, which keeps the whole spacecraft to about 7 t. Two early accidents (Soyuz 1 and Soyuz 11, Table 31.5) killed four cosmonauts. Upgraded through the T, TM, TMA and MS versions, Soyuz had made about 155 crewed launches by October 2026, the latest, Soyuz MS-29, on 14 July 2026.

The **Space Shuttle** (1981–2011) was a reusable winged orbiter carrying up to eight people and a 24 t payload and landing on a runway. Its two-deck crew module had a pressurised volume of about 66 m^3, six times that of the Apollo command module. The Shuttle flew 135 times, assembled most of the International Space Station (ISS) and serviced Hubble five times, but two orbiters and 14 astronauts were lost (Chapter 10).

31.1.4 Shenzhou and China's Programme

China's **Shenzhou** follows the Soyuz layout but is larger, with a 2.5 m descent module, and on its early flights the orbital module stayed in orbit as a satellite after the crew returned. Yang Liwei made China's first spaceflight on Shenzhou 5 on 15 October 2003, and since 2021 Shenzhou has ferried crews of three to the Tiangong space station, usually for six months.

Shenzhou also showed why a station needs a spare lifeboat. In November 2025, shortly before its crew was due home, a crack — probably from the impact of a debris particle smaller than a millimetre — was found in a window of Shenzhou-20. The crew returned in the newer Shenzhou-21 instead. China therefore launched an uncrewed **Shenzhou-22** from its standby rocket on 25 November to give the Shenzhou-21 crew a ride home. They landed in it on 29 May 2026 after a Chinese-record 210 days. Russia had met the same problem in December 2022, when a coolant leak attributed to a micrometeoroid emptied the radiator loop of Soyuz MS-22 and an uncrewed Soyuz MS-23 was sent as a replacement.

31.1.5 The Commercial Era: Crew Dragon and Starliner

After the Shuttle retired, the United States relied on Soyuz for nine years while NASA's **Commercial Crew Program** paid two companies to develop vehicles that they own and operate under fixed-price contracts. SpaceX's **Crew Dragon** launched Doug Hurley and Bob Behnken to the ISS on 30 May 2020. By October 2026 Dragon had flown 21 crewed missions. They were the Demo-2 test flight, 13 NASA crew rotations, four private Axiom missions to the ISS and three free-flying private missions. The free-flying missions included Polaris Dawn's spacewalk in 2024 and Fram2's polar orbit in 2025. In January 2026 Dragon made the first medical evacuation from the ISS, bringing the Crew-11 astronauts home early. Five capsules are in service; Endeavour has flown six times.

Boeing's **CST-100 Starliner** has struggled. Its first uncrewed test in 2019 failed to reach the ISS because of software errors, and the second succeeded in 2022. The **Crew Flight Test** launched on 5 June 2024 with Butch Wilmore and Suni Williams. During the approach to the ISS, five of the 28 small thrusters on the service module failed, and helium leaks appeared in the propulsion system (Figure 31.4). The thrusters, packed in enclosed "doghouse" pods, had overheated during repeated firings, deforming the Teflon seals in their valves. NASA brought Starliner home uncrewed in September 2024 — it landed safely — and returned its crew on a Crew Dragon in March 2025, after 286 days in space instead of eight. In February 2026 NASA reclassified the flight as a Type A mishap, its most serious category. As of October 2026, an uncrewed cargo flight, **Starliner-1**, with new thruster valves and seals was planned for December 2026–January 2027, and the first crew rotation flight for 2028.

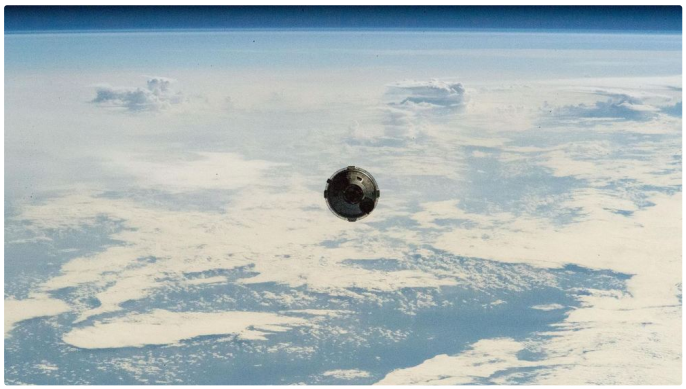


FIGURE 31.4 Boeing's Starliner approaching the ISS on 6 June 2024 during the Crew Flight Test, with Butch Wilmore and Suni Williams aboard. Thruster failures during this approach led NASA to return the spacecraft without its crew. Image: NASA/JSC.

31.1.6 Orion and Artemis II

NASA's **Orion** is the first new deep-space crew vehicle since Apollo. Its 5.0 m crew module, built by Lockheed Martin, carries four astronauts for up to 21 days. The **European Service Module**, built by Airbus for ESA, provides propulsion, about 11 kW from four solar wings and life-support consumables. After the uncrewed Artemis I flight around the Moon in 2022, **Artemis II** was launched on 1 April 2026 by the Space Launch System. Reid Wiseman, Victor Glover, Christina Koch and the Canadian Jeremy Hansen flew Orion *Integrity* on a free-return trajectory around the Moon (Figure 31.5). They passed about 6 500 km above the far side and reached 406 771 km from Earth — farther than any humans before them. They splashed down off San Diego on 10 April after an entry at about 11 km/s. Problems were minor, such as a frozen urine vent line.

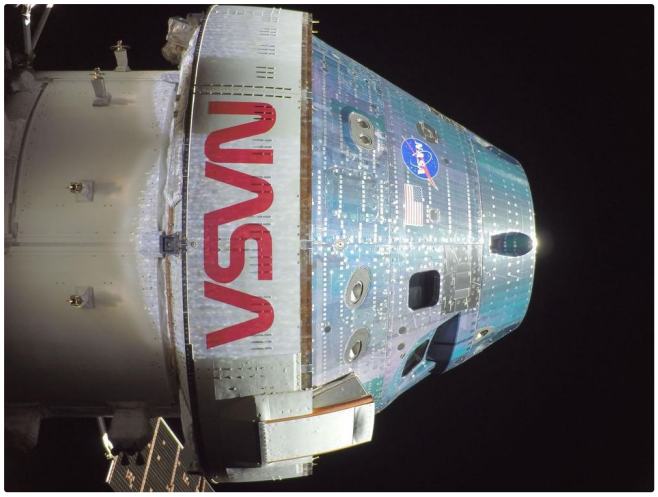


FIGURE 31.5 Orion photographed itself with a camera on a solar array wing on the second day of Artemis II, 3 April 2026. The crew module, with its windows, sits on the European Service Module. Image: NASA/JSC (NASA).

31.1.7 Coming Next: Gaganyaan, Mengzhou and the Lunar Landers

India's **Gaganyaan** is a three-person spacecraft of about 8 t (crew module plus service module), launched by the human-rated LVM3, for flights of up to seven days at about 400 km. As of October 2026 the first uncrewed orbital flight, G1, carrying the humanoid robot Vyommitra, had not flown. ISRO aimed for the end of 2026, though mid-2026 reports pointed to 2027. A crewed flight was not expected before 2027–28.

China's **Mengzhou** is a reusable successor to Shenzhou for Earth orbit (up to seven crew) and the Moon (three crew). After abort tests in 2025 and 2026, its first uncrewed orbital flight is expected around 2027. For lunar missions, one Long March 10 will launch Mengzhou and another the **Lanyue** lander. The lander completed a landing and take-off test in August 2025 while suspended from a tower rig that simulated lunar gravity. China's goal is a crewed landing by 2030.

The Artemis lunar landers are the largest crewed spacecraft yet designed. SpaceX's **Starship HLS** is a version of the 50 m Starship upper stage, with a crew cabin near the top reached by an elevator. It must be refuelled in orbit by a series of tanker flights. This had not yet been demonstrated when Starship first reached orbit in September 2026. Blue Origin's **Blue Moon MK2** is a 16 m, 45 t-class lander burning liquid hydrogen and oxygen. In 2026 NASA turned Artemis III into a crewed test in Earth orbit, no earlier than late 2027, in which Orion is to dock with one or both landers. The first landing moved to Artemis IV, no earlier than 2028.

TABLE 31.1 Crewed spacecraft compared (status in October 2026). Masses at launch without escape systems; free, habitable volume is roughly half the pressurised volume (5.9 m^3 in the Apollo CM, 8.9 m^3 in Orion).

Vehicle	Crew	Mass (t)	Pressurised volume (m^3)	Launcher	Landing	First crewed flight; status
Vostok (USSR)	1	4.7	—	Vostok	Land; crew ejects	1961; retired 1963
Mercury (USA)	1	1.4	2.8	Redstone, Atlas	Water	1961; retired 1963
Gemini (USA)	2	3.2–3.8	≈ 2.3	Titan II	Water	1965; retired 1966
Apollo CSM (USA)	3	28.8 (CM 5.6)	10.4	Saturn IB, V	Water	1968; retired 1975
Apollo LM (USA)	2	15–16	6.7	Saturn V	Moon	1969; retired 1972

Vehicle	Crew	Mass (t)	Pressurised volume (m³)	Launcher	Landing	First crewed flight; status
Soyuz MS (Russia)	3	7.1	8.5 (two modules)	Soyuz-2.1a	Land, retro-rockets	1967; flying
Space Shuttle (USA)	7–8	≈ 110 (orbiter)	66	Shuttle stack	Runway	1981; retired 2011
Shenzhou (China)	3	7.8	14 (two modules)	Long March 2F	Land, retro-rockets	2003; flying
Crew Dragon (USA)	4	≈ 12.5	9.3	Falcon 9	Water	2020; flying
Starliner (USA)	4	≈ 13	11	Atlas V	Land, airbags	2024; grounded
Orion (USA, Europe)	4	26.5 (with ESM)	19.6	SLS	Water	2026; next flight 2027
Gaganyaan (India)	3	≈ 8.2	—	LVM3	Water	Uncrewed G1 pending
Mengzhou (China)	3–7	14 (LEO), 21.6 (Moon)	13	Long March 10A, 10	Land, airbags	Uncrewed flight ≈ 2027

31.2 Anatomy of a Crewed Spacecraft

Most crewed spacecraft are **capsules** with a three-part layout (Figure 31.6). The **crew module**, the only part that returns, is a pressure vessel wrapped in a heat shield and thermal protection, containing the crew, the controls and the parachutes. The **service module** carries everything not needed for entry — main engine and most of the propellant, solar arrays, radiators and consumables — and burns up in the atmosphere. The **launch abort system** sits on top for the first minutes of flight.

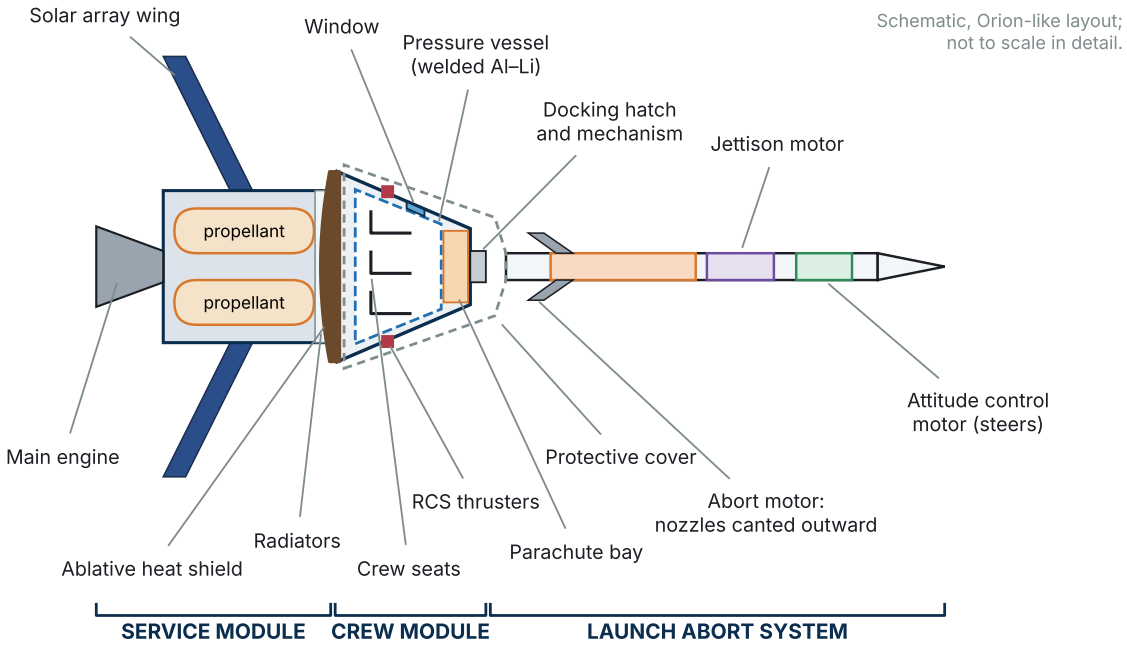


FIGURE 31.6 Anatomy of a capsule stack, based on Orion's layout. Only the crew module returns to Earth; the service module provides propulsion, power and heat rejection during the mission, and the abort system protects the crew during ascent.

31.2.1 Pressure Vessel and Atmosphere

The heart of the crew module is the **pressure vessel**, usually welded from aluminium or aluminium–lithium alloy. At sea-level pressure (101 kPa) every square metre of wall carries about 10 t, so the vessel is built from cones, cylinders and domes that carry pressure efficiently as tension (Chapter 5). It is designed for 1.5–2 times its operating pressure, proof-tested, and shielded from debris by the heat shield and backshell around it.

The cabin atmosphere has large consequences. Mercury, Gemini and Apollo flew pure oxygen at about 34–38 kPa, which kept the structure light. But on 27 January 1967 a fire broke out in the Apollo 1 command module, filled with pure oxygen above sea-level pressure for a ground test. The fire killed its three astronauts, whose inward-opening hatch could not be opened against the cabin pressure. Apollo was redesigned with a quick-opening hatch, non-flammable materials and a nitrogen–oxygen mixture on the pad. Soyuz, the Shuttle, Shenzhou, Dragon and Orion all fly an Earth-like nitrogen–oxygen atmosphere at about 101 kPa, maintained by the systems described in Chapter 30.

31.2.2 Windows

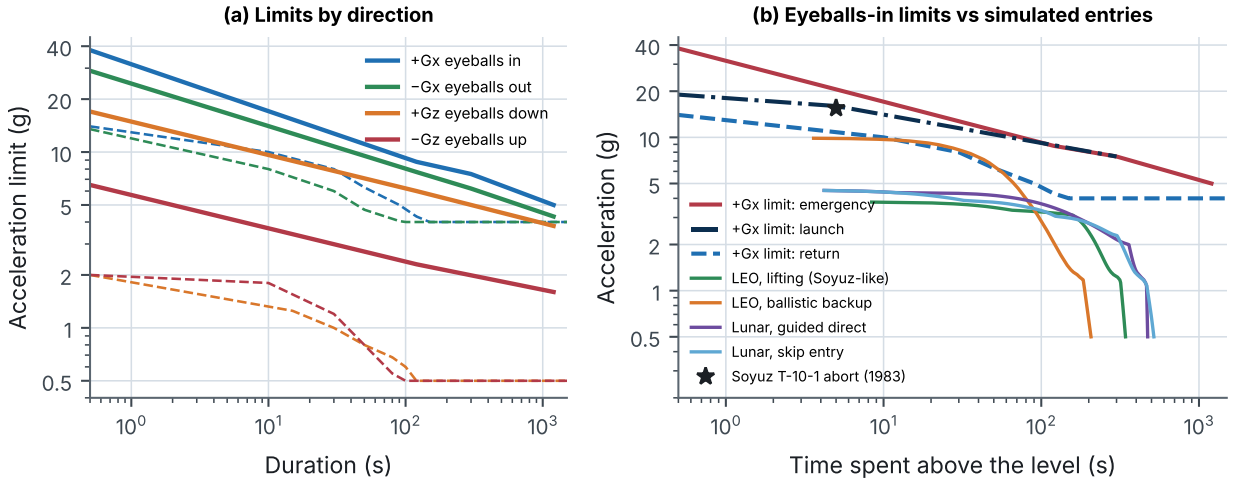
Windows are weak points in the pressure shell, but crews need them for docking and for seeing the horizon. They are typically stacked as three panes: an outer pane of fused silica that takes entry heating and debris impacts, and two inner pressure panes, either of which can hold cabin pressure on its own. Because glass fails by the slow growth of tiny flaws, windows have large safety factors. The crack in Shenzhou-20's window ran right through the outer, protective pane, but the two pressure panes behind it held — exactly what the redundancy is for.

31.2.3 Seats and Acceleration Limits

Capsule crews sit in contoured seats facing the heat shield, because the body tolerates acceleration very differently along its three axes (Figure 31.7). In the **eyeballs-in** direction (+Gx, chest to back) heart and brain are at the same level, so trained crews can withstand 15 g for several seconds and 4–5 g for many minutes. In the **eyeballs-down** direction (+Gz, head to foot) blood drains from the brain and, without protection, vision greys out at about 4 g. NASA's human-system standard, NASA-STD-3001, sets limits for each axis and duration, lower for crews deconditioned by months in weightlessness and higher for aborts and emergency entries.

In Figure 31.7 (b), a lifting entry from low Earth orbit (LEO) stays inside the nominal return limit. A ballistic backup entry, at almost 10 g, crosses it but stays within the emergency limit. This is why Soyuz and Shenzhou use ballistic entry only as a backup.

Seats spread the load over as much of the body as possible. Soyuz cosmonauts lie in individually moulded seat liners on seats that stroke on shock absorbers at touchdown; Dragon's four carbon-fibre seats sit side by side beneath the display panel (Figure 31.8).



Limits: NASA-STD-3001 Vol. 2 Rev. A, Figs 2–5 (sustained, > 0.5 s); in (a) solid = abort/emergency entry, dashed = nominal return. (b) Time each simulated entry of the entry-simulation figure spends above a given load; T-10-1: reported 14–17 g for about 5 s.

FIGURE 31.7 Human tolerance of sustained acceleration. (a) NASA-STD-3001 limits for four directions: eyeballs-in is by far the most tolerable. (b) Eyeballs-in limits compared with the time spent above each acceleration level in the simulated entries of Figure 31.13, and with the 1983 Soyuz T-10-1 pad abort.



FIGURE 31.8 The NASA SpaceX Crew-10 astronauts in their seats inside a Dragon capsule during a pre-flight equipment test at Kennedy Space Center, February 2025. The commander and pilot take the two central seats, facing the touchscreen display panel (out of view). Image: NASA/KSC (SpaceX).

31.2.4 Displays and Avionics

Avionics — the guidance, navigation and control computers, sensors and displays — show the largest generational change. The Apollo Guidance Computer had 2 048 words of erasable memory and was operated through a numeric keypad. The Shuttle had five computers, four running identical software and a fifth independently written backup software. Dragon's crew fly with three large touchscreens and a row of physical buttons for critical actions such as an emergency deorbit, parachute deployment or fire suppression. The vehicle normally rendezvouses and docks on its own. Orion uses airliner-derived flight computers with independent backup flight software, and its crew demonstrated manual piloting on Artemis II.

The governing principle is **fault tolerance**: no single failure — and for many functions no two — may cause loss of the crew. It is achieved with redundant, physically separated strings of equipment, voting computers that outvote a faulty member, dissimilar backups that cannot share a software error, and manual control of critical functions.

31.2.5 Propulsion, Power and Thermal Control

Crewed spacecraft carry a **main engine** for large manoeuvres (Orion's service module uses a refurbished 27 kN Space Shuttle orbital manoeuvring engine), **reaction-control thrusters** for attitude control and docking and, in pusher designs, **abort engines**. Almost all burn storable **hypergolic** propellants, which ignite on contact and can wait for months in orbit (Chapter 35). Dragon has 16 Draco thrusters of 400 N and 8 SuperDraco abort engines of about 71 kN, all fed from the same tanks. Thrusters are a recurring source of trouble. On Gemini 8 in 1966 a stuck thruster spun the spacecraft at about one revolution per second until Neil Armstrong regained control. Starliner's 2024 failures came from heat building up during long series of pulses. Qualification tests must therefore reproduce realistic firing sequences.

Gemini and Apollo introduced **fuel cells**, which combine hydrogen and oxygen to make electricity — on Apollo and the Shuttle also the crew's drinking water. Today's vehicles use **solar arrays** with batteries for eclipses and entry. Crew and electronics produce hundreds of watts to several kilowatts of heat, carried by pumped fluid loops to **radiators** on the service module or, on Dragon, the trunk (Chapter 5). The Soyuz MS-22 leak showed how exposed these external loops are.

31.2.6 Docking Systems

Docking joins two spacecraft so that crews can move between them through a pressurised tunnel. Early systems used a **probe** on the active vehicle and a cone-shaped **drogue** on the passive one, as Soyuz and the Russian ISS segment still do. **Androgynous** systems, with identical rings on both sides, let any two vehicles dock and leave the passage clear without removing a probe. The International Docking System Standard (IDSS) of 2010 defines a common interface for vehicles of different agencies and companies (Table 31.2). Modern systems first make a **soft capture**, with latches on a compliant, actuator-controlled ring absorbing the contact energy at closing speeds of about 0.1 m/s or less. They then make a **hard capture**, in which hooks pull the structures together and seal them.

TABLE 31.2 Docking systems for crewed spacecraft.

System	Type	Used by	Notes
Apollo probe and drogue	Probe on CSM, drogue on LM	Apollo, Skylab	Removed to open the tunnel
SSVP	Probe and drogue	Soyuz, Progress, Russian ISS modules	In use since 1971; 0.8 m hatch
APAS-75 / -89 / -95	Androgynous, peripheral	Apollo–Soyuz, Mir, Shuttle	Shuttle docked to Mir and ISS
Chinese mechanism	APAS-derived	Shenzhou, Tianzhou, Tiangong	Automated 2011, manual 2012
IDSS (NDS, SpaceX system)	Androgynous, low-impact	Dragon, Starliner, Orion; ISS adapters	Common standard; 0.8 m passage

31.3 Launch Abort

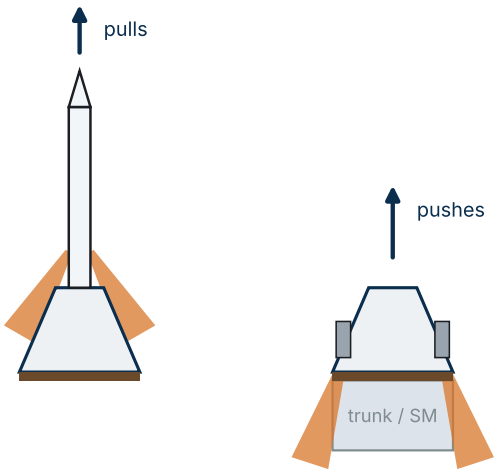
31.3.1 Why Crews Need to Escape

A launch vehicle is a controlled explosion of hundreds of tonnes of propellant. If it fails, the crew may have only seconds before blast overpressure, a fireball, flying fragments or, at altitude, aerodynamic forces reach them. The **launch abort system** must detect the failure, carry the crew module far enough away fast enough, stabilise it and hand it over to the normal parachute landing.

Vostok and Gemini used **ejection seats**, which work only at low altitude and speed. When Gemini 6A's engines shut down 1.2 s after ignition on 12 December 1965, Wally Schirra judged that the rocket had not lifted off. He chose not to eject, saving the mission and avoiding a risky ejection from a pure-oxygen cabin. Apollo and almost every crewed vehicle since — the Space Shuttle being the main exception — instead remove the whole crew module from the rocket.

31.3.2 Towers and Pushers

There are two ways to do this (Figure 31.9). A **tower**, or tractor, system mounts solid rocket motors above the capsule, with nozzles canted outwards so that the plumes clear it. It *pulls* the capsule away with a short, violent burn and is jettisoned once the rocket is high enough. Mercury, Apollo, Soyuz, Shenzhou, Orion, Gaganyaan and Mengzhou use towers. A **pusher** system builds liquid-propellant engines into the capsule or service module, like Dragon's SuperDracos and Starliner's four 178 kN RS-88 engines, and *pushes* it away.



(a) Tower (tractor) Mercury, Apollo, Soyuz, Shenzhou, Orion, Gaganyaan, Mengzhou

(b) Pusher (integrated engines) Crew Dragon, Starliner

	Tower	Pusher
Thrust-to-weight	≈ 7–13	≈ 4–6
Burn time	1–5 s	≈ 5–10 s
Available	pad → jettison	pad → orbit
Mass to orbit	jettisoned	carried
Re-use	lost each flight	with capsule
Propellant	solid	hypergolic
Risk	jettison event	propellant on board

(c) Trade-offs (typical values)

FIGURE 31.9 Launch escape systems. (a) A tower pulls the capsule away; (b) a pusher uses engines built into the capsule or service module. (c) Typical trade-offs between them.

Towers give very high thrust for their mass from simple solid motors, but they are dead weight after jettison and are discarded on every flight. Pushers are available all the way to orbit, and their propellant can be used for manoeuvring. However, they carry tonnes of toxic propellant next to the crew for the whole mission, and they give a slower escape.

31.3.3 Abort Modes

The abort method changes as the rocket climbs. On the pad, the escape motor must lift the capsule high enough to open its parachutes; through **maximum dynamic pressure** (max-Q) it must also overcome drag. After the tower is jettisoned, two to three minutes after launch, the capsule separates with its own engines (on Soyuz, with motors on the payload fairing) and makes a ballistic or lifting entry downrange. Close to orbital speed, it may be better to continue to a lower orbit (**abort to orbit**).

31.3.4 Aborts in Flight

Table 31.3 lists the aborts that occurred during crewed launches or, for Soyuz T-10-1, with the crew aboard on the pad; every crew survived. (The Shuttle also had five engine shutdowns on the pad, which needed no escape.) The two Soyuz in-flight aborts showed that the backup motors and ballistic entry worked — at the cost of high accelerations.

TABLE 31.3 Launch aborts with crews aboard.

Flight	Date	Phase	What happened	Outcome
Soyuz 18a	5 Apr 1975	Staging	Stages failed to separate; steep entry from 192 km at about 21 g	Crew injured; landed in the Altai mountains
Soyuz T-10-1	26 Sep 1983	On the pad	Booster fire; tower fired seconds before the explosion; 14–17 g for 5 s	Crew unhurt; landed 4 km away
STS-51F	29 Jul 1985	Ascent, 5 min 45 s	Main engine shut down by faulty sensors	Abort to orbit; mission completed
Soyuz MS-10	11 Oct 2018	Booster separation, ≈ 2 min	Strap-on struck the core (sensor pin deformed in assembly); fairing motors pulled the spacecraft clear; ballistic entry at 6–7 g	Crew landed 19 min 41 s after launch

31.3.5 Abort Tests and Abort Sizing

Abort systems are proven with uncrewed test flights. Orion's Pad Abort-1 in May 2010 accelerated a test capsule to about 200 m/s (445 mph) in three seconds. Ascent Abort-2 in July 2019 fired the abort motor at about 9.5 km and Mach 1.1 (Figure 31.10). SpaceX tested Dragon from the pad in 2015 and, in January 2020, near max-Q from a Falcon 9 whose engines were shut down at the abort command. Starliner's pad abort test in November 2019 succeeded although one of three main parachutes failed because of an improperly connected link (Figure 31.11). India tested Gaganyaan's system in flight in 2023, and China tested Mengzhou in 2025 and 2026.

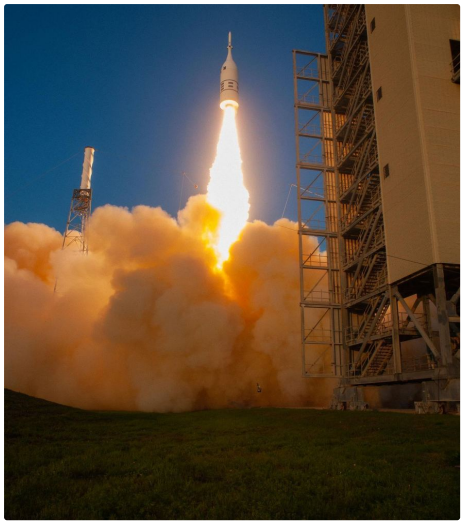
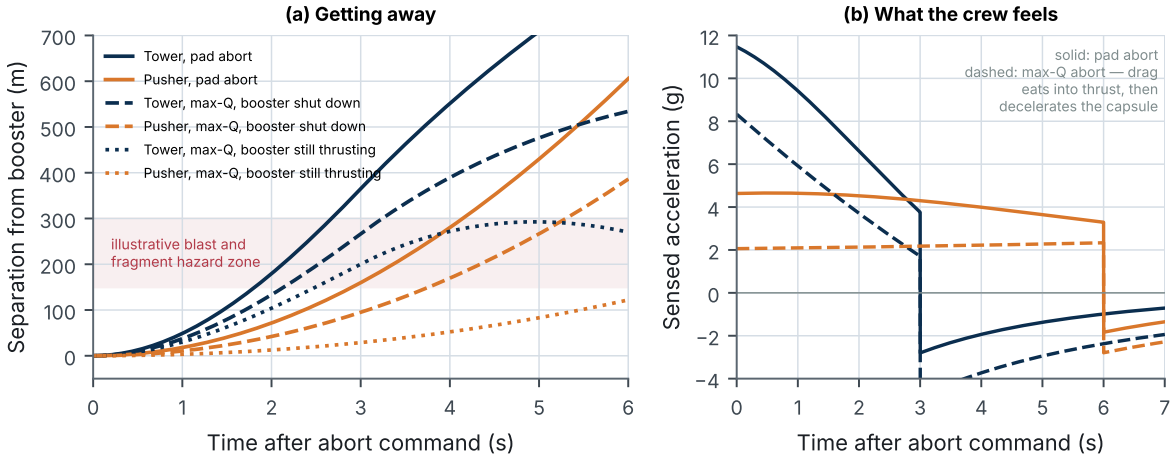


FIGURE 31.10 Orion's Ascent Abort-2 test lifts off from Cape Canaveral on 2 July 2019. Less than a minute later the abort motor fired at about 9.5 km altitude, pulling the test crew module off the booster.
Image: NASA/KSC (NASA/Tony Gray and Kevin O'Conne).



FIGURE 31.11 Starliner's pad abort test at White Sands, 4 November 2019: four abort engines and several manoeuvring thrusters push the spacecraft off the stand with a combined thrust of about 710 kN.
Image: NASA/KSC (NASA JSC/Boeing).

How much thrust is needed? That depends on how far the crew module must get from the booster, and how quickly; Figure 31.12 simulates both types of system.



1-D model along the flight path. Tower: 16 t abort vehicle, 1.8 MN regressive thrust for 3 s. Pusher: 12.5 t, 8 × 71 kN for 6 s. Max-Q: 420 m/s at 11 km (q ≈ 31 kPa), C_D ≈ 0.8; booster 300 t.

FIGURE 31.12 Simulated separation after an abort. (a) Distance from the booster for a tower and a pusher, from the pad and at maximum dynamic pressure. If the booster keeps thrusting, even a tower struggles to escape. (b) The acceleration the crew feels: at max-Q, drag first reduces the net push and then decelerates the capsule once the motor burns out.

EXAMPLE: SIZING AN ABORT MOTOR

Suppose the crew module must be 300 m from the booster within 3 s of the abort command, and the abort vehicle (crew module plus tower) has a mass of 16 t.

1. Required relative acceleration: $a_{rel} = 2s/t^2 = 2{\times}300/3^2$ = 66.7 m/s² (6.8 g).
2. **Pad abort.** The booster stands still and the capsule climbs against gravity: $T/W = (a_{rel} + g)/g$ = **7.8**, a thrust of 1.22 MN.
3. **Max-Q, booster shut down.** At 420 m/s and 11 km (ρ = 0.356 kg/m³), $q = \frac{1}{2}\rho v^2$ = 31 kPa. With $C_D A$ = 0.8 × 19.6 = 15.7 m², the drag is 0.49 MN (3.1 g). Gravity acts equally on capsule and booster and cancels in their relative motion: $T = m\,a_{rel} + D$ = 1.07 + 0.49 = 1.55 MN, **T/W = 9.9**.
4. **Max-Q, booster still thrusting** at 1.5 g: $T = m(a_{rel} + 1.5g) + D$ = 1.79 MN, **T/W = 11.4** — close to the 1.8 MN (400 000 lbf) of Orion's abort motor.
5. A Dragon-class pusher (8 × 71 kN on 12.5 t, T/W = 4.6) needs about 4.1 s to open 300 m from the pad and 5.3 s at max-Q (Figure 31.12). Pushers therefore depend on shutting the booster's engines down the instant an abort begins. A solid booster cannot be shut down, which favours towers on rockets with solid first stages.

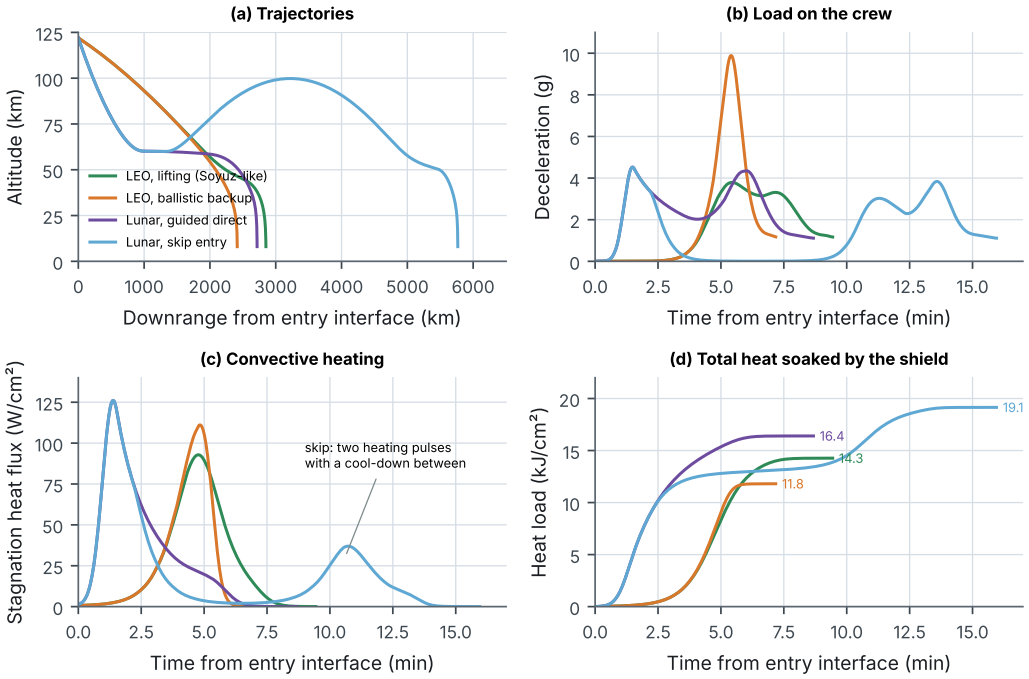
31.4 Coming Home: Entry, Descent and Landing

31.4.1 Capsule Shapes and Lift

A capsule returning from orbit must dissipate about 30 MJ of kinetic energy per kilogram, and from the Moon about 60 MJ (Chapter 10). Placing the centre of mass slightly off the axis makes the capsule fly at an angle of attack of 15–25°. It then produces a **lift-to-drag ratio** (L/D) of 0.25–0.4 — about 0.37 for Apollo and 0.25–0.3 for Soyuz, Shenzhou and Orion. By rolling to point this lift up, down or sideways (**bank-angle modulation**), the guidance controls both deceleration and landing point; Artemis II splashed down 0.95 km from its predicted point. A broad cone like Orion's has a low **ballistic coefficient** $\beta = m/(C_D A)$, so it decelerates high in thin air. But it must stay stable from hypersonic to subsonic speeds, and it carries uprighting bags in case it floats apex-down.

31.4.2 Ballistic, Lifting and Skip Entries

Figure 31.13 compares four ways of coming home. A **lifting entry from LEO** (Soyuz-like, at 7.6 km/s and −1.35°) peaks at 3.8 g. The **ballistic backup** of the same capsule is flown when the guidance fails, with the capsule spinning slowly to cancel its lift. It peaks near 10 g in the model, compared with the roughly 8 g reported on the ballistic Soyuz entries of 2007 and 2008. Returning from the Moon at 11 km/s, a capsule must enter more steeply (about −6°) so as not to skip out; with lift and guidance the model peaks at 4.5 g. A **skip entry** dips into the atmosphere, climbs back to about 100 km and enters again. This roughly doubles the downrange distance, so that the capsule can reach a fixed landing zone from a wider range of return trajectories.



Model: 3-DOF point mass, US-76 density, Sutton–Graves convective heating only (radiative heating, significant above 10 km/s, omitted). LEO: 7.6 km/s, −1.35°, β = 610 kg/m², L/D 0.3, bank 55°. Lunar: 11.0 km/s, −6.0°, β = 350 kg/m², L/D 0.27, simple bank-angle guidance. Entry interface 122 km. Illustrative, not flight trajectories.

FIGURE 31.13 Simulated capsule entries. (a) Trajectories; (b) deceleration; (c) convective heat flux at the stagnation point; (d) total heat absorbed per unit area. Lift cuts the peak load from about 10 g to about 4 g; a lunar return raises the peak heating, and a skip entry splits it into two pulses and increases the total heat load.

EXAMPLE: WHY LUNAR RETURN NEEDS LIFT

The Allen–Eggers formula (Chapter 10) gives the peak deceleration of a ballistic body entering on a straight path at speed V_E and angle γ_E, with scale height H = 7.2 km:

$$n_{max} = \frac{V_E^2 \sin|\gamma_E|}{2\,e\,H\,g_0}$$

1. **LEO return**, 7.6 km/s at −1.35°: n_{max} = 7 600² × 0.0236 / (2 × 2.718 × 7 200 × 9.81) = **3.5 g**.
2. **Lunar return**, 11.0 km/s at −6.0°: n_{max} = 11 000² × 0.105 / 383 900 = **33 g** — a factor (11.0/7.6)² = 2.1 from speed and 4.4 from the steeper angle.
3. **Simulation** (Figure 31.13). The LEO ballistic case actually peaks at 9.9 g: on such a shallow path gravity bends the trajectory down, steepening it to about −5.5° by the time of peak deceleration. The lunar ballistic case at −6.0° peaks at 9.7 g, because above orbital speed the curving path lifts the capsule — but entering just 0.5° steeper raises it to 15.5 g.
4. **Lesson.** The formula shows the scaling — deceleration grows with the square of speed and the steepness of entry — but real entries must be integrated numerically. A lunar-return capsule is so sensitive to its entry angle that it needs lift and closed-loop guidance; with L/D = 0.27 the simulated peak falls to 4.5 g.

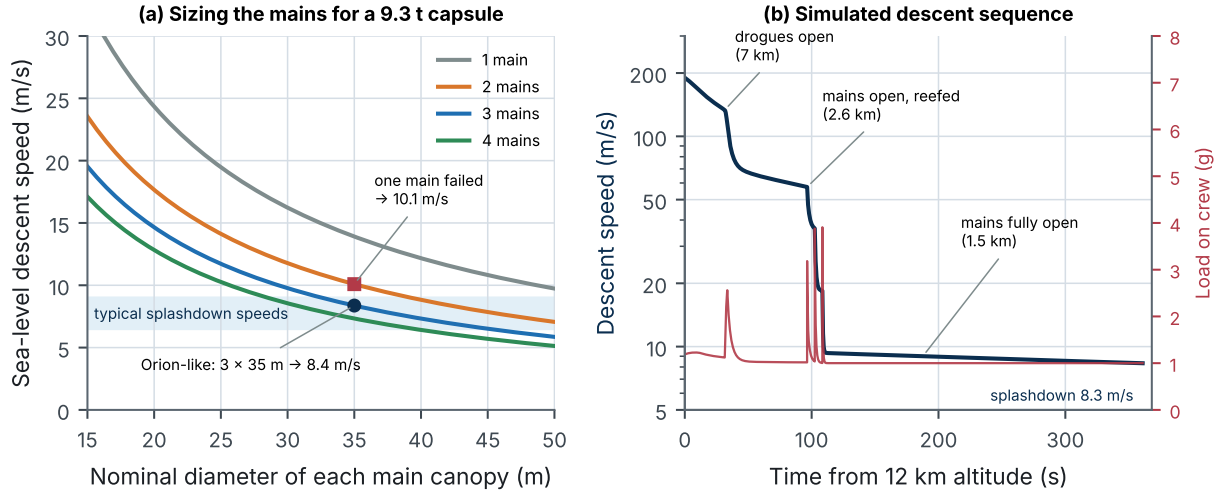
31.4.3 Heat Shields

A crewed heat shield cannot be tested at full scale before flight. Nearly all capsules use **ablators**, which char and erode, carrying heat away in the gases they release (Chapter 10). Apollo's **Avcoat**, an epoxy-novolac resin with silica fibres filling a fibreglass honeycomb, was revived for Orion, whose 5 m shield is made of 186 bonded blocks. Dragon uses SpaceX's **PICA-X**, a low-density carbon–phenolic ablator derived from NASA's PICA. Soyuz and Shenzhou jettison their heat shields shortly before landing.

After **Artemis I**, engineers found that charred Avcoat had broken away from more than a hundred places on Orion's heat shield. NASA traced the cause to the skip entry. As the heating fell between the two pulses — the dip in Figure 31.13 (c) — the Avcoat kept generating gas inside. But it formed permeable char too slowly to let the gas escape, so pressure built up and cracked off pieces of char. Rather than replace the Artemis II shield, NASA flew a modified entry with a shorter maximum downrange distance, reducing the time spent in the troublesome heating regime. Artemis II's shield, whose surface exceeded 2 200 °C, showed only nine spots of char loss against 103, and the Artemis III shield uses Avcoat of more uniform permeability.

31.4.4 Parachutes

Below about 10 km a capsule falls at 100–200 m/s, and parachutes take over in stages that keep opening loads within the strength of the structure and the tolerance of the crew. Pilot parachutes or mortars deploy **drogues**, which stabilise the capsule and slow it to 50–90 m/s. The drogues then pull out the **mains**. Their **reefing** lines hold each canopy partly closed until pyrotechnic cutters release them a few seconds later (Figure 31.14 b). Redundancy is essential: Apollo 15 landed safely in 1971 although one of its three mains collapsed, and modern systems are designed to land safely with one main failed. Soyuz has a single large main parachute and a complete reserve.



v = √(2mg / ρ C_D S); ringsail C_D ≈ 0.8 on nominal area, cluster efficiency 0.90–0.95. (b) drogues 2 × 7 m from 12 km at 190 m/s (vertical), mains reefed to 6 % then 25 % of area.

FIGURE 31.14 Parachute design. (a) Sea-level descent speed versus main-canopy diameter for a 9.3 t capsule with one to four mains; Orion's three 35 m mains give about 8 m/s, and about 10 m/s with one failed. (b) A simulated descent: drogues at 7 km, reefed mains at 2.6 km, full mains below 1.5 km.

At terminal velocity, drag equals weight, so the descent speed is $v = \sqrt{2mg/(\rho C_D S)}$ for total canopy area S: halving the landing speed needs four times as much parachute.

EXAMPLE: SIZING THE MAIN PARACHUTES

A 9.3 t crew module must splash down at no more than 10 m/s at sea level (ρ = 1.225 kg/m^3) even if one of its three mains fails. Ringsail canopies have $C_D \approx 0.80$ on their nominal area, and two canopies in a cluster lose about 5 % of their drag through interference.

1. Drag area needed: $S = 2mg/(\rho v^2 C_D)$ = 2 × 9 300 × 9.81 / (1.225 × 10^2 × 0.80) = 1 860 m^2.
2. Two working canopies: each needs 1 860 / (2 × 0.95) = 980 m^2, a nominal diameter $D_0 = \sqrt{4 \times 980/\pi}$ = **35.3 m** — the size of Orion's mains.
3. With all three open (cluster efficiency 0.92), v = **8.3 m/s** (19 mph), and the capsule meets the water with about 320 kJ of kinetic energy.
4. **Why reef?** The mains open at 2.6 km (ρ = 0.94 kg/m^3) and about 57 m/s, where q = 1.55 kPa. Opened fully at once, three canopies would decelerate the capsule at about 36 g; reefed to 6 % of their area, about 2 g. The peak in the simulated sequence is 3.9 g.

31.4.5 Water or Land?

Water landings are gentle but need ships and divers, and seawater makes reuse harder. American capsules from Mercury to Orion have splashed down (Figure 31.15); since mid-2025 Dragon has usually returned to the Pacific off California, where its trunk, jettisoned after the deorbit burn, falls into the ocean.



FIGURE 31.15 Orion Integrity descends under its three main parachutes before splashdown in the Pacific off San Diego on 10 April 2026, ending Artemis II. Image: NASA/KSC (NASA/Josh Valcarcel).

Landing on land makes recovery and reuse easier but needs a soft touchdown. Soyuz and Shenzhou fire **soft-landing rockets**, six small solid motors triggered by a gamma-ray altimeter about a metre above the ground, which cut the speed from about 7 m/s to about 1.5 m/s (Figure 31.16). Starliner inflates six **airbags** beneath its jettisoned heat shield and lands at White Sands, New Mexico; Mengzhou will also use airbags (Table 31.4).



FIGURE 31.16 Soyuz MS-08 lands near Zhezkazgan, Kazakhstan, on 4 October 2018. The flash beneath the descent module is the soft-landing rockets firing just before touchdown. Image: NASA (NASA/Bill Ingalls).

TABLE 31.4 Descent and landing systems of capsules.

Vehicle	Drogues	Mains	Speed under mains	Final attenuation
Apollo CM	2 × 5.0 m	3 × 25.5 m ringsail	≈ 10 m/s (22 mph)	Water
Soyuz	1 (≈ 24 m²) after pilots	1 × ≈ 1 000 m², plus reserve	≈ 7 m/s	Soft-landing rockets → ≈ 1.5 m/s
Crew Dragon	2, at ≈ 5.5 km	4, at ≈ 1.8 km	≈ 7 m/s (15–16 mph)	Water
Starliner	2	3	—	Six airbags, on land
Orion	2 × 7 m	3 × 35 m ringsail	< 9 m/s (< 20 mph)	Water

At sea, Dragon's recovery ship lifts the capsule onto its deck within about an hour, and the crew are helped out while their balance readjusts to gravity (Figure 31.17). The Artemis II crew instead stepped onto a raft and were flown to their ship by helicopter. The winged alternative — the Shuttle's runway landing after a gliding entry with up to 2 000 km of cross-range — gave the gentlest deceleration of all. The cost was carrying wings, landing gear and a fragile thermal protection system to orbit and back (Chapter 10).



FIGURE 31.17 The recovery team opens the hatch of Dragon Endeavour on the SpaceX recovery ship after the Crew-11 splashdown off California on 15 January 2026. The discolouration is from entry heating. Image: NASA (NASA/Bill Ingalls).

31.5 Reliability and Safety

31.5.1 What Has Gone Wrong

Four accidents in flight have killed 18 people in orbital spacecraft, and the Apollo 1 fire killed three on the ground (Table 31.5). Each occurred where the crew could not escape: during entry and landing (Soyuz 1, Soyuz 11, Columbia), or during ascent in a vehicle without an escape system (Challenger).

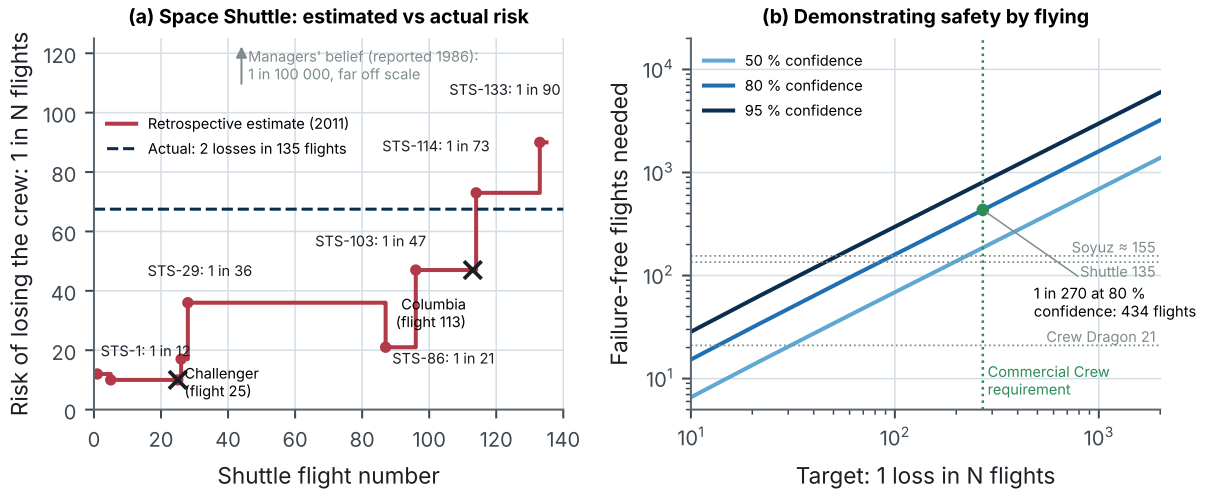
TABLE 31.5 Fatal accidents in orbital crewed spacecraft.

Accident	Date	Phase	Cause	Main changes afterwards
Apollo 1	27 Jan 1967	Ground test	Fire in pure oxygen; inward-opening hatch; 3 killed	N_2–O_2 on the pad, new hatch, materials control
Soyuz 1	24 Apr 1967	Landing	Main parachute failed, reserve tangled; 1 killed	Redesigned parachutes
Soyuz 11	30 Jun 1971	Entry	Valve opened at module separation; cabin depressurised; 3 killed	Pressure suits for launch and landing
Challenger (STS-51L)	28 Jan 1986	Ascent	Booster joint seal failed in cold weather; 7 killed	Booster redesign, bail-out system
Columbia (STS-107)	1 Feb 2003	Entry	Wing holed by foam at launch; 7 killed	In-orbit inspection; Shuttle retirement

31.5.2 Estimating Risk

How safe is a spacecraft? The best answer comes from **probabilistic risk assessment (PRA)**. This is a model of the vehicle built from fault trees and event sequences, using failure rates from tests and experience. It estimates the probability of **loss of crew (LOC)** per flight. Investigating the Challenger accident in 1986, the physicist Richard Feynman found that NASA managers believed the chance of losing a Shuttle was about 1 in 100 000, while engineers estimated about 1 in 100. A 2011 NASA study recomputed the risk as it really was through the programme (Figure 31.18 a). It was about 1 in 12 on the first flight and 1 in 10 at the time of Challenger. It improved to about 1 in 90 by the end as hazards were found and fixed. The actual record — 2 losses in 135 flights, or 1 in 68 — agreed with this retrospective estimate, not with the managers' belief.

Figure 31.18 (b) shows why flight experience alone cannot prove a vehicle safe. After n flights without loss, a vehicle whose true risk is p per flight would show that record with probability $(1-p)^n$. Claiming with confidence C that the risk is no worse than p requires $n = \ln(1-C)/\ln(1-p)$ successful flights. Showing 1 in 270 at 80 % confidence needs 434 flights — more than any crewed vehicle has made. Safety must be built in by design, analysis and test.



(a) Hamlin et al., NASA Shuttle risk progression (2011), loss-of-crew-and-vehicle odds at selected flights. (b) n = ln(1 − C) / ln(1 − 1/N): flights with no loss needed to show risk ≤ 1/N.

FIGURE 31.18 Measuring safety. (a) Retrospective estimates of the Space Shuttle's loss-of-crew risk at selected flights, compared with the actual loss rate. (b) The number of consecutive successful flights needed to demonstrate a risk target with a given statistical confidence.

31.5.3 Human-Rating

NASA certifies a spacecraft for crews (**human-rating**) under NPR 8705.2, *Human-Rating Requirements for Space Systems*, together with the technical standards of NASA-STD-3001. Among other things, the system must:

- tolerate failures, so that no single failure — and for many functions no combination of two — causes a catastrophic event;
- provide abort and crew survival capability in every phase where it is feasible;
- give the crew insight into, and manual control of, critical functions, within human limits for acceleration, noise, vibration, atmosphere and workload;
- be verified by test and analysis, including flight tests of the integrated system before crews fly.

For Commercial Crew, NASA required a loss-of-crew risk no worse than **1 in 270** over a 210-day ISS mission, with no more than 1 in 500 from ascent and entry. This is three to five times safer than the Shuttle at its best. In 2019 the US Government Accountability Office found that debris strikes during the long docked period dominated the risk. Starliner showed the limits of certification: thruster failures had also occurred on its 2022 uncrewed flight, but their cause was not fully understood before a crew flew.

AI IN CREWED SPACECRAFT

Crewed vehicles already rely on autonomy. Soviet and Russian spacecraft have docked automatically using radio systems since 1967 (Igla, then Kurs from 1986). Crew Dragon flies its whole rendezvous and docking on its own with lidar, thermal and visible cameras while the crew monitors. Machine learning is being added cautiously, where it can be verified. Uses include vision-based pose estimation for docking and inspection, and anomaly detection that compares thousands of telemetry channels with learned normal behaviour. Such detection could flag slow degradations such as Starliner's thruster heating. There are also crew assistants, such as the voice assistant demonstrated on Artemis I. On Mars missions, light-time delays will make onboard fault diagnosis essential. Chapters 49 and 51 discuss these applications and how they can be qualified for human spaceflight.

KEY TAKEAWAYS

- In October 2026 Soyuz, Shenzhou, Crew Dragon and Orion were in service, Starliner was grounded, and Gaganyaan and Mengzhou were preparing for uncrewed test flights.
- A capsule stack has a crew module (pressure vessel, heat shield, parachutes), a service module and a launch abort system; only the crew module returns.
- Crews tolerate acceleration best eyeballs-in: about 15 g for seconds and 4–5 g for minutes. Lift ($L/D \approx 0.3$) cuts entry loads from about 10 g to about 4 g and lets capsules steer to a precise landing.
- Abort systems must open hundreds of metres of separation within seconds: towers give $T/W \approx 7$–13; pushers ($T/W \approx 4$–6) work all the way to orbit but rely on shutting the booster down.
- Lunar return doubles the energy per kilogram and demands a precise entry angle, guided lift and robust heat shields.
- Parachute descent speed falls only with the square root of canopy area; systems are designed to land safely with one main parachute failed.
- Flight experience alone cannot demonstrate safety: the Shuttle's 1-in-68 record matched retrospective risk models, not its managers' 1-in-100 000 belief.

REVIEW QUESTIONS

1. Why do capsule crews lie facing the heat shield? Use Figure 31.7 to compare +Gx and +Gz limits for 60 s.
2. List the three elements of a capsule stack and their functions. Which are reused on Crew Dragon?
3. Compare tower and pusher abort systems. Why might a rocket with a solid-fuel first stage favour a tower?
4. An abort vehicle of 10 t must open 200 m of separation from a stationary booster within 2.5 s. What thrust is required?
5. Use the Allen–Eggers formula to estimate the peak deceleration of a ballistic capsule entering at 11 km/s and −6.5°. Why is the real value lower, and why do lunar-return capsules still need lift?
6. How does a skip entry differ from a direct entry, and how did the skip lead to char loss on Orion's Artemis I heat shield?
7. A 6 t capsule must splash down at no more than 9 m/s with one of its two mains failed. What nominal canopy diameter is needed ($C_D = 0.8$)?
8. Why do Soyuz and Shenzhou need soft-landing rockets, while Dragon and Orion do not?
9. What do the Shenzhou-20 window damage (2025) and the Soyuz MS-22 leak (2022) teach about lifeboats for space stations?
10. Why can flight experience alone not prove a crewed vehicle safe? How many successful flights would show a risk of 1 in 500 with 90 % confidence?

32 CHAPTER Space Stations and Habitats



Image: NASA/JSC

A space station is a spacecraft that people live in for months or years, supplied from Earth and maintained in orbit. This chapter traces stations from Salyut 1 and Skylab through Mir to the International Space Station and China's Tiangong, then takes the ISS apart system by system — power, thermal control, attitude control, reboost, communications and robotics — and explains the engineering that every station must solve: choosing an orbit against drag, sizing arrays and radiators, shielding against micrometeoroids and debris, providing enough habitable volume, and keeping cargo flowing. It covers expandable habitats and artificial gravity, the plan to deorbit the ISS around 2030, and the commercial and national stations competing to replace it as of October 2026.

IN THIS CHAPTER

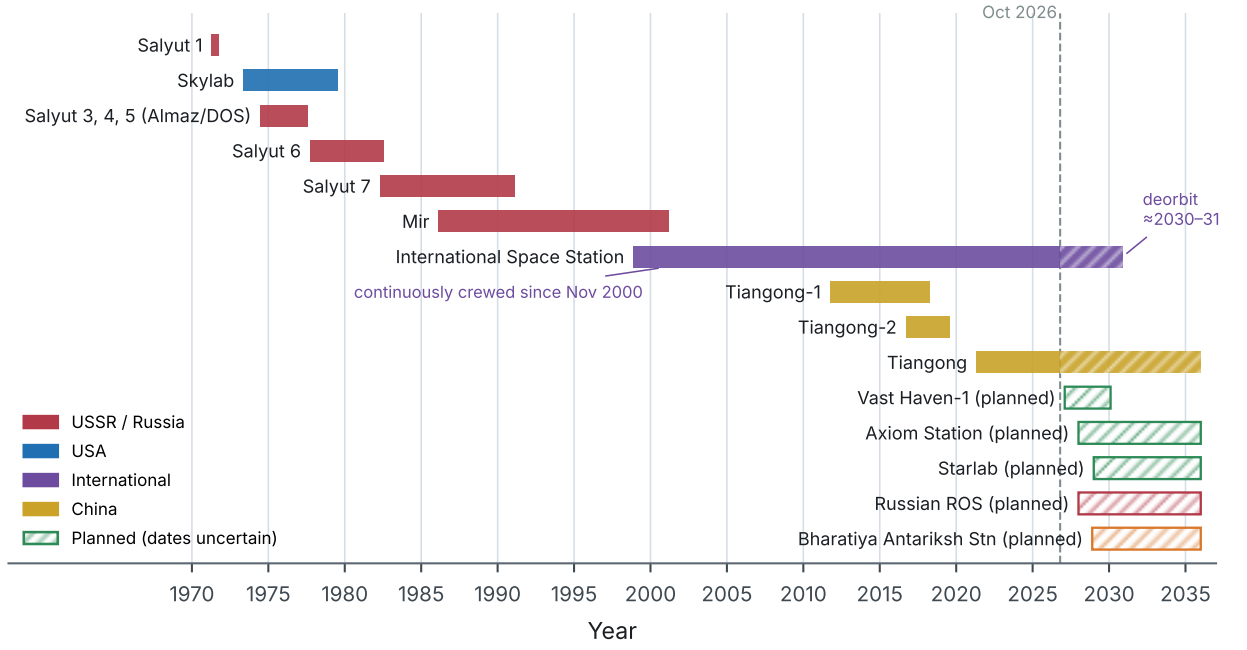
32.1 Fifty-Five Years of Space Stations
32.2 The International Space Station in Detail
32.3 Engineering a Station
32.4 Supplying a Station
32.5 Expandable Habitats and Artificial Gravity
32.6 After the ISS

A satellite can be designed once and left alone. A space station cannot: it is a house, a laboratory, a port and a factory in one, occupied by people who must be kept healthy and brought home safely. It must be supplied with tonnes of food, water, gas and spare parts every year, and boosted against atmospheric drag. It must also be repaired when things break, and protected against impacts from particles moving at ten times the speed of a rifle bullet. In return it offers something nothing else can — months of continuous human presence in weightlessness, for research, technology testing and preparation for journeys to the Moon and Mars.

People have lived aboard the **International Space Station (ISS)** without a break since November 2000, and Chinese crews have occupied the **Tiangong** station continuously since mid-2022. This chapter explains how stations work and how they are designed; the human body and life support are covered in Chapter 30, crewed spacecraft and docking systems in Chapter 31, and spacewalks in Chapter 33.

32.1 Fifty-Five Years of Space Stations

Each station since 1971 has taught how to keep people in orbit longer, with less help from the ground (Figure 32.1, Table 32.1).



Bars show time in orbit (launch to re-entry). Planned dates as reported in October 2026; Salyut 2 (failed 1973) omitted.

FIGURE 32.1 Space stations from Salyut 1 to the stations planned for the late 2020s. Each bar runs from launch to re-entry; hatched bars are planned, with uncertain dates.

32.1.1 Salyut and Skylab

The Soviet Union launched the first station, **Salyut 1**, on 19 April 1971: a single 18-tonne cylinder with about 100 m^3 of pressurised volume. Its first crew, on Soyuz 10, docked but could not enter. The second, aboard Soyuz 11, spent more than three weeks on board. But on 30 June 1971 Georgy Dobrovolsky, Vladislav Volkov and Viktor Patsayev died during their return, when a pressure-equalisation valve was jolted open as the descent module separated. The valve vented the module's air to space. They are the only people to have died above the atmosphere, and since then Soyuz crews have worn pressure suits for launch and re-entry.

The United States converted a Saturn V third stage into **Skylab**, launched on 14 May 1973. It was a 77-tonne station with 361 m^3 of volume, still the most spacious station ever launched in one piece (Figure 32.2). Sixty-three seconds into the launch its micrometeoroid shield tore away; one solar wing was lost and debris jammed the other. The first crew, Pete Conrad, Joe Kerwin and Paul Weitz, put up a parasol sunshade to cool the overheating workshop and cut free the jammed wing during a spacewalk — the first major in-orbit repair. Three crews spent 28, 59 and 84 days on

board, operating the Apollo Telescope Mount solar observatory. High solar activity then expanded the upper atmosphere, and Skylab re-entered on 11 July 1979 over the Indian Ocean and Western Australia, before the Space Shuttle could boost it.

Salyut 6 (1977–82) and **Salyut 7** (1982–91) introduced the feature that made long stays possible: a second docking port. Uncrewed **Progress** freighters, first flown in January 1978, could deliver food, water and propellant while a crew was on board, and short "visiting" crews could exchange the Soyuz lifeboat. Stays grew to 237 days in 1984. In 1985 Vladimir Dzhanibekov and Viktor Savinykh docked manually with a dead, frozen Salyut 7 and brought it back to life — a demonstration that crews can repair what automation cannot.

32.1.2 Mir

Mir grew from a core module launched on 19 February 1986 into a 130-tonne complex of seven modules by 1996 (Figure 32.3), the first station assembled in orbit. It was occupied for 4 592 days, including almost ten years without a break (1989–99), and in 1994–95 the physician Valeri Polyakov stayed 437 days, still the record for a single spaceflight. Between 1995 and 1998 nine Space Shuttle missions docked with Mir, and seven NASA astronauts lived on board — the rehearsal for the ISS. Mir also taught hard lessons. In February 1997 a burning oxygen-generating canister filled it with smoke. In June 1997 a remotely piloted Progress struck and punctured the Spektr module, which the crew had to seal off, losing much of the station's power. Mir was deliberately de-orbited into the South Pacific on 23 March 2001.

TABLE 32.1 Space stations that have been crewed. Masses exclude visiting vehicles.

Station	Operator	In orbit	Mass (t)	Pressurised volume (m³)	Notable for
Salyut 1	USSR	1971	18.4	≈ 100	First station
Skylab	USA	1973–79	77	361	Largest module
Salyut 6, 7	USSR	1977–91	19–20	≈ 90	Progress resupply
Mir	USSR, Russia	1986–2001	≈ 130	≈ 350	First modular station
ISS	5 agencies	1998–	≈ 420	1 005	Crewed since 2000
Tiangong-1, -2	China	2011–19	8.5	≈ 15	Docking, refuelling tests
Tiangong	China	2021–	≈ 70	≈ 340	Crewed since 2022



FIGURE 32.2 Skylab photographed by the departing Skylab 4 crew in February 1974. The gold twin-pole sunshade, erected by the second crew over the first crew's parasol, covers the workshop, which lost one of its two large solar wings during launch; the windmill-shaped arrays belong to the Apollo Telescope Mount. Image: NASA/JSC.

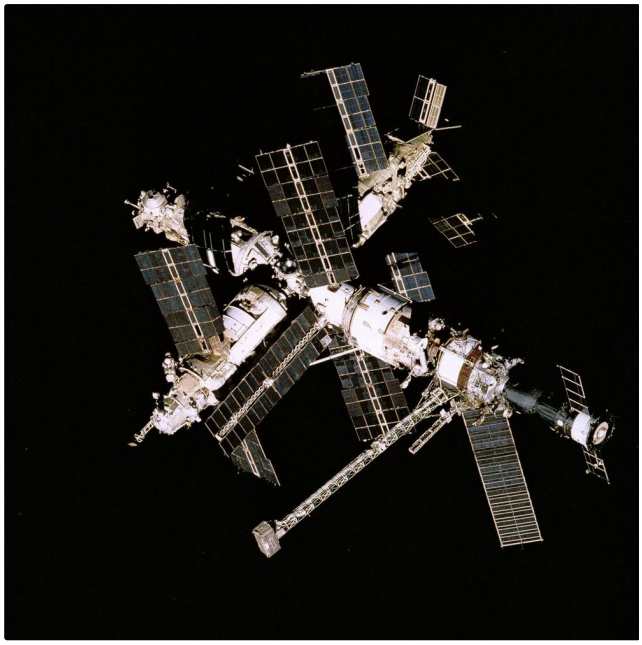


FIGURE 32.3 Mir photographed from the approaching Space Shuttle Atlantis on 29 June 1995, during the first Shuttle–Mir docking mission. The modules radiate from the docking node at the front of the core. Image: NASA/JSC.

32.1.3 The International Space Station

The ISS combined the American Freedom station design with the planned Mir-2. Five agencies — NASA, Roscosmos, the European Space Agency (ESA), the Japan Aerospace Exploration Agency (JAXA) and the Canadian Space Agency (CSA) — representing 15 countries are partners. The Russian-built, US-funded Zarya module was launched on 20 November 1998 and the US Unity node two weeks later; the first resident crew, Expedition 1, arrived on 2 November 2000. Assembly took 42 flights — 37 Space Shuttle missions and five Russian launches — and the last major Russian module, the Prichal docking node, arrived in November 2021. On 2 November 2025 the station marked 25 years of continuous occupation, with more than 290 people from 26 countries having visited.

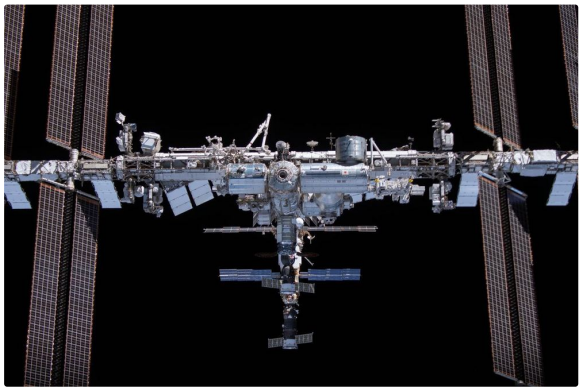


FIGURE 32.4 The ISS seen from Crew Dragon Endeavour after undocking in November 2021: the pressurised modules run down the centre from the 109 m truss and its solar-array wings to the Russian segment. Image: NASA/JSC.

32.1.4 China's Tiangong

China followed a step-by-step path. **Tiangong-1** (2011), an 8.5-tonne laboratory, received the uncrewed Shenzhou 8 and two crews in 2012 and 2013, then re-entered uncontrolled in April 2018. **Tiangong-2** (2016) hosted a 30-day mission and tested in-orbit refuelling by the Tianzhou-1 freighter in 2017. The modular **Tiangong** station began with the 22.5-tonne Tianhe core module on 29 April 2021. The Wentian and Mengtian laboratory modules (July and October 2022) completed a T-shaped station of about 70 tonnes — roughly 90–100 tonnes with docked spacecraft. It has about 340 m^3 of pressurised volume. It orbits at about 390 km and 41.5° inclination, normally carries three astronauts (six during crew handovers), and was the first crewed station to use **Hall-effect electric thrusters** for reboost. In November 2025, when the Shenzhou-20 return capsule was found to have a cracked window, China launched the uncrewed Shenzhou-22 as a replacement lifeboat within weeks.

32.2 The International Space Station in Detail

The ISS is 109 m from end to end and has a mass of about 420 tonnes. It offers 1 005 m^3 of pressurised volume — about as much as a Boeing 747 — of which 388 m^3 is habitable. It circles Earth every 93 minutes, about 15.5 times a day, at about 415–420 km altitude and 51.6° inclination, passing over more than 90 % of the world's population.

32.2.1 Modules and Segments

The station has two halves (Figure 32.5). The **Russian Orbital Segment** includes the Zvezda service module, which provides propulsion, crew quarters and life support for the Russian crew. It also includes Zarya, which was the first element; the docking and research modules Poisk, Rassvet, Nauka and Prichal; and ports for Soyuz and Progress. The **US Orbital Segment** includes the nodes Unity, Harmony and Tranquility. Its laboratories are the US Destiny laboratory, ESA's Columbus and JAXA's Kibo. It also has the Quest and commercial Bishop airlocks, the Leonardo storage module, the seven-window Cupola and the expandable BEAM. Modules connect through standard ports: Russian probe-and-drogue docking ports, and **Common Berthing Mechanisms (CBMs)** with 1.27 m square hatches wide enough to pass equipment racks. There are also **International Docking Adapters (IDAs)** for Crew and Cargo Dragon and Starliner.

32.2.2 Power

Electrical power comes from eight **solar-array wings**, each about 35 m long, mounted in pairs on rotating joints at the ends of the truss so that they can track the Sun. Their silicon cells, designed for 15 years, have degraded; by about 2020 the arrays generated roughly 160 kW in sunlight. Since 2021 NASA has installed **ISS Roll-Out Solar Arrays (iROSAs)**, flexible blankets that unroll from a compact canister and each produce more than 20 kW. They are mounted in front of the old wings, partly shading them (Figure 32.6). Six iROSAs installed by 2023 raised the output to about 215 kW, and the final pair arrived in October 2026 aboard the CRS-35 Cargo Dragon for installation by spacewalk. Part of the power charges 24 lithium-ion batteries (which replaced 48 nickel–hydrogen batteries in 2017–21) for the eclipse. So the power available to the station's systems and

experiments over a whole orbit is much lower than the sunlit output. NASA's fact sheet quotes 75–90 kW. The US segment distributes power at 160 V direct current, converted to 124 V for users; the Russian segment uses 28 V.

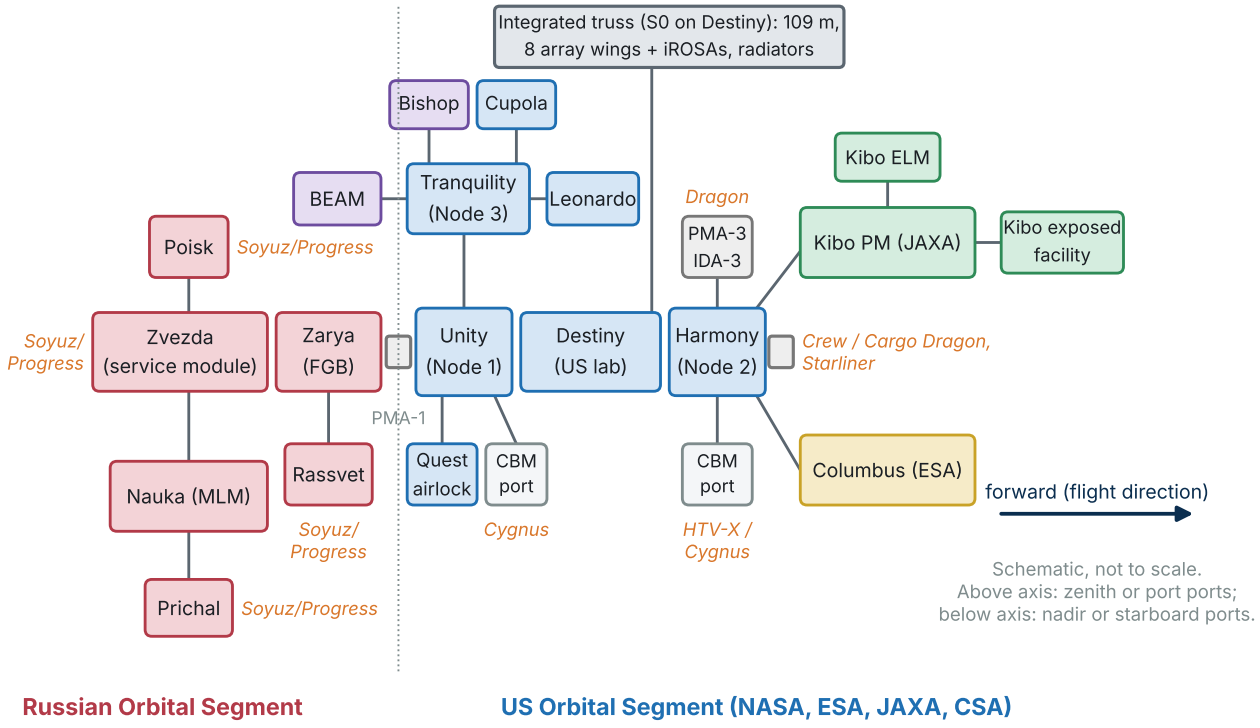


FIGURE 32.5 Schematic connection diagram of the ISS pressurised modules and the ports used by visiting vehicles in 2026.

32.2.3 Thermal Control

Every watt the station uses ends up as heat that must be radiated to space. Inside the US segment, two water loops — a low-temperature loop at about 4 °C and a moderate-temperature loop at about 17 °C — collect heat from cold plates and air heat exchangers. Water is safe in the cabin but would freeze outside, so heat exchangers pass the heat to the **External Active Thermal Control System**. This consists of two independent loops of liquid ammonia that circulate through six radiator assemblies on the truss. There are three assemblies on each side; each has eight panels and rejects about 11.7 kW, 70 kW in all. The radiators rotate to keep edge-on to the Sun. Separate radiators cool the power electronics at the array roots. Ammonia leaks and pump failures have required several repair spacewalks.

32.2.4 Attitude Control and Reboost

The station normally flies with one axis along its velocity and another pointing at Earth, turning once per orbit. Small torques from gravity gradient, drag and venting would slowly twist it out of this attitude, so four **control moment gyroscopes (CMGs)** on the Z1 truss absorb them. Each is a 100 kg wheel spinning at 6 600 rpm with an angular momentum of 4 760 N·m·s; tilting the wheel's axis on motorised gimbals produces a torque on the station without using propellant. When the CMGs approach saturation, or for large manoeuvres, Russian thrusters take over. The station is flown slightly tilted, in a **torque-equilibrium attitude** where aerodynamic and gravity-gradient torques cancel on average.

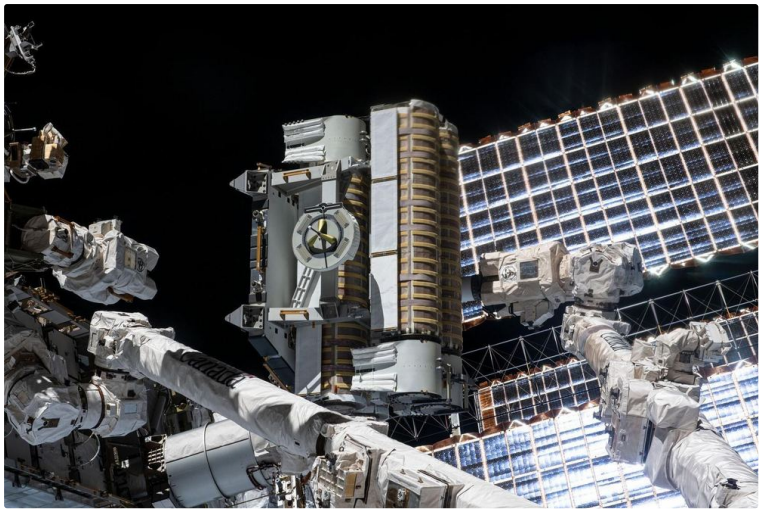


FIGURE 32.6 Spacewalkers work with the first ISS Roll-Out Solar Array (iROSA) on the station's truss in June 2021, during the Expedition 65 spacewalks that installed it. Each iROSA unrolls like a carpet over an older, gold-coloured array wing. Image: NASA/JSC.

Drag lowers the station by roughly 1–5 km per month, depending on solar activity. It is periodically **reboosted** — raised by a few kilometres — by the engines of Zvezda and docked Progress freighters. Since November 2024 Cargo Dragon has also reboosted it, and one mission (CRS-33, 2025–26) carried a propulsion kit in its trunk. ESA's ATV and some Cygnus missions provided reboosts in the past. Thrusters also perform **debris-avoidance manoeuvres** when a tracked object's collision probability exceeds about 1 in 10 000: about 40 between 1999 and 2024. In July 2021, shortly after Nauka docked, its thrusters fired unexpectedly and turned the whole station about one and a half revolutions before controllers regained control.

32.2.5 Communications, Data and Robotics

Voice, telemetry and commands travel over S-band, and high-rate data and video over Ku-band. Both go through NASA's **Tracking and Data Relay Satellites (TDRS)** in geostationary orbit, which give almost continuous coverage (Chapter 7). The Russian segment uses ground stations and Luch relay satellites. More than 50 computers run about 1.5 million lines of flight software and monitor about 350 000 sensors.

Canadarm2, a 17.6 m arm with seven joints and a latching "hand" at each end, can move end over end between grapple fixtures and ride a mobile base along the truss. It assembled much of the station, captures visiting cargo vehicles and moves spacewalkers. Its two-armed attachment **Dextre** handles smaller replaceable units (Figure 32.7), and Kibo and Nauka carry their own arms. After a wrist joint failed in May 2026, spacewalkers replaced it with a spare on 30 June 2026.



FIGURE 32.7 Canadarm2 with Dextre attached, extending from the Harmony module above the Sahara in August 2025. Image: NASA/JSC.

32.2.6 Crew, Research and Cost

The station normally hosts seven people — four launched on Crew Dragon and three on Soyuz, with seats swapped between the vehicles so that each segment always has trained crew. Nearly 3 000 investigations from more than 108 countries have used its racks, freezers, glove boxes, furnaces and external platforms, from protein crystal growth and combustion to Earth observation and the Alpha Magnetic Spectrometer. Total spending by all partners is often estimated at more than US$ 150 billion. NASA's Inspector General put NASA's own cost at about US$ 3 billion per year for operations and transport. Including research and related activities, it is about US$ 4.1 billion per year — about 16 % of NASA's budget.

TABLE 32.2 International Space Station key figures (2026).

Parameter	Value
Mass; dimensions	≈ 420 t; 109 m × 73 m
Pressurised / habitable volume	1 005 m³ / 388 m³
Orbit	≈ 415–420 km, 51.6°, 93 min
Crew	7 (more during crew handovers)
Power	8 wings + 6 iROSAs (≈ 215 kW in sunlight); 24 Li-ion batteries
Heat rejection	Two ammonia loops; 6 radiators, ≈ 70 kW
Attitude control	4 CMGs (4 760 N·m·s each) + Russian thrusters
Communications	S- and Ku-band via TDRS; UHF for spacewalks and approaching vehicles
Robotics	Canadarm2 (17.6 m) with Dextre; Kibo and European arms
Docking ports	Up to 8 vehicles at once (Soyuz, Progress, Dragon, Cygnus, HTV-X)

32.2.7 An Ageing Station

The oldest ISS modules have been in orbit since 1998. Since 2019 air has been leaking from the **PrK**, a small tunnel between Zvezda's main compartment and its aft docking port, through hairline cracks. In April 2024 the leak reached about 1.7 kg of air per day; NASA rated it its highest-level safety risk, and the hatch is normally kept closed. NASA and Roscosmos have not

agreed on the root cause. The tunnel was reported to be holding pressure in January 2026, but the leak returned in May. On 5 June 2026 five crew members briefly sheltered in their Crew Dragon while Roscosmos began, then paused, a structural repair. Press reports later said Roscosmos would stop pressurising the tunnel. Such ageing problems are a main reason for retiring the station around 2030.

32.3 Engineering a Station

32.3.1 Choosing the Orbit

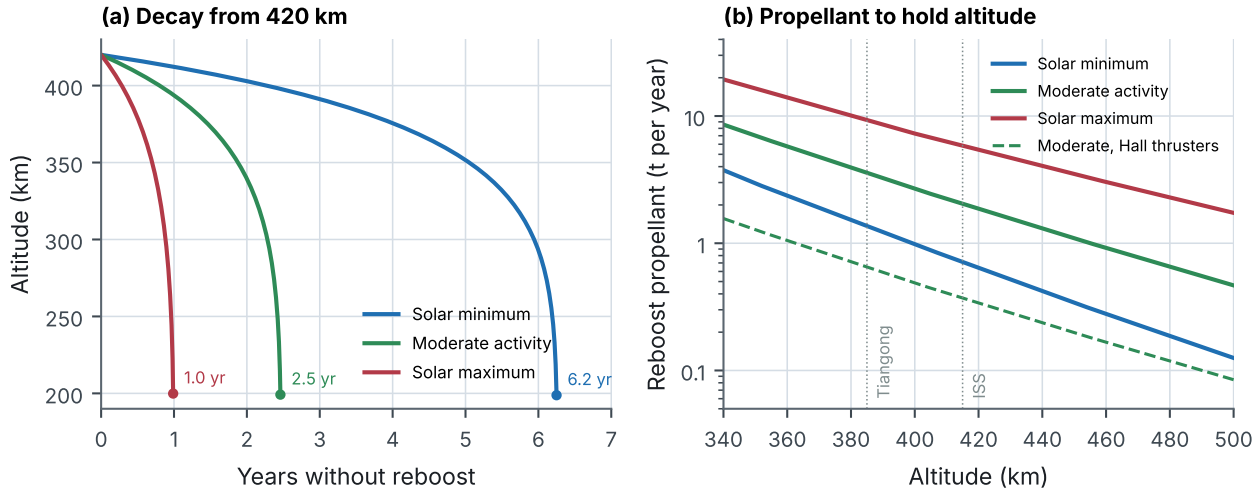
A station's **inclination** — the angle between its orbit and the equator — is usually set by its launch sites. A rocket launched due east reaches an inclination equal to the latitude of its launch site, and higher inclinations cost payload. The ISS flies at 51.6° because Russian vehicles from Baikonur (46° N) cannot launch due east without dropping stages on China; American vehicles from Florida (28.5° N) pay a payload penalty to reach it. Tiangong's 41.5° suits the Jiuquan launch site.

Altitude is a compromise. Lower is cheaper to reach — every kilometre of altitude costs payload for each crew and cargo flight — but the air is denser, so the station must spend more propellant fighting drag (Table 32.3). Higher reduces drag but increases radiation exposure, as orbits approach the inner Van Allen belt, and the density of orbital debris, and makes Earth-observation instruments less sharp. During the Shuttle era the ISS flew as low as 330–350 km so that the Shuttle could deliver the heaviest modules; since 2011 it has been kept near 400–420 km.

Drag decays the orbit as described in Chapter 2:

$$\frac{da}{dt} = -\frac{\rho \sqrt{\mu a}}{B}, \qquad B = \frac{m}{C_D A}$$

where ρ is the air density, a the orbit radius, μ Earth's gravitational parameter and B the ballistic coefficient. Holding altitude requires a velocity increment each year equal to the drag deceleration integrated over time, $\Delta v = \int \rho v^2 / (2B)\, dt$. Because thermospheric density changes by a factor of about ten over the 11-year solar cycle, the reboost budget does too (Figure 32.8). Without reboosts, an ISS-class station released at 420 km would re-enter within about one year at solar maximum and about six years at solar minimum.



Model: da/dt = −ρ√(μa)/B with B = m/(C_DA) = 150 kg/m² (450 t, C_DA ≈ 3 000 m²); MSIS-like density table of Chapter 2; chemical I_{sp} = 290 s, Hall I_{sp} = 1 600 s.

FIGURE 32.8 (a) Altitude of an ISS-class station left without reboost, starting from 420 km. (b) Propellant needed each year to hold a constant altitude with chemical thrusters, and with Hall-effect electric thrusters at moderate solar activity.

TABLE 32.3 Drag penalty of altitude for a 450 t station with B = 150 kg/m², computed with the model of Figure 32.8 (specific impulse 290 s).

Altitude (km)	Decay, moderate activity (km/month)	Δv per year: min / moderate / max (m/s)	Propellant per year, moderate (t)	Life without reboost, moderate (years)
350	6.5	19 / 45 / 106	7.0	0.6
380	3.7	10 / 25 / 65	3.9	1.1
400	2.5	6 / 17 / 46	2.7	1.7
420	1.7	4 / 12 / 34	1.9	2.5
450	1.0	2 / 7 / 22	1.1	4.3
500	0.4	1 / 3 / 11	0.5	11

EXAMPLE: ANNUAL REBOOST FOR AN ISS-CLASS STATION

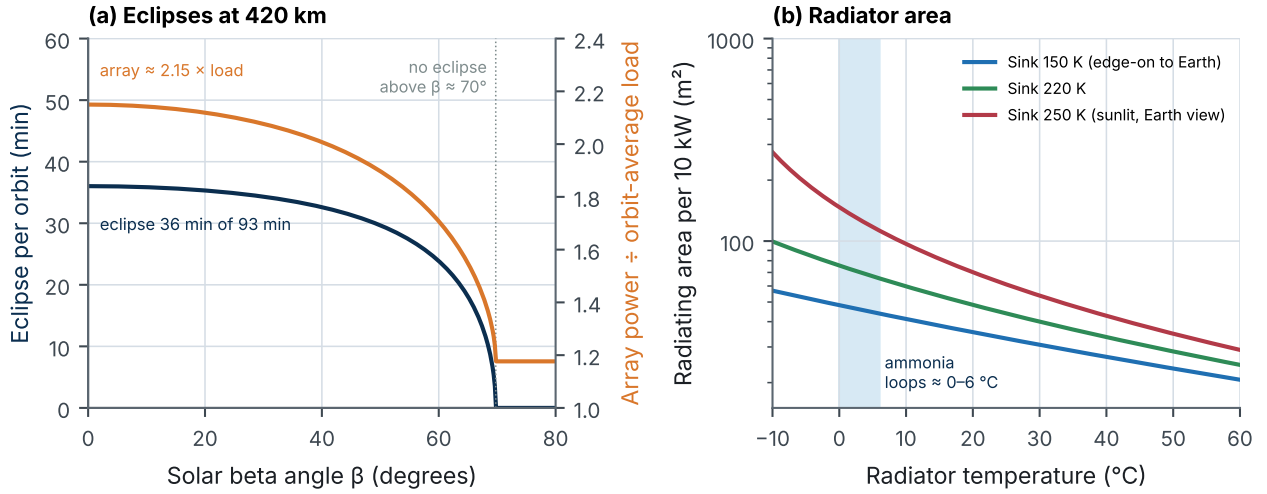
Take a station of 450 t, including docked vehicles, with an average drag area $C_DA \approx 3\,000$ m², so that $B = 150$ kg/m². It flies at 410 km in moderate solar activity ($\rho \approx 2.3 \times 10^{-12}$ kg/m³).

1. Orbital speed: $v = \sqrt{\mu/a} = \sqrt{3.986 \times 10^{14} / 6.788 \times 10^{6}} = 7\,663$ m/s.
2. Drag deceleration: $\rho v^2/(2B) = 2.3 \times 10^{-12} \times 7\,663^2 / 300 = 4.5 \times 10^{-7}$ m/s² — about 50 billionths of g.
3. Over a year (3.16×10^7 s) this accumulates to **Δv ≈ 14 m/s**, and the orbit would sink by about 68 m per day (2.1 km per month) if not corrected.
4. With thrusters of $I_{sp} = 290$ s ($v_e = 2\,844$ m/s), the propellant is $m(1 - e^{-\Delta v/v_e}) = 450\,000 \times (1 - e^{-14.1/2\,844}) \approx$ **2.2 t per year**.
5. At solar maximum the density is nearly three times higher: Δv ≈ 40 m/s and **6.3 t per year**. Hall thrusters ($I_{sp} \approx 1\,600$ s) would need only about 0.4 t at moderate activity.

The reported ISS average of about 7 t per year is higher because it also covers attitude control and debris avoidance.

32.3.2 Power and Thermal Budgets

The angle between the orbit plane and the direction of the Sun is the **beta angle (β)**. It changes continuously as the orbit plane precesses and Earth moves round the Sun. For the ISS it swings between about −75° and +75° over roughly two months. At β = 0 the station spends 36 of every 93 minutes in shadow; above β ≈ 70° it is in permanent sunlight (Figure 32.9). The arrays must supply both the load and the battery charge for the next eclipse. Because storing energy in a battery and recovering it wastes about a third of it, the arrays must produce about 2.2 times the orbit-average load at low β. At high β, constant sunlight overheats some surfaces, so the attitude may be changed and visiting-vehicle traffic restricted.



(a) Path efficiencies 0.65 (via battery) and 0.85 (direct). (b) $q = \varepsilon \eta \sigma(T^4 - T_{sink}^4)$, ε = 0.85, fin efficiency η = 0.85; one radiating face.

FIGURE 32.9 (a) Eclipse duration at 420 km and the ratio of array output to orbit-average load needed to recharge batteries, as functions of beta angle. (b) Radiator area needed to reject 10 kW as a function of radiator temperature for three effective sink temperatures.

The radiators must reject the whole electrical load plus the heat produced by the crew (about 100–150 W per person). Radiated flux follows the Stefan–Boltzmann law, $q = \varepsilon\sigma(T^4 - T_{sink}^4)$. So a radiator near 0 °C — cold enough to cool the cabin through heat exchangers — emits only about 200–250 W/m² even with a cold view of space. It emits much less when it sees the warm Earth or sunlight. That is why the ISS radiators are among its largest structures. Power, heat rejection and drag are coupled: larger arrays bring more heat to reject, larger radiators and more drag.

32.3.3 Protecting Against Micrometeoroids and Debris

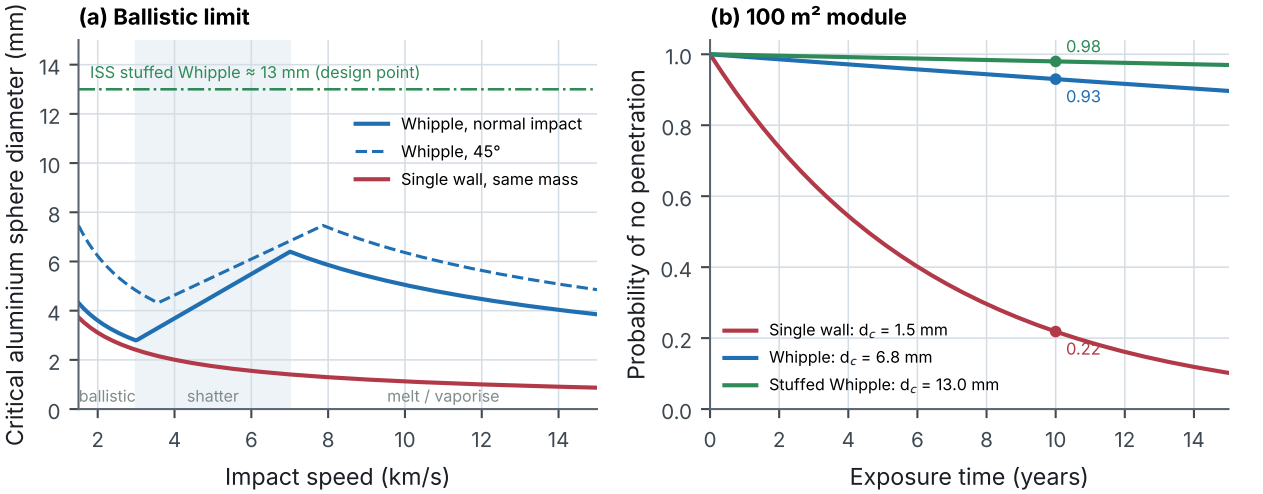
A station is a large target: the ISS presents more than 1 000 m² of pressurised-module surface to the particle environment described in Chapters 2 and 21. Its crew modules are protected by **Whipple shields**. In these, a thin aluminium bumper spaced in front of the pressure wall breaks up an incoming particle into a spreading cloud of fragments. Modules are also protected by **stuffed Whipple shields**, with layers of Nextel ceramic cloth and Kevlar between bumper and wall. The ISS uses more than 100 different shield configurations, the heaviest on forward-facing surfaces, which see the most debris impacts.

The performance of a shield is described by its **ballistic limit**: the diameter d_c of the largest aluminium sphere it can stop, as a function of speed and impact angle (Figure 32.10). At low speeds (below about 3 km/s) a projectile stays intact and the shield behaves like a thick plate.

Between about 3 and 7 km/s the projectile shatters on the bumper, and the shield becomes *more* effective with speed. Above 7 km/s the fragments melt and vaporise, and d_c falls slowly with speed. NASA's ballistic limit equation for the high-speed regime is

$$d_c = 3.918\, t_w^{2/3}\, \rho_p^{-1/3}\, \rho_b^{-1/9}\, (V\cos\theta)^{-2/3}\, S^{1/3} \left(\frac{\sigma}{70}\right)^{1/3}$$

in cm, with wall thickness t_w and spacing S in cm, projectile and bumper densities ρ_p, ρ_b in g/cm^3, speed V in km/s, impact angle θ and wall yield strength σ in ksi. Doubling the spacing increases d_c by 26 %; a single wall of the same mass stops particles four to five times smaller. The ISS stuffed Whipple shields are designed to stop particles of about 1.3 cm at typical impact conditions. Larger, catalogued objects are avoided by manoeuvring, but objects between about 1 and 10 cm can be neither shielded against nor tracked (Chapter 21).



Shield: 1.6 mm Al bumper, 10 cm gap, 3.2 mm Al 2219 wall (single wall 4.8 mm). Christiansen/Cour-Palais equations; flux from Chapter 2 model; (b) at 9 km/s, 45°.

FIGURE 32.10 (a) Ballistic limit of a Whipple shield and of a single aluminium wall of equal mass, from NASA's empirical equations. (b) Probability that a 100 m² module suffers no penetration, for the three wall designs, using the illustrative flux model of Chapter 2.

EXAMPLE: PENETRATION RISK FOR A MODULE OVER TEN YEARS

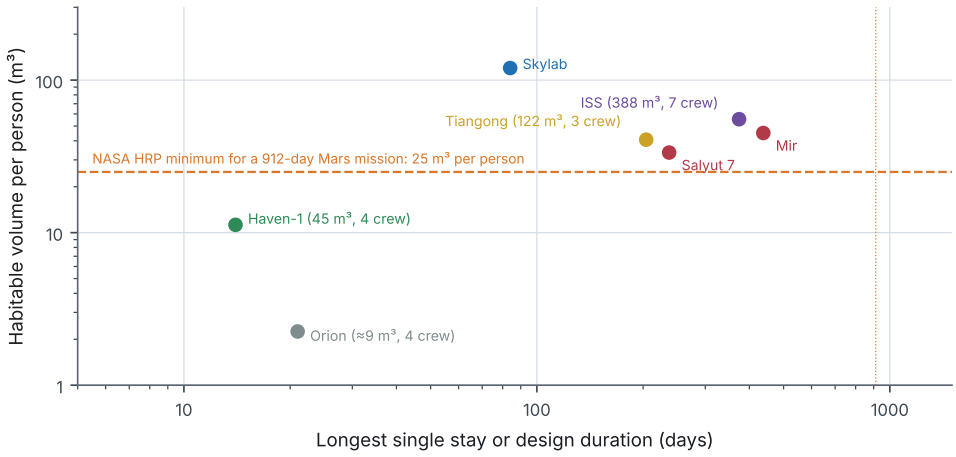
A laboratory module exposes 100 m² of effective area to impacts. Typical debris strikes at 9 km/s and 45°.

1. **Shield.** A Whipple shield with a 1.6 mm aluminium bumper, a 10 cm gap and a 3.2 mm aluminium-alloy wall has $d_c \approx 6.8$ mm at these conditions. A single 4.8 mm wall of the same mass has $d_c \approx 1.5$ mm. The ISS stuffed Whipple stops about 13 mm.
2. **Flux.** The cumulative flux of particles larger than d_c is read from the illustrative ISS-altitude curve of Chapter 2, which follows the shape of NASA's ORDEM debris model. It is about 1.5×10^{-3} per m² per year above 1.5 mm, 7×10^{-5} above 6.8 mm and 2×10^{-5} above 13 mm.
3. **Expected penetrations** in ten years: $N = FAT$ = 1.5, 0.073 and 0.021 respectively.
4. Impacts are random, so the **probability of no penetration** is $P = e^{-N}$: **0.22** for the single wall, **0.93** for the Whipple shield and **0.98** for the stuffed Whipple.
5. Ten such modules multiply N by ten: PNP falls to about 0.81 even with the best shields. Not every penetration is catastrophic: a small hole leaks slowly enough to be found and sealed. NASA therefore also estimates the risk of evacuation or crew loss with tools such as BUMPER and MSCSurv.

The ISS has suffered damage. Impacts chip its windows, which is why the Cupola has external shutters. A debris strike punched a 5 mm hole in Canadarm2's thermal blanket in 2021. In December 2022 the coolant radiator of a docked Soyuz, MS-22, was punctured, probably by a micrometeoroid. The capsule had to return uncrewed, and a replacement Soyuz was launched. After Russia destroyed its Kosmos 1408 satellite in an anti-satellite test on 15 November 2021, the crew sheltered in their docked spacecraft while the station passed through the debris cloud.

32.3.4 Habitable Volume and Layout

Volume is measured in three ways. **Pressurised volume** is everything inside the hull. The **habitable volume** excludes the space taken by racks and fixed equipment. The **net habitable volume** is what remains free for people to move in once stowage and loose equipment are counted. In 2014 a NASA Human Research Program panel of experts recommended a minimum net habitable volume of 25 m^3 per person for a Mars mission lasting about 912 days. Its comparison listed about 120 m^3 per person for Skylab, 45 for Mir and 34 for Salyut. By comparison, today's ISS offers about 55 m^3 (388 m^3 for seven people) (Figure 32.11). The ISS crew quarters — private sleeping cabins — are about 2.1 m^3 each, the size of a telephone box.



Skylab, Salyut, Mir per-person values from the NASA HRP consensus report (2014); others computed from habitable volume ÷ nominal crew. Durations: record stays or design values.

FIGURE 32.11 Habitable volume per person against the longest stay on several stations and spacecraft, with the minimum recommended by NASA's Human Research Program for a long Mars mission.

In weightlessness there is no natural up or down. Even so, designers give every module a consistent "floor" and "ceiling", with lighting overhead and labels the right way up. This helps people orient faster and make fewer mistakes. Handrails and foot restraints replace chairs, and fan and pump noise is a persistent habitability problem. Windows matter for morale as much as for operations: the Cupola, installed in 2010, gives a panoramic view used for robotics and photography (Figure 32.12).

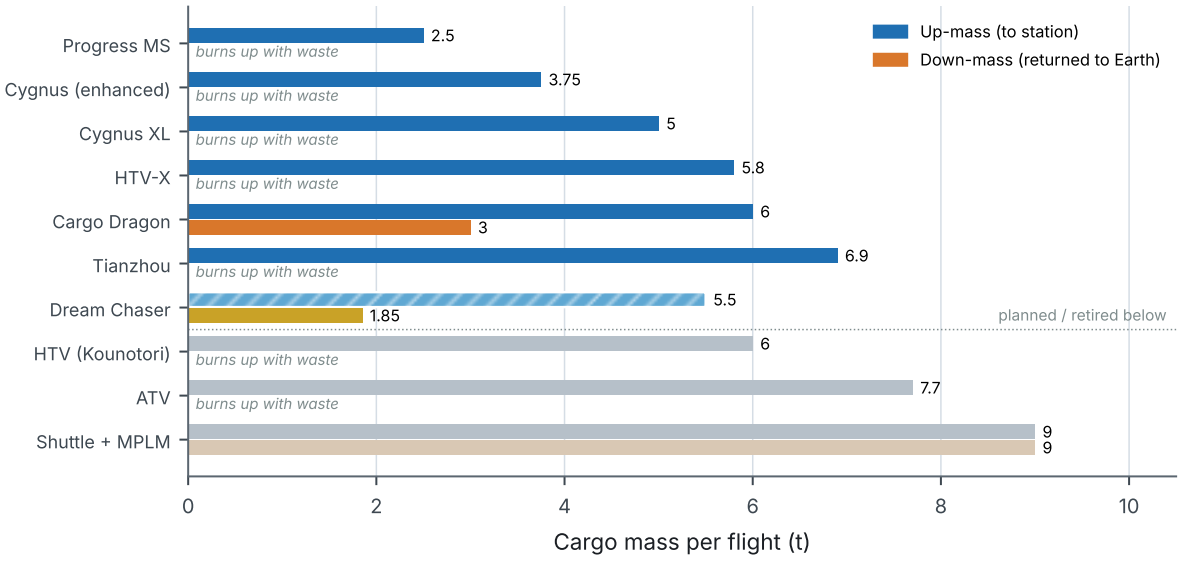


FIGURE 32.12 ESA astronaut Sophie Adenot and NASA astronaut Jack Hathaway in the Cupola in May 2026, monitoring the automated approach of a Cargo Dragon. Image: NASA/JSC.

32.4 Supplying a Station

32.4.1 Cargo Vehicles

Even recycling about 98 % of its water (Chapter 30), a crew of seven consumes several tonnes of food, gases, clothing and hygiene items a year, and the station also needs propellant, spares and experiments. Cargo vehicles differ in how much they carry up (**up-mass**) and whether they can carry unpressurised cargo such as pumps and batteries outside. They also differ in whether they can refuel and reboost the station, and whether anything comes back (**down-mass**) (Figure 32.13, Table 32.4). Most freighters are loaded with rubbish before they leave and burn up on re-entry. Apart from small items carried in crew capsules, only Dragon returns experiments and failed hardware to Earth. Dream Chaser could do so if NASA later orders ISS flights.



Maximum or typical capacities from operator data; Progress includes propellant and water; ATV up to 7.7 t including refuelling propellant. Dream Chaser not yet flown.

FIGURE 32.13 Cargo capacity per flight of station supply vehicles. Most freighters burn up with the station's waste; Cargo Dragon is currently the only vehicle that returns cargo.

JAXA's **HTV-X**, successor to the HTV, made its first flight on an H3 rocket on 26 October 2025; it was captured by Canadarm2 on 29 October (Figure 32.14). Northrop Grumman's larger **Cygnus XL**, about a third roomier than its predecessor, first flew in September 2025 (Figure 32.15). Sierra

Space's winged **Dream Chaser** has repeatedly slipped. In September 2025 NASA removed its guaranteed ISS flights. Its first mission is a free-flying demonstration on a Vulcan rocket, planned for late 2026 at the earliest.

TABLE 32.4 Station cargo vehicles (October 2026).

Vehicle	Operator	Since	Arrival	Up / down (t)	Notes
Progress MS	Roscosmos	1978 (MS: 2015)	Automatic docking	≈ 2.5 / 0	Refuels and reboosts Russian segment
Cygnus / Cygnus XL	Northrop Grumman	2013 / 2025	Berthed by Canadarm2	3.75–5 / 0	XL first flew Sept 2025
Cargo Dragon	SpaceX	2012 (Dragon 2: 2020)	Automatic docking	≈ 6 / 3	Ocean splashdown; reboost kit
HTV-X	JAXA	Oct 2025	Berthed by Canadarm2	≈ 5.8 / 0	Can fly up to 1.5 years after leaving
Tianzhou	CMSA	2017	Automatic docking	≈ 6–7 / 0	Supplies Tiangong; refuels it
Dream Chaser	Sierra Space	Not yet flown	(Berthing planned)	≈ 5.5 / 1.85	Winged; runway landing; free-flyer demo first
ATV; HTV	ESA; JAXA	2008–14; 2009–20	Docking; berthing	7.7; 6 / 0	Retired



FIGURE 32.14 JAXA's first HTV-X in the grip of Canadarm2 on 29 October 2025, carrying about 5.8 t of supplies. Image: NASA/JSC.

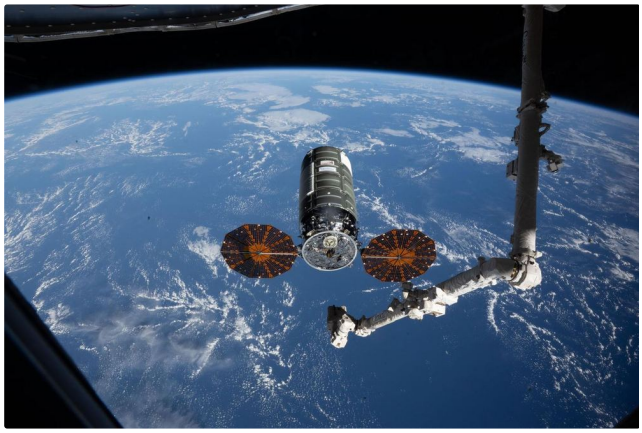


FIGURE 32.15 A Cygnus XL freighter waiting at its capture point, 12 m from the station, as Canadarm2 prepares to grapple it in April 2026. Image: NASA/JSC.

32.4.2 Docking and Berthing

Vehicles join a station in two ways. In **docking**, the visiting vehicle flies itself into contact. Soyuz and Progress use the Russian Kurs radar system and probe-and-drogue ports, while Dragon uses its own sensors and the androgynous International Docking System Standard (Chapter 31). In **berthing**, the vehicle flies to a capture point 10–12 m below the station and holds position with its thrusters disabled. While it holds, an astronaut grapples it with Canadarm2 and installs it on a CBM port, where powered bolts pull the two rings together. Berthing needs no precision docking system on the visitor and gives a wide hatch for racks, but requires the arm and a trained crew.

Approaches follow strict rules. There is an **approach ellipsoid** of 4 × 2 × 2 km around the station, and a **keep-out sphere** of 200 m radius. A vehicle may enter the sphere only after a "go" from mission control, along a defined corridor and with an automatic abort capability. Solar arrays are usually turned edge-on to protect them from thruster plumes.

32.5 Expandable Habitats and Artificial Gravity

32.5.1 Expandable Modules

A rigid module can be no wider than the rocket fairing that launches it — about 4.5 m for most modules so far. An **expandable** (or inflatable) module is launched folded and then pressurised in orbit, giving several times more volume per launch. Its wall is built from layers of fabric. Inner gas bladders (usually redundant) hold the air, and a **restraint layer** of woven Kevlar or Vectran straps carries the pressure load. Many spaced layers of ceramic cloth and foam act as a multi-shock micrometeoroid shield. Then come multilayer thermal insulation and an outer cover against atomic oxygen.

The restraint layer must resist a hoop load per unit length of $N = p r$. For a module 8.2 m across at one atmosphere (101 kPa), that is about 415 kN per metre of wall — the weight of 42 tonnes hanging from each metre. The fabric must withstand about four times that. NASA's TransHab technology of the 1990s passed to Bigelow Aerospace, which flew the uncrewed Genesis I and II prototypes in 2006–07. The **Bigelow Expandable Activity Module (BEAM)**, about 1.4 t, was berthed to the ISS in April 2016 and expanded on 28 May 2016 to about 16 m³ (Figure 32.16). Planned as a two-year test, it is still attached a decade later and used for storage; its ownership passed to NASA in 2021 after Bigelow ceased operations. Sierra Space's **LIFE** habitat is about 8 m in diameter, with about 285 m³ inside. In full-scale tests in December 2023 and June 2024 it burst at 77 psi and 74 psi (530 and 510 kPa). These pressures are 22–27 % above the 60.8 psi (four times operating pressure) that NASA recommends.

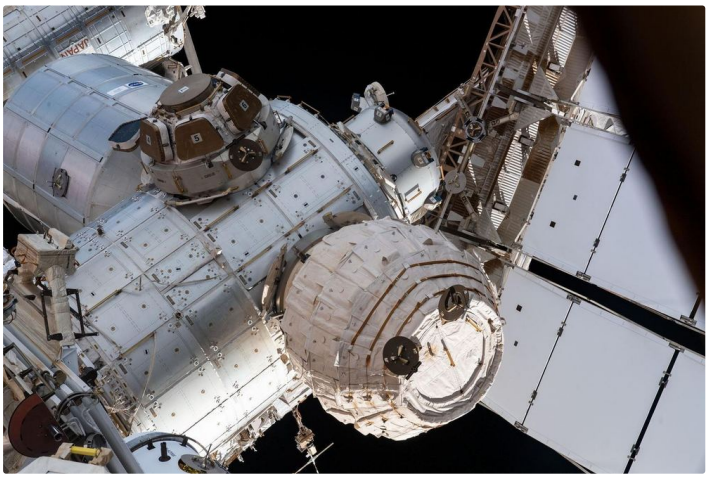


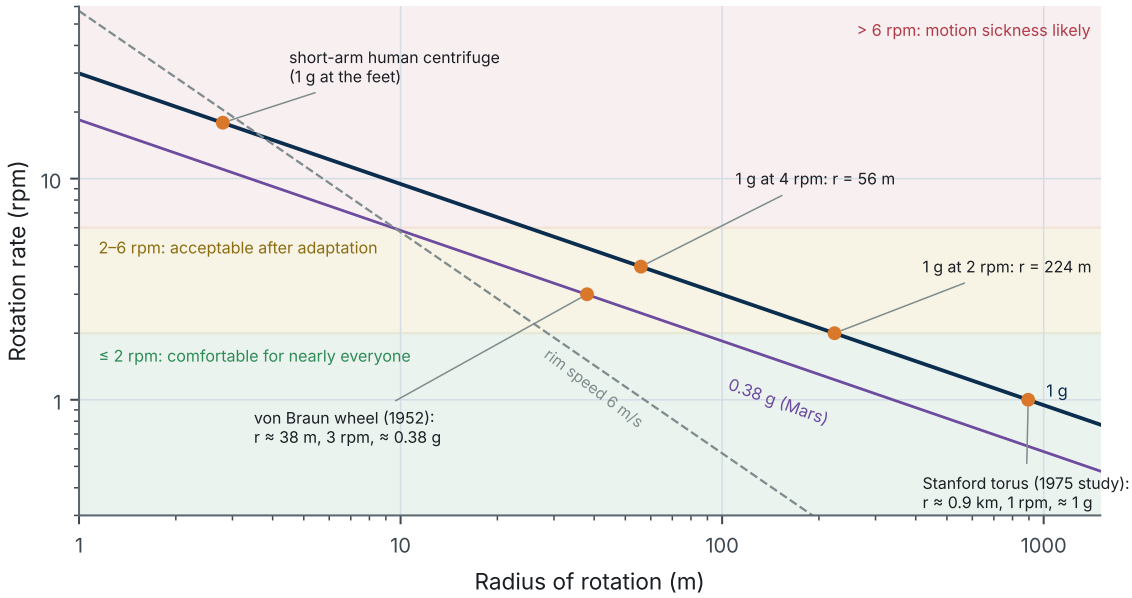
FIGURE 32.16 BEAM (the fabric-covered module, lower centre), attached to the Tranquility node next to the shuttered Cupola and the Bishop airlock, photographed from the Russian segment in August 2022. Image: NASA/JSC.

32.5.2 Artificial Gravity

Weightlessness weakens bones, muscles and the heart and affects the eyes (Chapter 30). Exercise reduces but does not prevent these effects, and a rotating habitat could replace weight with centripetal acceleration:

$$a = \omega^2 r, \qquad a_{Cor} = 2\,\omega\, v_{rel}$$

where ω is the rotation rate, r the radius and v_{rel} the speed of a person moving inside the habitat. The **Coriolis acceleration** a_{Cor} makes moving objects veer sideways, and turning the head in a rotating room confuses the inner ear, causing motion sickness. Experiments in rotating rooms suggest that almost everyone is comfortable at up to about 2 rpm, and that people can adapt to about 4–6 rpm (Figure 32.17). A low rotation rate requires a large radius: for 1 g at 2 rpm, the radius is 224 m. A **gravity gradient** between head and feet is a further issue for small radii.



ω = √(a/r). Comfort bands follow limits commonly quoted in the artificial-gravity literature (compiled by T. W. Hall); individual tolerance varies.

FIGURE 32.17 Rotation rate against radius for 1 g and Mars-level artificial gravity, with commonly quoted comfort limits and some historical concepts.

EXAMPLE: A ROTATING HABITAT FOR 1 G AT 2 RPM

1. Rotation rate: ω = 2 × 2π/60 = 0.209 rad/s.
2. Radius for 1 g: $r = g/\omega^2$ = 9.81/0.209² = **224 m** — a structure almost half a kilometre across. Rim speed: ωr = 47 m/s.
3. Coriolis acceleration on a crew member walking at 1 m/s: 2 × 0.209 × 1 = 0.42 m/s² — about 4 % of their weight, pushing sideways.
4. Walking 1 m/s in the direction of rotation raises apparent weight by 4.3 %; walking the other way lowers it by 4.2 %.
5. Gravity gradient over a 1.8 m person: 1.8/224 = **0.8 %** — negligible. At 4 rpm the radius shrinks to 56 m, but the Coriolis acceleration doubles and the gradient reaches 3 %.

Mars gravity (0.38 g) at 2 rpm needs 85 m. A cheaper option is to spin two masses — a habitat and a spent stage or counterweight — connected by a tether.

No crewed station has yet used artificial gravity. In 1966 Gemini 11 spun itself and an Agena stage on a 30 m tether, producing a tiny artificial gravity. On the ISS, JAXA's Multiple Artificial-gravity Research System centrifuge has kept mice at controlled artificial-gravity levels for comparison with weightless animals, and short-arm human centrifuges have been tested on the ground in bed-rest studies.

32.6 After the ISS

32.6.1 Retiring and Deorbiting the ISS

The partners plan to operate the ISS until 2030; Roscosmos said in April 2026 that it would begin rounding off its ISS work in 2028 and expected the deorbit to be completed in 2030. A 420-tonne station cannot be left to re-enter uncontrolled: tens of tonnes of dense parts — truss segments, docking rings, tanks — would survive and fall anywhere under its 51.6° ground track. It must instead be steered into a remote area of the South Pacific.

In June 2024 NASA awarded SpaceX a contract worth up to US$ 843 million for the **US Deorbit Vehicle (USDV)**, a heavily modified Dragon with 46 Draco thrusters and about 16 t of propellant; with launch, NASA puts the total at about US$ 1.5 billion. Launched about 18 months before the end, the USDV will dock at the forward port while the station sinks. The last crew will leave at about 330 km, some six months before re-entry. At about 225 km the USDV will begin a series of burns. The last burn, about four days later, aims the debris at the South Pacific — in early 2031 on current plans. Lowering the perigee from a 225 km circular orbit to about 50 km requires about 52 m/s. For 450 t, this is about 8 t of propellant at I_{sp} = 300 s. Much of the remaining propellant is needed to control the station's attitude as aerodynamic torques grow during the final orbits.

32.6.2 Commercial Stations

NASA wants to buy services on commercial stations rather than own one. Its **Commercial LEO Destinations (CLD)** programme gave Space Act Agreements worth US$ 415.6 million in December 2021. They went to Blue Origin, Nanoracks (now part of Voyager, for Starlab) and Northrop Grumman (which later joined Starlab). Axiom Space separately holds a contract to attach modules to the ISS. The programme was restructured repeatedly in 2025–26. An August 2025 directive lowered the minimum capability NASA would require. A proposal in March 2026 to attach a government-owned core module to the ISS was dropped in June. On 9 October 2026 NASA issued its final request for proposals for firm-fixed-price contracts covering development, certification and services. Proposals are due on 8 December 2026, and two or more awards are expected in spring 2027. Because a commercial station must be certified and crewed before the ISS is deorbited, and schedules keep slipping, a gap in continuous US presence in orbit is possible (Table 32.5).

Vast's **Haven-1** would host four-person crews arriving by Crew Dragon for about two weeks at a time; Vast proposes a larger multi-module **Haven-2** for the NASA competition. **Axiom Space** has flown private crews to the ISS since 2022. In December 2024 it changed its plan so that its first module, a power and thermal module, will attach to the ISS. This module will later depart to join a habitat module as a free-flying station. **Starlab** is a single 8 m-diameter station to be launched fully outfitted on one Starship.

32.6.3 National Stations: China, Russia and India

If the ISS ends on schedule, China's Tiangong will be the only government station in low Earth orbit, and its planned six-module expansion would double it. Russia's **Russian Orbital Station (ROS)** was redesigned in December 2025 to use the ISS's 51.6° orbit rather than a near-polar one, with its first modules attached to the ISS from about 2028. India's **Bharatiya Antariksh Station (BAS)** builds on the Gaganyaan crew vehicle and the SpaDeX docking experiment; its first module, of about 10 t, is to be launched on an LVM-3.

TABLE 32.5 Successors and new stations (status as of October 2026).

Station	Developer	First launch (planned)	Configuration	Status, October 2026
Haven-1	Vast	NET Q1 2027	14 t, 45 m³; 4 crew for ≈ 2 weeks	Integration and tests; Haven Demo flew Nov 2025
Axiom Station	Axiom Space	≈ 2028	Power module first on ISS, then joins Hab-1 to fly free	Modules in production
Starlab	Voyager, Airbus and partners	NET 2029, one Starship launch	8 m diameter, ≈ 450 m³, 4 crew	Critical design review Feb 2026
Orbital Reef	Blue Origin, Sierra Space	Not announced	Multi-module, LIFE habitats	Few public milestones
ROS	Roscosmos	2028	First modules attached to ISS, 51.6°	Redesigned Dec 2025
Bharatiya Antariksh Station	ISRO	BAS-01 by 2028	Five modules by 2035	First module approved 2024
Tiangong expansion	CMSA	Not dated	Six modules, ≈ 180 t	Plan shown 2023, confirmed 2026

32.6.4 Gateway and Surface Habitats

The **Lunar Gateway**, a small station planned for a near-rectilinear halo orbit around the Moon, was to begin with a Power and Propulsion Element (PPE) and a Habitation and Logistics Outpost (HALO). The FY2026 budget request proposed cancelling it, and on 24 March 2026 NASA paused Gateway "in its current form" in favour of a lunar surface base. By August 2026 the HALO hardware was being repurposed for lunar-surface demonstration missions. The PPE was reported to be under study for other uses. Its future as a station is uncertain.

On the surface, NASA's Artemis Base Camp concept foresaw a pressurised rover and a fixed habitat near the lunar south pole; its Moon Base programme (March 2026) starts with demonstrations of surviving the lunar night. Surface habitats gain gravity, local materials and a planet that blocks half the sky, but must cope with abrasive dust, temperature swings of almost 300 °C on the Moon and no magnetic field. A 1 m layer of regolith (density about 1.5 g/cm³) provides about 150 g/cm² of shielding — enough against solar particle events, though several metres are needed to remove most of the cosmic-ray dose. Proposals include covering inflatable habitats with bagged regolith, sintering or 3D-printing regolith into walls, and using lava tubes (Chapters 30 and 41).

AI IN STATION OPERATIONS

Station operations are moving towards autonomy. The ISS already uses automated fault detection and procedure execution. NASA's **Astrobee** — three cube-shaped free-flying robots that navigate Kibo with cameras and fans (Figure 32.18) — has been used since 2019 to test autonomous inventory, inspection and mapping. DLR's **CIMON**, an AI-powered floating assistant with voice interaction, flew from 2018, and JAXA's Int-Ball camera drone films crew activities. Machine-learning models are being trained on years of station telemetry to predict failures of pumps, fans and CMGs before they happen. Commercial stations plan to operate with very small ground teams and no permanent crew. Chapter 51 discusses AI for human spaceflight in detail.

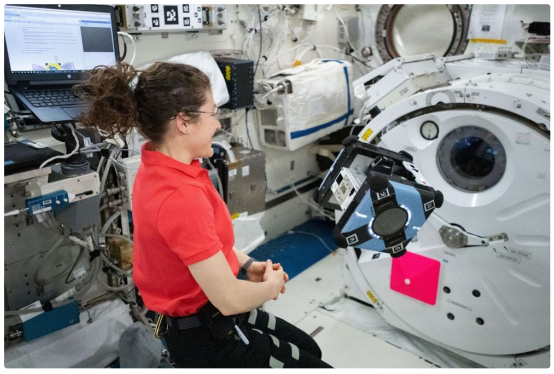


FIGURE 32.18 NASA astronaut Christina Koch testing the mobility of an Astrobee free-flying robot in the Kibo laboratory in July 2019. Image: NASA/JSC (Christina Koch).

KEY TAKEAWAYS

- Stations evolved from single modules (Salyut 1, Skylab) through resupply by freighters (Salyut 6) and modular assembly (Mir). This led to the 420 t ISS, crewed continuously since November 2000, and China's Tiangong, crewed continuously since 2022.
- A station is a coupled system: solar arrays sized for eclipses, radiators sized for the full load, CMGs for attitude, and propellant for reboost. Reboost needs about 4–40 m/s per year at ISS altitude, depending on the solar cycle.
- Orbit selection trades launch cost against drag: lowering the orbit from 420 to 350 km almost quadruples the reboost propellant, and solar maximum roughly triples it compared with average conditions.
- Whipple and stuffed Whipple shields stop particles up to about 1.3 cm; combining ballistic-limit equations with a flux model gives the probability of no penetration, which falls with area and time.
- Logistics depend on several freighters, most of them destroyed on re-entry; Cargo Dragon is the only vehicle returning cargo, and HTV-X and Cygnus XL entered service in 2025.
- Expandable habitats offer several times more volume per launch; artificial gravity at comfortable rotation rates needs radii of hundreds of metres.
- The ISS is to be deorbited by a SpaceX-built vehicle around 2030–31; commercial stations (Haven-1, Axiom, Starlab), Russia's ROS, India's BAS and an expanded Tiangong are planned, while Gateway has been paused.

REVIEW QUESTIONS

1. What design feature introduced on Salyut 6 made long-duration stations possible, and why?
2. Why did Skylab re-enter earlier than NASA had planned? Link your answer to the solar cycle.

3. Explain how a control moment gyroscope produces torque without propellant, and why the ISS still needs thrusters for attitude control.
4. Using Table 32.3, estimate how much more propellant per year a station at 380 km needs than one at 420 km, at moderate solar activity. What are the advantages of the lower altitude?
5. At $\beta = 0$, an ISS-like station is in eclipse for 36 of 93 minutes. With path efficiencies of 0.65 through the battery and 0.85 directly, what array output is needed for a 100 kW orbit-average load?
6. Why is a Whipple shield more effective at 6 km/s than at 3 km/s? Sketch its ballistic-limit curve.
7. A module of 60 m^2 has a shield that stops particles up to 8 mm, and the flux above 8 mm is 5×10^{-5} per m^2 per year. What is the probability of no penetration over 15 years?
8. Compare docking and berthing. Why do Cygnus and HTV-X berth rather than dock?
9. What radius is needed for 0.38 g at 3 rpm? What Coriolis acceleration acts on a crew member walking at 1.2 m/s?
10. List the main candidates to succeed the ISS and the status of each as of October 2026. What is the risk of a "gap" in continuous human presence in orbit?

33 CHAPTER Extravehicular Activity Systems



Image: NASA/JSC

A spacesuit is a one-person spacecraft. It must hold a breathable atmosphere against vacuum, remove carbon dioxide and up to several hundred watts of body heat, shield its wearer from sunlight, cold and micrometeoroids, and still let a gloved hand turn a bolt for seven hours. This chapter follows sixty years of spacewalks — from Leonov's ballooning suit in 1965 through Apollo, the Space Shuttle, Hubble and the 284 spacewalks at the International Space Station to China's record nine-hour EVA and the first commercial spacewalk in 2024. It explains the engineering of pressure garments, joints, gloves and portable life support systems, the physiology of decompression sickness and prebreathe, how spacewalks are trained and run, and what suits for the Moon and Mars must do differently.

IN THIS CHAPTER

33.1 Sixty Years of Spacewalks
33.2 Anatomy of a Spacesuit
33.3 The Portable Life Support System
33.4 Decompression Sickness and Prebreathe
33.5 Working Outside
33.6 Spacewalking on Other Worlds

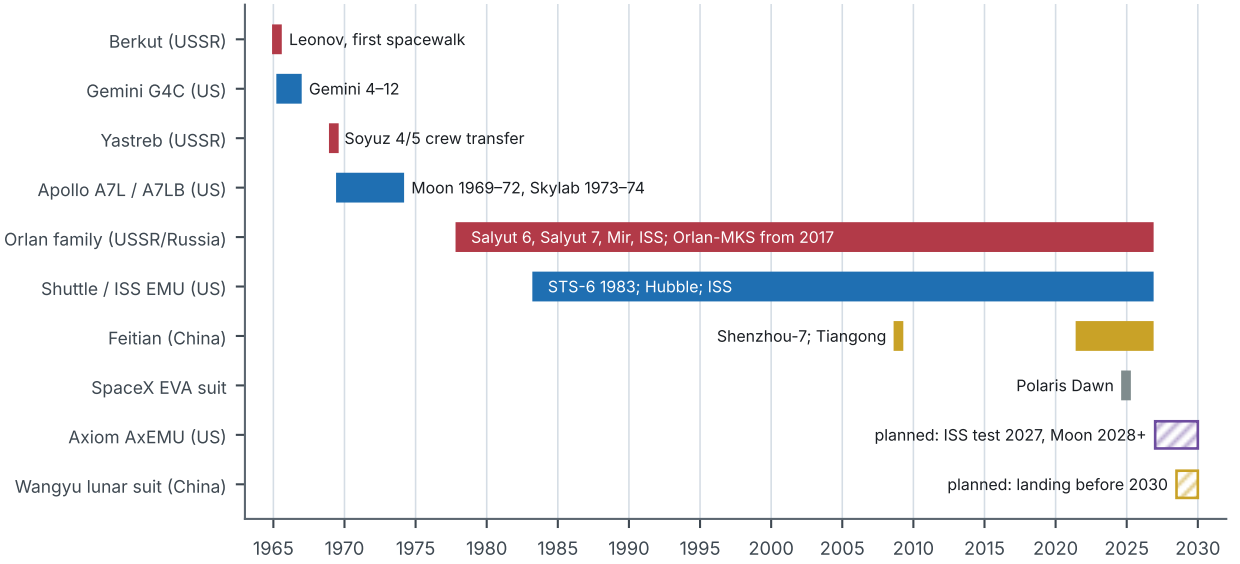
Extravehicular activity (EVA) — a spacewalk — is any time a crew member leaves the protection of a pressurised vehicle and works in a spacesuit in vacuum or on another world's surface. A spacesuit used this way is the smallest crewed spacecraft there is. Wrapped closely around one person, it must provide pressure, oxygen, carbon dioxide removal, cooling, power and communications. It must protect against radiation, temperature extremes and micrometeoroids, and give enough mobility for hands to do precise work for a full working day.

People go outside because some jobs still need hands and eyes on the spot. Spacewalkers built the International Space Station (ISS), upgraded the Hubble Space Telescope five times, saved Skylab and collected 382 kg of rock on the Moon. By October 2026 there had been 284 spacewalks

at the ISS alone, and the longest single EVA had reached 9 h 06 min. This chapter covers the history, the suit and its backpack, decompression, operations and the suits being built for the Moon and Mars. Life support in general is covered in Chapter 30, crewed vehicles in Chapter 31, stations and airlocks in Chapter 32, and robotic servicing in Chapter 40.

33.1 Sixty Years of Spacewalks

The history of EVA is a history of learning how hard it is. Figure 33.1 shows the suits that have been used outside and their years of service, and Table 33.1 compares the main ones.



Bars: years each suit type was used for spacewalks (hatched: planned, status October 2026). Red: Soviet/Russian; blue: NASA; gold: China.

FIGURE 33.1 Spacesuits used for spacewalks since 1965, with the years in which each type was used outside. The Orlan and the Shuttle/ISS EMU have each served for more than four decades; the Axiom lunar suit and China's Wangyu are in development.

33.1.1 First Steps Outside: Voskhod and Gemini

On 18 March 1965 Alexei Leonov left Voskhod 2 through an inflatable airlock and floated for about 12 minutes at the end of a tether. His Berkut suit, pressurised to 0.4 atmospheres, ballooned so much in vacuum that his feet and hands pulled away from the boots and gloves, and he could not fit back into the airlock. He switched the suit to its reserve setting of about 0.27 atmospheres, accepting a risk of decompression sickness, and struggled back inside. His memoir's account of going in head first is more dramatic than his post-flight report. The first lesson of EVA: a pressurised suit is a balloon whose shape must be controlled.

Ed White followed on Gemini 4 on 3 June 1965 (Figure 33.2) and spent about 20 minutes outside; his hand-held gas gun ran dry after about three minutes, leaving him to pull on his tether. Most of the Gemini spacewalks that followed proved far harder than expected. On Gemini 9A in June 1966 Eugene Cernan spent most of a two-hour spacewalk struggling to reach and put on a manoeuvring backpack stowed on the spacecraft's adapter. With nothing to brace against, every push moved his body rather than the task; his heart rate reportedly reached about 180 beats per minute and his visor fogged until he could barely see. Richard Gordon on Gemini 11 was similarly exhausted.



FIGURE 33.2 Ed White during the first American spacewalk, Gemini 4, 3 June 1965. The gold-coated visor protects his eyes from sunlight; the umbilical supplies oxygen and a tether; he holds the hand-held manoeuvring unit. Image: NASA/JSC.

The solution came on Gemini 12 in November 1966. Buzz Aldrin trained underwater beforehand, and his spacecraft carried handrails, foot restraints ("golden slippers"), waist tethers and rest periods written into the plan. He worked for about 5.5 hours over three EVAs without overexertion. Restraints, neutral-buoyancy training and paced timelines are still the foundation of EVA practice.

33.1.2 Apollo and Skylab

The Apollo **A7L** suit, built by ILC Dover, worked at 25.8 kPa (3.75 psi) of pure oxygen. For the Moon it was combined with a **portable life support system (PLSS)** backpack. This supplied oxygen, scrubbed carbon dioxide with lithium hydroxide and cooled the astronaut by sublimating water to vacuum — the architecture of every backpack since. On the Moon, twelve astronauts spent about 80 hours outside in 14 EVAs between July 1969 and December 1972 (see the section on other worlds below).

In May 1973 the Skylab space station reached orbit with its micrometeoroid shield torn away and one solar wing missing and the other jammed. On 7 June Pete Conrad and Joe Kerwin, in Apollo-type suits fed by umbilicals, cut the restraining strap and freed the remaining wing in a 3 h 25 min EVA. It was the first demonstration that spacewalkers could rescue a crippled spacecraft.

33.1.3 The Shuttle, the MMU and Hubble

The Space Shuttle brought the **Extravehicular Mobility Unit (EMU)**, built by Hamilton Standard and ILC Dover, first used by Story Musgrave and Don Peterson on STS-6 in April 1983. Unlike the custom-tailored Apollo suits, the EMU is assembled from standard-sized parts, so a fleet of parts can fit many astronauts. Its PLSS is mounted on the hard upper torso, and it works at 29.6 kPa (4.3 psi).

On 7 February 1984 Bruce McCandless flew the **Manned Manoeuvring Unit (MMU)**, a nitrogen-gas jet backpack with 24 thrusters, up to about 98 m from Challenger with no tether at all (Figure 33.3). MMUs were used on two more flights in 1984, to help repair the Solar Maximum Mission satellite and to capture two stranded communications satellites. They were then retired, because the Shuttle's arm and tethered crews could do the same work with less risk.

The **Hubble Space Telescope** turned EVA into a precision trade. Hubble was designed to be serviced: its instruments slide out on rails, its electronics boxes have captive bolts, and handrails and foot-restraint sockets cover its exterior. Five servicing missions between 1993 and 2009 made 23 spacewalks totalling 166 hours, replacing instruments, gyroscopes, batteries and solar arrays and fixing the telescope's flawed optics (Figure 33.4). On the final mission astronauts repaired two instruments never designed to be opened, removing more than a hundred tiny screws.

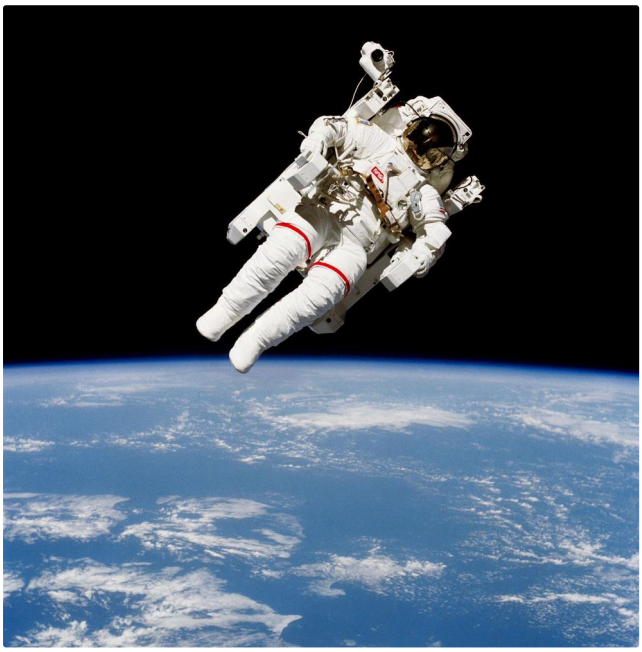


FIGURE 33.3 Bruce McCandless flying the Manned Manoeuvring Unit a few metres from the Space Shuttle Challenger on 7 February 1984, the first untethered spacewalk. Image: NASA/JSC.

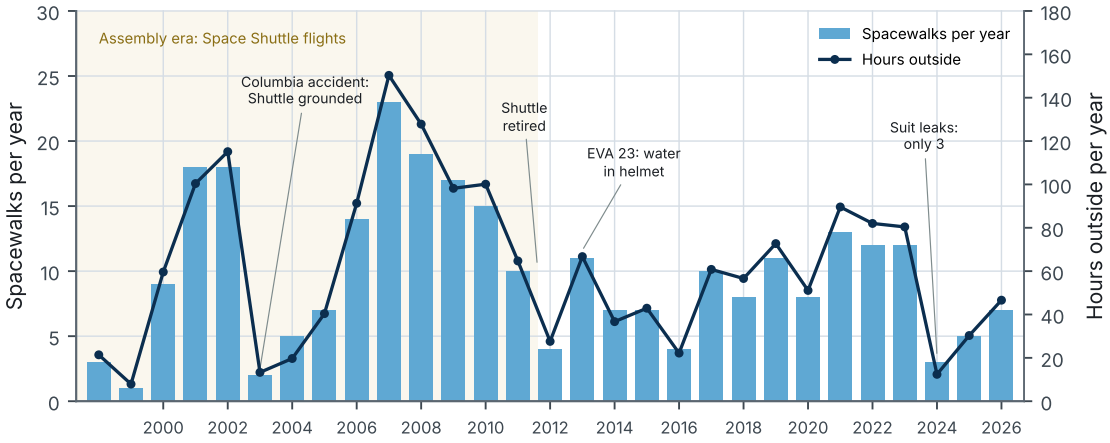


FIGURE 33.4 John Grunsfeld and Andrew Feustel at work on Hubble during the final servicing mission, STS-125, on 14 May 2009. One rides a foot restraint on the Shuttle's robotic arm. Image: NASA/JSC.

33.1.4 Building and Keeping the Space Station

The ISS was assembled in orbit between 1998 and 2011, and spacewalks connected every module, truss segment and cable that robots could not. Figure 33.5 shows the result: from December 1998 to September 2026, there were 284 spacewalks totalling about 1 790 hours; NASA's itemised list shows a peak of 23 in 2007, when Shuttle assembly flights and station crews worked outside

almost monthly. Since the Shuttle's retirement, spacewalks have mostly been maintenance — replacing pumps, batteries and failed electronics, installing new roll-out solar arrays — at a rate of about eight a year. The longest ISS EVA, by Jim Voss and Susan Helms in March 2001, lasted 8 h 56 min.



Data: NASA itemised list of ISS spacewalks (US and Russian), accessed October 2026, corrected for duplicates: 283 of NASA's total of 284, ≈1 790 h, December 1998 – September 2026.

FIGURE 33.5 Spacewalks at the International Space Station per year and hours spent outside, December 1998 to September 2026. The 2024 dip followed water leaks and other faults in the ageing US suits; US spacewalks resumed in 2025–26.

In 2024 only three spacewalks took place: one US EVA in June ended after 31 minutes when water leaked from a service-and-cooling umbilical, and NASA paused others while suit faults were investigated. Seven spacewalks followed in 2026 up to September, including a seven-hour repair of a wrist joint of the station's Canadarm2 robotic arm.

33.1.5 Russia, China and the First Commercial Spacewalk

The Soviet **Orlan** suits, first used outside Salyut 6 in December 1977, are semi-rigid. They have a single-piece aluminium torso with a rear hatch formed by the hinged backpack. So a cosmonaut climbs in from behind in a few minutes without help. The current **Orlan-MKS**, in use on the ISS since 2017, works at 40 kPa and has about 7 hours of autonomy. It has an onboard computer and automatic thermal control, and is rated for 15–20 spacewalks over five to six years (Figure 33.6). Russia delivered new Orlan-MKS suits to the ISS in 2025 and 2026.

China's **Feitian** suit was derived from the Orlan and, according to Chinese state media, also works at about 40 kPa. Zhai Zhigang wore the first version for China's first spacewalk, about 20 minutes long, from Shenzhou-7 on 27 September 2008. The second generation, in use at the Tiangong station since July 2021, supports up to 8 hours outside; Chinese sources report that its design life has been extended from 15 to 20 spacewalks. On 17 December 2024 Cai Xuzhe and Song Lingdong of Shenzhou-19 stayed outside for 9 h 06 min, the longest spacewalk ever. By September 2026 the second-generation suits had supported 27 spacewalks at Tiangong, many of them to fit debris shields to the station.



FIGURE 33.6 Cosmonaut Alexey Zubritsky in an Orlan suit during a Russian spacewalk on 28 October 2025. The blue stripes distinguish the two cosmonauts; the bulky backpack is also the suit's entry hatch. Image: NASA/JSC.

The first commercial spacewalk took place on 12 September 2024 on the privately funded Polaris Dawn mission. Crew Dragon has no airlock, so after about two days of prebreathe with the cabin at reduced pressure, all four crew members suited up and the whole cabin was vented to vacuum. Jared Isaacman and SpaceX engineer Sarah Gillis in turn leaned out of the hatch to test the mobility of the new SpaceX EVA suit. The suit worked at about 34 kPa (5 psi) and drew oxygen, power and data through 3.7 m umbilicals rather than a backpack. From depressurisation to repressurisation the EVA lasted 1 h 46 min. No second commercial spacewalk had flown by October 2026.

TABLE 33.1 Spacesuits for EVA. Masses are Earth masses of suit plus life support; the AxEMU is derived from NASA's xEMU, designed for pressures from 29.6 to 56.5 kPa.

Suit	Operator	Pressure	Mass (approx.)	Time outside	First EVA
A7L / A7LB + PLSS	NASA (Apollo)	25.8 kPa	82–96 kg	4 h; 8 h from Apollo 15	1969
Shuttle/ISS EMU	NASA	29.6 kPa	≈ 145 kg (ISS)	7 h + 1 h reserve	1983
Orlan-MKS	Roscosmos	40 kPa	≈ 110 kg	≈ 7 h	2017 (Orlan 1977)
Feitian (2nd gen.)	CMSA	≈ 40 kPa (reported)	120–130 kg	up to 8 h	2021 (1st gen. 2008)
SpaceX EVA suit	SpaceX	≈ 34 kPa	not published	umbilical, no backpack	2024
Axiom AxEMU	Axiom / NASA	variable (xEMU heritage)	not published	≥ 8 h (goal)	planned 2027–28

33.2 Anatomy of a Spacesuit

A spacesuit has two main parts. The **pressure garment** holds the atmosphere around the body, and the life support system keeps that atmosphere breathable and cool. Figure 33.7 shows the main elements of the US EMU and the layers of its wall.

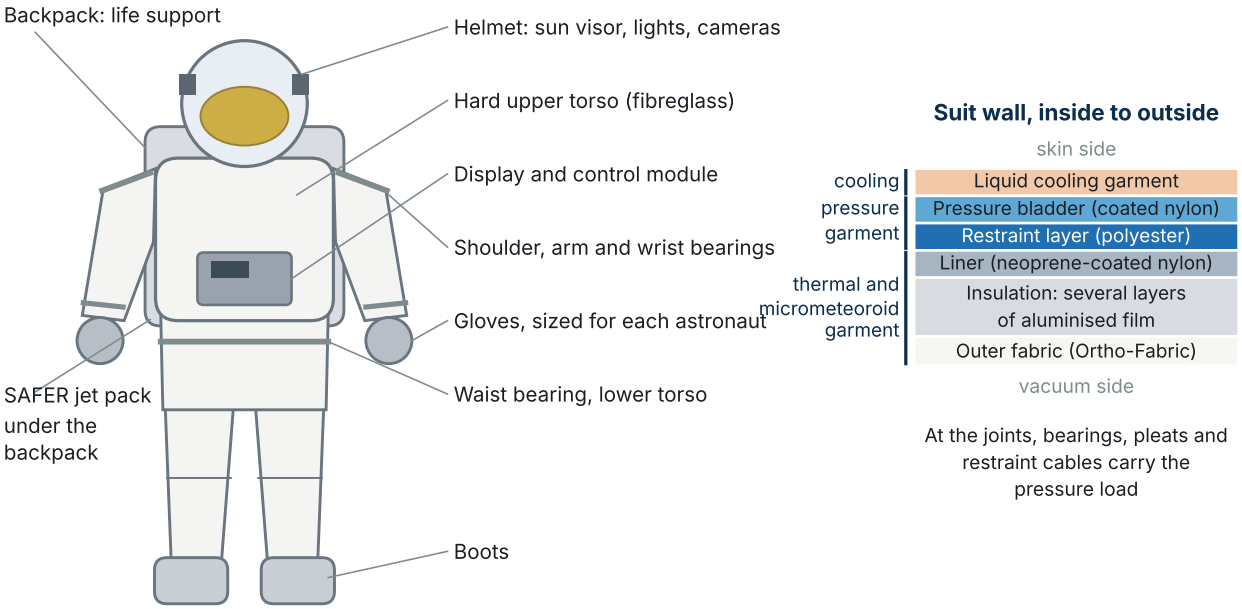


FIGURE 33.7 Main elements of the US Extravehicular Mobility Unit and, right, the layers of the suit wall from the skin outwards. The bladder holds the gas, the restraint carries the pressure load, and the thermal and micrometeoroid garment protects both.

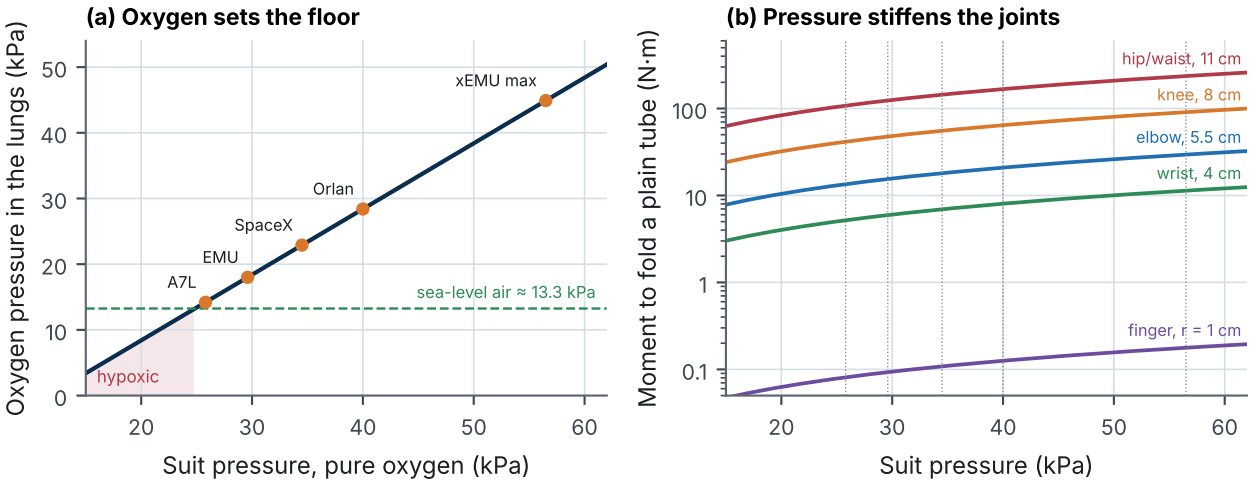
33.2.1 Why Suits Work at Low Pressure

The suit pressure is the most important number in its design. It must be high enough to supply oxygen: with pure oxygen, the oxygen pressure in the lungs is the suit pressure minus about 6.3 kPa of water vapour and 5.3 kPa of carbon dioxide. Matching the 13.3 kPa of a person breathing sea-level air needs about 25 kPa — almost exactly the Apollo suit pressure (Figure 33.8 a). Below about 6.3 kPa (the **Armstrong limit**) body fluids boil at body temperature, so even an emergency suit pressure must stay well above it.

Every kilopascal above the minimum makes the suit stiffer. An inflated fabric tube resists bending because bending it reduces its volume, and the gas must be compressed to make room. A simple model gives the moment needed to fold an unjointed tube of radius r at pressure p:

$$M \approx \pi p r^3$$

For an elbow-sized tube, 5.5 cm in radius, this is about 15 N·m at 29.6 kPa and 30 N·m at 56.5 kPa; for a knee it is three times larger (Figure 33.8 b). An astronaut flexing an arm hundreds of times a day against such moments would tire quickly. Low pressure is therefore the first defence of mobility, and clever joints are the second. The price of low pressure is decompression sickness, discussed below.



(a) Alveolar gas equation with 100 % O_2, water vapour 6.3 kPa, CO_2 5.3 kPa. (b) Collapse moment of an inflated fabric tube, $M = \pi\, p\, r^3$; real joints use pleats, bearings and constant-volume shapes to cut this by a large factor. Dotted lines: suit pressures of panel (a).

FIGURE 33.8 Why suits are pressurised to 26–57 kPa. (a) Oxygen pressure in the lungs for a pure-oxygen suit: about 25 kPa matches breathing sea-level air. (b) Moment needed to fold a plain inflated tube of different radii, which grows in proportion to pressure.

33.2.2 Layers

The wall of a suit has three functional groups (Figure 33.7). Next to the skin is a **liquid cooling and ventilation garment (LCVG)**. In the EMU it is a stretch-mesh undergarment threaded with about 90 m of thin plastic tubing. It carries cool water over the body and collects ventilation gas from the limbs. Over it the **pressure garment** has two layers. One is a gas-tight **bladder** of urethane-coated nylon. The other is a **restraint layer** of strong polyester fabric that carries the pressure load and fixes the suit's shape, so the bladder never stretches. Outside is the **thermal micrometeoroid garment (TMG)**: a neoprene-coated nylon liner, several layers of aluminised film acting as multilayer insulation (Chapter 5), and an outer shell of Ortho-Fabric. This fabric is a weave of Gore-Tex, Kevlar and Nomex that resists abrasion, fire and small particles. The suit's surroundings range from about +120 °C in sunlight to −120 °C in shadow.

33.2.3 Joints and Mobility

The aim of joint design is a **constant-volume joint**: one whose internal volume, and therefore the work done against the gas, stays nearly the same as it bends. Three techniques are used. **Convolutes** are bellows-like folds that fill on the inside of the bend as the outside opens. **Flat-pattern joints** use sewn gores and restraint cables or webbing along the axis of the joint, so the fabric on one side shortens as the other lengthens. **Rotary bearings** — sealed rings at the shoulder, upper arm, wrist, waist and hip — let a limb rotate with almost no change in volume. Hard parts carry loads. The EMU's **hard upper torso** is a fibreglass shell to which the arms, helmet, lower torso and backpack attach. The Orlan, Feitian and AxEMU extend this to a rear-entry hard torso.

Fit matters as much as design: in a suit that is too large the joints of suit and body no longer coincide; one too small presses on shoulders and knees. In a small NASA survey, 14 of 22 EVA astronauts reported EVA-related shoulder injuries, most of them during training in the pool, where the hard torso restricts shoulder movement, especially when working upside down.

33.2.4 Gloves and Hands

Gloves are the hardest part of a suit to design: the hand has many small joints and weak muscles, and a glove must be pressurised, insulated and tough enough for sharp edges. EMU gloves are custom-sized from casts or scans of each astronaut's hands. They have a bladder and restraint with finger joints and a wrist bearing, heated fingertips for cold work, and rubberised palms. Even so, grasping with a pressurised glove takes much more effort than with a bare hand and fatigues the forearms within hours. In one NASA study, 770 suited training events produced 352 reported symptoms, 47 % of them in the hands. The most common was **fingernail delamination**, the loss of nails, attributed to moisture in the gloves and loads on the fingertips. Gloves are also vulnerable to damage: after a cut glove ended a spacewalk early in 2007, crews began photographing their gloves during spacewalks, and gloves are inspected after every EVA (Figure 33.9).



FIGURE 33.9 Astronauts Mike Foreman (left) and Rick Linnehan in the ISS Quest airlock after a spacewalk in March 2008, still wearing their EMUs. Gloves are photographed and inspected for cuts after each spacewalk. Image: NASA/JSC.

33.2.5 Helmet and Visors

The EMU helmet is a clear polycarbonate bubble. Over it fits the **extravehicular visor assembly**, with a clear protective visor and a gold-coated sun visor that reflects visible light and infrared, plus adjustable eyeshades. Lights and video cameras are mounted on the sides. Inside are a microphone and earphones in a fabric "Snoopy cap", a drink bag of about 0.95 L and a foam block against which the astronaut can press the nose to clear the ears. The SpaceX EVA suit added a head-up display and a copper and indium tin oxide visor coating.

33.3 The Portable Life Support System

The **portable life support system (PLSS)** turns a pressure garment into a spacecraft. Figure 33.10 shows its two loops. In the **ventilation loop**, a fan blows oxygen into the helmet at the face. The gas sweeps over the head and body, collecting carbon dioxide, moisture and heat. It is drawn back from the wrists and ankles through a canister that removes carbon dioxide and odours. It then passes through the sublimator, which cools it and condenses moisture, and a water

separator, before returning to the helmet. In the **cooling-water loop**, a pump circulates water through the LCVG tubes and the sublimator. The EMU's fan, pump and separator are a single rotating machine.

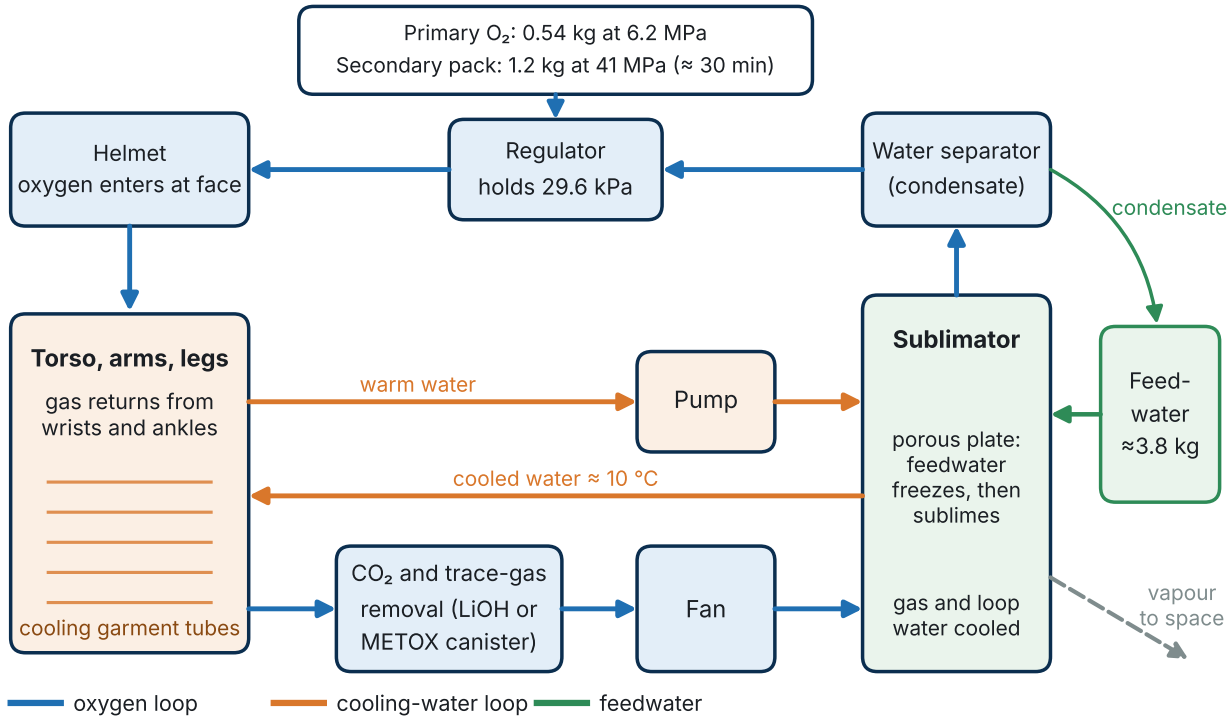


FIGURE 33.10 Simplified flow diagram of the EMU portable life support system. Oxygen is recirculated through a carbon dioxide canister and a sublimator; cooling water circulates through the garment; feedwater is lost to space as vapour.

33.3.1 Oxygen and Pressure

The suit is a closed loop, so oxygen is needed only to replace what the astronaut consumes and what leaks out. An astronaut working at 250 W consumes about 63 g of oxygen per hour. The EMU's primary tanks hold about 0.54 kg at 6.2 MPa; a regulator keeps the suit at 29.6 kPa. A **secondary oxygen pack** of about 1.2 kg at 41 MPa can maintain pressure for about 30 minutes in **purge mode**. If the fan fails, it flushes oxygen straight through the suit to remove carbon dioxide and heat. This is the "get-back" time: EVA plans keep the astronaut within about 30 minutes of the airlock.

33.3.2 Removing Carbon Dioxide

Apollo backpacks and early Shuttle EMUs absorbed carbon dioxide in disposable **lithium hydroxide (LiOH)** cartridges (the chemistry is described in Chapter 30). The ISS EMU uses regenerable **metal oxide (METOX)** canisters based on silver oxide, which absorb about 0.54 kg of carbon dioxide. They are then regenerated by baking in an oven on the station, which takes more than 1 kW for up to 14 hours. The next generation, used in NASA's xEMU prototype and the AxEMU, uses a **rapid cycle amine (RCA)** swing bed. Two small beds of amine sorbent alternate every few minutes. One absorbs carbon dioxide and water while the other is exposed to vacuum to release them. Because the beds regenerate continuously, CO_2 capacity no longer limits EVA duration (Table 33.2).

EXAMPLE: CARBON DIOXIDE REMOVAL FOR A 7-HOUR EVA

An astronaut works for 7 hours at an average metabolic rate of 250 W, with a respiratory quotient (CO_2 produced ÷ O_2 consumed) of 0.85.

1. Oxygen: at RQ 0.85 each litre of oxygen releases about 20.4 kJ (Weir equation). So 250 W needs 250 × 3 600 / 20 400 = **44.1 L/h** of O_2 (at 0 °C, 1 atm), or 63 g/h.
2. Carbon dioxide: 0.85 × 44.1 = 37.5 L/h = **73.6 g/h**; over 7 hours **515 g**, or 589 g if a 1-hour reserve is added.
3. Lithium hydroxide absorbs at most 0.92 kg of CO_2 per kg (Chapter 30), so 589 g needs at least **0.64 kg**; real beds, never fully used, need about twice that. The Apollo 15–17 backpacks carried 1.42 kg.
4. A METOX canister with 0.54 kg capacity lasts 0.54/0.0736 = **7.3 h** at 250 W, but only 6.3 h at 293 W — one reason heavy work shortens EVAs (Figure 33.11 b).
5. Ventilation: the EMU fan moves about 0.17 m^3/min (6 cubic feet per minute) of gas at 29.6 kPa, or 0.034 mol/s. Carbon dioxide is added at 4.6×10^{-4} mol/s, so the returning gas carries about 1.4 % more CO_2 than the gas entering the helmet — a rise of **0.40 kPa (3 mmHg)**. Fresh gas is therefore aimed at the face, where exhaled CO_2 is concentrated.

33.3.3 Rejecting Heat

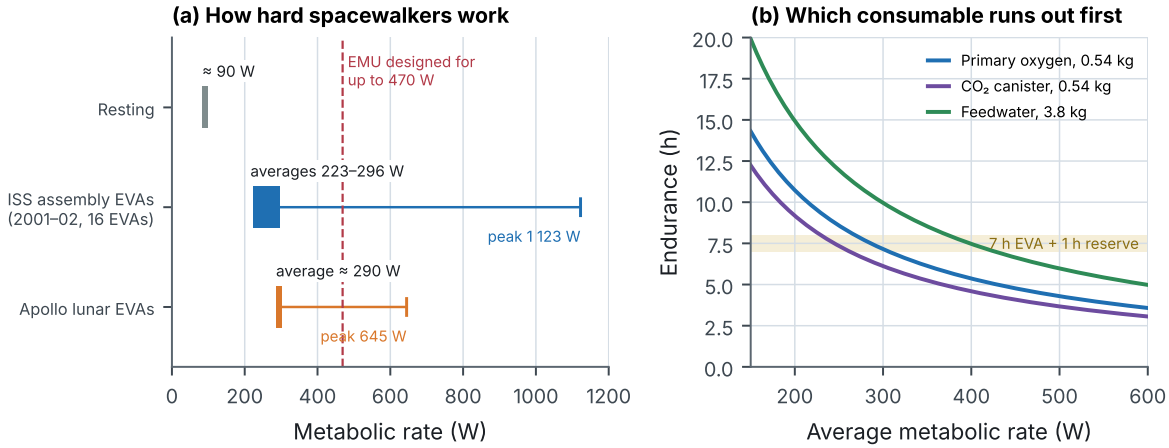
The astronaut's metabolic heat, typically 200–300 W and sometimes much more, must be removed without air to carry it away. A radiator alone would be impractical (see the example below), so the PLSS throws away water. In a **sublimator**, feedwater flows to one side of a porous metal plate whose other side faces vacuum. Water in the pores freezes, and the ice sublimates — turns directly to vapour — absorbing 2.83 MJ per kilogram. The process regulates itself: more heat sublimates ice faster. NASA's newer **spacesuit water membrane evaporator (SWME)**, used in the xEMU design, evaporates water through hydrophobic hollow fibres and is less sensitive to contamination, which has blocked sublimator plates in the past.

EXAMPLE: WATER FOR A SUBLIMATOR

The PLSS must reject 300 W for a 7-hour EVA using a sublimator (latent heat 2.83 MJ/kg).

1. Water rate: 300 / 2.83×10^6 = 1.06×10^{-4} kg/s = **0.38 kg/h**.
2. For 7 hours: **2.7 kg**. The EMU carries about 3.8 kg of feedwater, enough for 10 hours at 300 W but only 6.4 hours at 470 W (1 600 Btu/h), the highest rate for which the EMU was designed.
3. Apollo's extended backpacks carried 5.2 kg of water for 8-hour lunar EVAs. This gives an average capacity of $5.2 \times 2.83 \times 10^6$ / 28 800 ≈ **510 W** — more than the EMU, for the heavier work of lunar surface EVAs.
4. A suit radiator would need 300 / ($0.9 \times 5.67 \times 10^{-8} \times 280^4$) ≈ **0.96 m^2** of surface viewing only cold space. That is roughly half the outer area of a suit, much of which faces the Sun, the Earth or the spacecraft at any moment.

Figure 33.11 puts these numbers together. Average metabolic rates on 16 ISS assembly spacewalks in 2001–02 were 223–296 W, with peaks above 1 000 W; Apollo lunar EVAs averaged about 290 W. At such rates the CO_2 canister runs out first, then the oxygen, and the water last; the first two give roughly the 7 hours plus reserve for which the EMU is sized. A 2024 NASA review concluded that EVAs of 6.5 hours are reasonable while average metabolic rates stay below about 290 W (1 000 Btu/h).



(a) ISS assembly EVAs on STS-104 to STS-113 (Katuntsev, Osipov and Gernhardt 2004, via NASA) and Apollo data. (b) Weir equation, RQ 0.85, 20.4 kJ per litre O_2; all metabolic heat rejected by sublimation (2.83 MJ/kg); suit leakage and heat leaks ignored. Capacities approximate those of the ISS EMU.

FIGURE 33.11 Metabolic rates and consumables. (a) Measured averages and peaks for ISS assembly and Apollo lunar spacewalks. (b) Endurance set by each consumable for EMU-like capacities as a function of average metabolic rate.

TABLE 33.2 Portable life support systems compared. xEMU and AxEMU values are design goals.

Function	Apollo PLSS (1969–72)	Shuttle/ISS EMU	xEMU / AxEMU (2020s)
Oxygen storage	9.9 MPa tank (Apollo 15–17)	0.54 kg at 6.2 MPa + 1.2 kg secondary	high-pressure tanks, variable-pressure regulator
CO_2 removal	LiOH, 1.42 kg	LiOH or METOX (regenerable in oven)	rapid cycle amine, vented to vacuum
Cooling	sublimator, 5.2 kg water	sublimator, 3.8 kg water	water membrane evaporator
Emergency supply	oxygen purge system, ≥ 30 min	secondary oxygen pack, ≈ 30 min	secondary oxygen supply
Time outside	4 h, later 8 h	7 h + 1 h reserve	≥ 8 h (goal)

33.3.4 Power, Communications and Displays

A rechargeable battery powers the fan, pump, radios, sensors, lights and cameras. The EMU's **display and control module** on the chest carries the oxygen and cooling controls and a small text display; astronauts read its labels through a wrist mirror, and checklists are worn on the cuff. A caution-and-warning system monitors pressure, oxygen, carbon dioxide, battery and water, and suit data are radioed to the ground with voice and video. Newer suits add colour and head-up displays and record health data such as heart rate.

33.3.5 SAFER: the Self-Rescue Jet Pack

A broken tether could leave an astronaut drifting away from the station with no one able to follow. The **Simplified Aid For EVA Rescue (SAFER)** is a small nitrogen-jet backpack, about 38 kg, that latches under the EMU's PLSS. It has 24 small thrusters, a hand controller that folds out from its lower edge and an automatic attitude hold that stops tumbling at the push of a button. Its 1.4 kg of nitrogen gives a velocity change of only about 3 m/s — enough to stop, turn and fly back a short distance, but not to explore. SAFER was first tested on STS-64 in September 1994 (Figure 33.12), is worn on every US spacewalk at the ISS, and has never had to be used in an emergency.



FIGURE 33.12 Astronaut Mark Lee testing SAFER above the Earth during the STS-64 mission in September 1994, the first untethered US spacewalk in ten years. Image: NASA/MSFC.

33.4 Decompression Sickness and Prebreathe

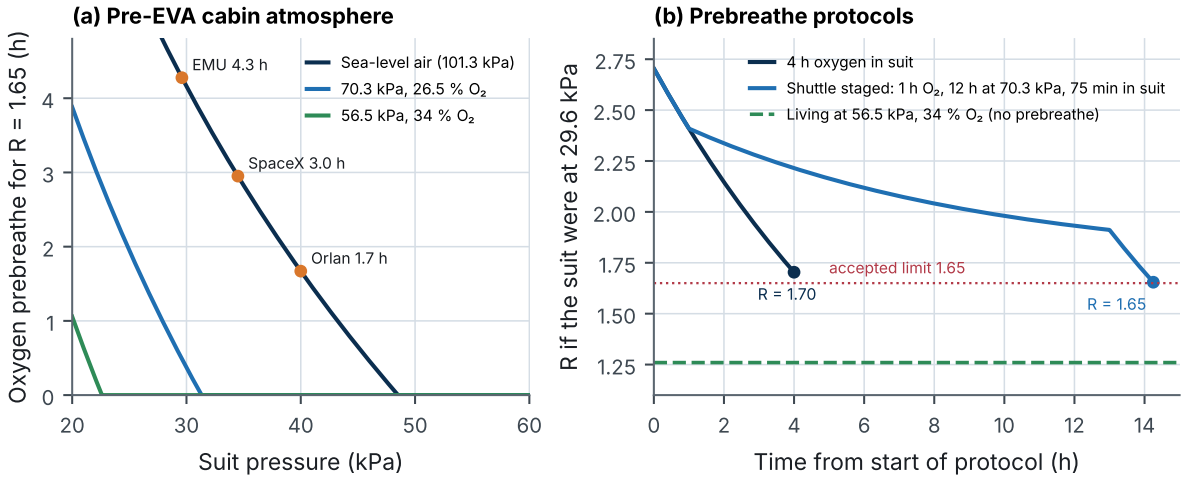
33.4.1 Nitrogen in the Tissues

The ISS cabin holds sea-level air, so the body's tissues are saturated with nitrogen at about 80 kPa. If an astronaut moves straight to a suit at 29.6 kPa, the nitrogen becomes supersaturated and can form bubbles — **decompression sickness (DCS)**, introduced in Chapter 30. Mild DCS is pain in the joints; serious ("Type II") DCS affects the brain, spinal cord or lungs. NASA uses the **tissue ratio** R — tissue nitrogen pressure divided by suit pressure — as an index of risk:

$$R = \frac{P_{N_2,\,\text{tissue}}}{P_{\text{suit}}}, \qquad P_{N_2,\,\text{tissue}}(t) = P_{N_2,\,\text{amb}} + \left(P_{N_2,0} - P_{N_2,\,\text{amb}}\right) 2^{-t/t_{1/2}}$$

The tissue relaxes towards the nitrogen pressure it breathes with a half-time $t_{1/2}$ of 360 minutes, the value that best matched NASA and US Air Force chamber tests. Breathing pure oxygen drops the inspired nitrogen to almost zero, so nitrogen leaves the tissues. For the EMU at 29.6 kPa, protocols with $R \leq 1.65$ have historically been accepted. NASA's standard requires that any protocol be shown in ground trials to give less than a 4 % probability of DCS with 95 % confidence.

Without prebreathe, R is 2.70 for the EMU, 2.32 for a 34.5 kPa suit and 2.00 for the 40 kPa Orlan. Figure 33.13 a shows the oxygen prebreathe needed to reach R = 1.65. It is 4.3 hours from sea-level air for the EMU, 3.0 hours at 34.5 kPa and 1.7 hours for the Orlan. Above 48.5 kPa none is needed. Russian practice accepts a shorter prebreathe, and so a higher R: cosmonauts breathe oxygen in the Orlan for only about 30 minutes before going out. A low-pressure, oxygen-rich cabin changes the picture completely. Living at 70.3 kPa with 26.5 % oxygen, the "campout" atmosphere, leaves only 52 kPa of nitrogen in the tissues. NASA's 56.5 kPa, 34 % oxygen **exploration atmosphere** leaves 37 kPa, R = 1.26. By this criterion it needs no prebreathe at all, although NASA plans about 15 minutes of in-suit oxygen as a margin (Chapter 30).



Single 360-min tissue (NASA R-value model): tissue N_2 relaxes exponentially towards the inspired N_2 pressure; R = tissue N_2 / suit pressure. Initial tissue N_2 = 0.79 × 101.3 kPa. Real protocols were validated in chamber trials; exercise during prebreathe, not modelled here, speeds nitrogen removal.

FIGURE 33.13 Decompression and prebreathe in the 360-minute tissue model. (a) Oxygen prebreathe needed to reach R = 1.65 as a function of suit pressure, for three cabin atmospheres. (b) R during two Shuttle-era protocols, compared with living in the exploration atmosphere.

33.4.2 Prebreathe Protocols

Four hours in a suit breathing oxygen wastes crew time, so shorter protocols were developed (Table 33.3). The Shuttle's **staged decompression** lowered the whole cabin to 70.3 kPa for at least 12 hours before EVA days; in practice crews spent an average of more than 40 hours at this pressure. The ISS cannot lower its whole cabin, so other approaches are used. In the **campout**, the EVA crew sleep in the closed Quest airlock at 70.3 kPa for up to 8 h 40 min before the EVA. In **exercise prebreathe**, the astronaut breathes oxygen while pedalling vigorously; exercise increases blood flow and speeds nitrogen removal, cutting the required time to about two hours. The **in-suit light exercise (ISLE)** protocol replaces hard exercise with light movement in the suit after 40 minutes on a mask and a 20-minute partial depressurisation. NASA has never had a Type II DCS case in spaceflight, and no DCS was reported in more than 140 Shuttle EVAs.

TABLE 33.3 Ways to prepare for a spacewalk. Durations are approximate and protocols have been revised over time.

Protocol	Main steps	Typical duration	Used on
In-suit prebreathe	O_2 in suit at 101 kPa	4 h	ISS, contingency
Staged decompression	1 h O_2; ≥ 12 h cabin at 70.3 kPa; 75 min O_2 in suit	1–2 days of cabin time	Space Shuttle
Campout	night in airlock at 70.3 kPa (≤ 8 h 40 min); O_2 in suit	overnight + ≈ 1 h	ISS
Exercise prebreathe	O_2 on mask with cycle exercise; in-suit prebreathe	≈ 2 h of O_2	ISS
ISLE	40 min O_2 mask; 20 min depress to 70.3 kPa; ≈ 100 min in suit with light exercise	≈ 2.7 h	ISS (from 2011)
Orlan	O_2 in suit at reduced pressure	≈ 30 min	Russian segment
Polaris Dawn	≈ 2 days with cabin at ≈ 59 kPa	≈ 45 h	Crew Dragon, 2024

EXAMPLE: R-VALUE FOR THE SHUTTLE STAGED PROTOCOL

Start from tissue nitrogen in equilibrium with sea-level air: 0.79 × 101.3 = 80.0 kPa. Use $t_{1/2}$ = 360 min and a suit at 29.6 kPa.

1. No prebreathe: R = 80.0/29.6 = **2.70** — unacceptable.
2. 60 min of pure oxygen at 101.3 kPa: $80.0 \times 2^{-60/360}$ = **71.3 kPa**.
3. 12 h in a cabin at 70.3 kPa with 26.5 % O_2, where inspired nitrogen is 0.735 × 70.3 = 51.7 kPa: $51.7 + (71.3 - 51.7) \times 2^{-720/360}$ = **56.6 kPa**.
4. 75 min of oxygen in the suit: $56.6 \times 2^{-75/360}$ = **49.0 kPa**, so R = 49.0/29.6 = **1.65** — exactly the accepted limit.
5. By comparison, 4 hours of in-suit oxygen from sea-level air gives $80.0 \times 2^{-240/360}$ = 50.4 kPa, R = **1.70** (Figure 33.13 b); the real 4-hour protocol was validated by chamber tests rather than by R alone.

33.4.3 The Pressure–Mobility Trade

The choice of suit pressure is a three-way trade between oxygen supply, mobility and decompression (Figure 33.8 and Figure 33.13). A higher-pressure suit needs no prebreathe but is stiffer, heavier and leakier. The answer adopted for NASA's xEMU, and inherited by the AxEMU, is a **variable-pressure suit** that can run at 29.6 kPa for mobility or up to 56.5 kPa to shorten prebreathe or treat suspected DCS. For frequent lunar or Martian surface EVAs, a reduced-pressure, oxygen-enriched habitat combined with a suit at 29.6–34 kPa is the preferred direction, limited mainly by the fire risk of oxygen-rich cabins (Chapter 30).

33.5 Working Outside

33.5.1 Tools, Tethers and Restraints

The first rule of EVA is never to let go. Each astronaut is clipped to the structure by a retractable **safety tether** of about 26 m on a reel, and moves hand over hand along handrails, attaching a **waist tether** at each worksite. Every tool and part is tethered too; lost objects become orbital debris (Chapter 21). For two-handed work the astronaut locks the boots into an **articulating portable foot restraint**, which fits sockets on the structure or the end of the robotic arm. The station's Canadarm2 (Chapter 32) can carry an astronaut and a large unit, such as a battery or pump, from one worksite to another, as the Shuttle arm did at Hubble (Figure 33.4).

Tools are designed for bulky gloves. The **pistol grip tool**, a battery-powered torque-limited wrench, does most bolt work; bolts are captive, connectors have large tabs, and replaceable units have guide pins and handles placed for a gloved hand (Figure 33.14).

33.5.2 Training in Water

The best simulation of weightlessness on Earth is neutral buoyancy. NASA's **Neutral Buoyancy Laboratory (NBL)** in Houston, built in the mid-1990s, is a pool 62 m long, 31 m wide and 12 m deep. It holds 23.5 million litres of water and full-size mockups of ISS modules. Suited astronauts are weighted until they neither float nor sink and rehearse every task with support divers, for

several hours per hour of planned EVA (Figure 33.15). Water drag and gravity inside the suit make the pool imperfect, so it is complemented by virtual reality, air-bearing floors and, for lunar suits, reduced-gravity rigs. Russia and China have similar hydrolabs.



FIGURE 33.14 NASA astronaut Chris Williams during a 7 h 20 min spacewalk on 30 June 2026 to replace a faulty wrist joint of the Canadarm2 robotic arm. Image: NASA/JSC.



FIGURE 33.15 Astronauts Loral O'Hara and Stan Love during the first two-suit test of Axiom's lunar spacesuit at the Neutral Buoyancy Laboratory in September 2025. Image: NASA/JSC (NASA).

33.5.3 Timelines and Workload

A typical ISS spacewalk lasts 6–7 hours, plus about as much again for suit checks, prebreathe, depressurisation and post-EVA servicing. The timeline is planned to the minute, with "get-ahead" tasks and pre-agreed stopping points. Flight controllers estimate metabolic rate from oxygen use and cooling-water temperatures, because consumables and heat limit the day (Figure 33.11). NASA measures **EVA efficiency** as EVA time divided by the crew time spent preparing for it and cleaning up afterwards; this was only about 0.4 for ISS protocols and 0.5 for the Shuttle's.

33.5.4 When Things Go Wrong

Serious EVA incidents have been few, but each taught a lesson (Table 33.4). The most dangerous at the ISS came on 16 July 2013, during Luca Parmitano's EVA 23. About 44 minutes in he reported water at the back of his head. It spread over his eyes, nose and ears. He found his way back to the airlock largely by feel before the EVA ended at 92 minutes. Between 1 and 1.5 litres of water had entered the helmet from the ventilation loop. The investigation traced it to contaminants blocking holes in the fan/pump/separator, and found that a smaller leak on the previous EVA had been misdiagnosed as a leaking drink bag. NASA added absorbent pads and a snorkel to the helmet and tightened the handling of suit anomalies.

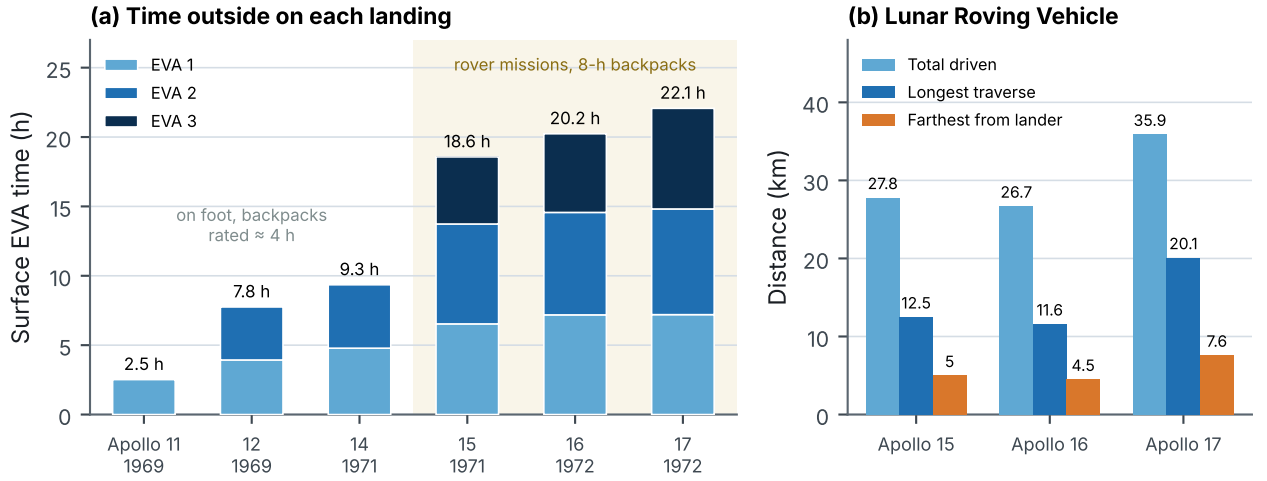
TABLE 33.4 Notable spacewalk incidents and what they taught.

Year	Event	Lesson
1965	Leonov's suit balloons; he lowers pressure to re-enter the airlock	control of suit shape; airlock sizing
1973	Skylab solar wing freed with improvised tools	EVA as a contingency capability
2007	Torn ISS solar array repaired from the end of a boom extension	improvised tools; rehearse with ground teams
2013	EVA 23: about 1.5 L of water in Parmitano's helmet	treat suit anomalies as life-threatening
2024	ISS EVA ends after 31 min: water leak in suit umbilical	ageing suits limit operations

33.6 Spacewalking on Other Worlds

33.6.1 Lessons from Apollo

Gravity on the Moon is one sixth of Earth's, so the 82–96 kg Apollo suit and backpack felt like only 14–16 kg, but the suit's stiff legs and the low traction made normal walking awkward. Astronauts found that a **loping** gait, bounding from foot to foot, or a two-footed hop, was faster and less tiring, and falls were common but usually harmless. Apollo 11–14 crews walked, with backpacks rated for about 4 hours. Apollo 15–17 carried the **Lunar Roving Vehicle (LRV)** and 8-hour backpacks. Figure 33.16 shows how surface time and range grew. The Apollo 17 crew spent 22 h 04 min outside over three EVAs and drove 35.9 km, up to 7.6 km from their lander.



Data: NASA mission reports; NSSDC Lunar Roving Vehicle summary. Total ≈ 80 h of lunar surface EVA by 12 astronauts.

FIGURE 33.16 Apollo surface EVAs. (a) Time outside per landing, by EVA. (b) Distances driven with the Lunar Roving Vehicle on Apollo 15–17; the farthest point from the lander was limited by the need to be able to walk back if the rover failed.

The rover was a safety system as well as a vehicle. Traverses were planned so that, if the rover broke down at the farthest point, both astronauts could walk back to the lander on the consumables left in their backpacks. This **walkback limit** is the surface equivalent of the 30-minute get-back time in orbit, and it will constrain Artemis traverses too until pressurised rovers are available.

Lunar dust was the biggest surprise. The regolith is made of sharp, glassy fragments never rounded by wind or water, and it clings electrostatically to everything. After three days on the Moon, Apollo 17 suits were grey to the knees, their bearings were harder to turn and their outer fabric was abraded. Dust in the cabin irritated the crew's eyes and lungs (Figure 33.17). Gloves wore fastest, and astronauts' hands and fingers were sore after hours of gripping tools.

33.6.2 Artemis Suits

NASA decided in 2022 to buy lunar and station suits as a service rather than build them, under its **Exploration Extravehicular Activity Services (xEVAS)** contracts. Collins Aerospace withdrew in 2024, leaving Axiom Space as the sole provider. Axiom's **AxEMU** is derived from NASA's xEMU prototype (Figure 33.18). It is a rear-entry suit with a hard upper torso and interchangeable parts intended to fit at least 90 % of the US male and female population. It has mobility joints at the

hips and knees that let the wearer kneel and squat, and an xEMU-style backpack. Its goals include EVAs of at least 8 hours and at least two hours in the permanently shadowed craters near the lunar south pole, where temperatures fall below −200 °C.

By early 2026 Axiom had completed its own design review and more than 850 hours of crewed pressurised tests. Qualification was planned to run to mid-2027, with a first suit flying in Earth orbit in 2027. In April 2026 NASA's Office of Inspector General warned that the schedule had little or no margin for the first landing, now planned for Artemis IV in 2028. It also warned that delays in line with recent spaceflight programmes (about 8.7 years on average from contract to test flight) could push suit demonstrations to 2031. In September 2026 NASA and Axiom were reported to be studying a simpler, lighter suit configuration for the first landing. China, meanwhile, unveiled its **Wangyu** lunar suit in 2024 for a crewed landing planned before 2030.



FIGURE 33.17 Gene Cernan in the lunar module after the second Apollo 17 EVA in December 1972, his suit coated with lunar dust. Image: NASA/JSC.



FIGURE 33.18 The flight design of Axiom Space's AxEMU lunar spacesuit, shown in 2024 with its white outer layer. Image: NASA/JSC (Axiom Space).

33.6.3 Suits for Mars

Mars adds problems that the Moon does not (Table 33.5).

TABLE 33.5 How the environment changes spacesuit design.

Factor	ISS orbit	Moon	Mars
Gravity	≈ 0	0.17 g	0.38 g
Ambient pressure	vacuum	vacuum	≈ 0.6 kPa CO_2
Heat rejection	sublimator	sublimator	evaporator or radiator; sublimator unreliable
Dust	none	sharp, electrostatic	fine, windblown, perchlorates
Return to safety	≈ 30 min to airlock	walkback limit	walkback limit; no quick return to Earth
Spacewalks	≈ 8 a year	a few per early landing	many over a long stay

Its gravity, 0.38 g, makes suit mass more than twice as burdensome as on the Moon, so a Mars suit must be much lighter than the AxEMU. Its thin carbon dioxide atmosphere, about 0.6 kPa, carries heat by convection, so insulation that works in vacuum works less well. The surface pressure is close to the triple point of water, 0.61 kPa, so a sublimator will not work reliably, and evaporative coolers lose efficiency. An amine scrubber that regenerates by exposure to "vacuum" must vent against an atmosphere that is itself 95 % carbon dioxide, so it needs heating or a different sorbent. Martian dust is finer than lunar dust, is lifted by wind and contains perchlorates, which are toxic. Suits must also be maintained and repaired on site for months.

33.6.4 Suitports and Rear-Entry Suits

Every trip through an airlock costs gas and time and brings dust inside. A **suitport** avoids this: the suit is stored outside the habitat or rover with its rear hatch sealed against a port in the wall, and the crew member climbs in through the back and detaches. NASA tested the idea on its Space Exploration Vehicle rover prototype and the Z-1 prototype suit in the early 2010s, and rear-entry suits such as the Orlan and AxEMU make it practical. Suitports would allow quick EVAs from pressurised rovers on long traverses, if the seals survive dust and repeated use.

AI FOR SPACEWALKS

AI is starting to support EVA in three ways. *Suit health monitoring:* models trained on suit telemetry can watch pressures, temperatures, fan speed and carbon dioxide. They can flag early signs of faults like the water intrusion of EVA 23, or a rising metabolic rate, faster than controllers watching many screens. *EVA planning and support:* optimisation tools can schedule tasks against consumables, lighting and walkback limits. Voice assistants in the helmet can read procedures, log progress and answer questions. This helps when the light-time delay to Mars — up to 22 minutes one way — makes ground support slow. *Robotic assistants:* free-flying robots and dexterous manipulators can fetch tools, hold lights and inspect worksites, and may eventually do routine external maintenance so that people go outside only when judgement and dexterity are needed. Chapters 48 and 51 describe these systems and how they can be verified for human spaceflight.

KEY TAKEAWAYS

- A spacesuit is a one-person spacecraft: a pressure garment (bladder, restraint, thermal and micrometeoroid layers) plus a portable life support system that supplies oxygen, removes carbon dioxide and rejects heat.
- Suits run at 26–57 kPa: about 25 kPa of pure oxygen matches sea-level breathing, and lower pressure means more mobility, because the moment to bend a pressurised joint grows as pr^3.
- Constant-volume joints, bearings, hard torsos and custom gloves make work possible; hands and shoulders still suffer most.
- The PLSS of every suit since Apollo recirculates oxygen through a CO_2 scrubber and cools the astronaut by sublimating or evaporating water: about 0.38 kg of water per hour for 300 W.
- Moving from sea-level air to a 29.6 kPa suit gives a tissue ratio R = 2.70; oxygen prebreathe, exercise or a reduced-pressure campout brings it to about 1.65. Variable-pressure suits and low-pressure, oxygen-rich habitats can cut prebreathe to minutes or remove it altogether.
- EVA success depends on operations: tethers, restraints, neutral-buoyancy training, paced timelines and a 30-minute get-back or walkback reserve.
- In October 2026 the ISS still relied on suits designed in the 1970s; Axiom's AxEMU was in qualification for a first flight in 2027 and a landing planned for 2028, with little schedule margin.

REVIEW QUESTIONS

1. What went wrong on Leonov's spacewalk and on Gemini 9A, and which design and operational changes followed?
2. Use the alveolar gas equation to show why a pure-oxygen suit must be pressurised to at least about 25 kPa. What is the oxygen pressure in the lungs at 40 kPa?
3. Estimate the moment needed to bend an unjointed knee section (radius 8 cm) of a suit at 29.6 kPa and at 56.5 kPa. How do constant-volume joints reduce it?
4. Describe the functions of the bladder, restraint layer and thermal micrometeoroid garment. Why must the restraint, not the bladder, carry the load?
5. An astronaut works for 6 hours at 350 W. Estimate the oxygen consumed, the carbon dioxide produced (RQ 0.85) and the sublimator water needed.
6. Why does a sublimator regulate itself? Why would it be unreliable on the surface of Mars?
7. Calculate R for an astronaut who breathes oxygen for 2 hours at 101.3 kPa before entering a 29.6 kPa suit ($t_{1/2}$ = 360 min). How much longer would be needed to reach R = 1.65?
8. Compare the campout and exercise prebreathe protocols. Why can the ISS not use the Shuttle's staged decompression?
9. What caused the water intrusion on EVA 23, and why was the earlier warning missed? What does this suggest about anomaly handling?
10. List three differences between designing a suit for the ISS and one for Mars, and explain how a suitport would help a Martian rover crew.